

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 2

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\401-E-31006- RP.pvdb

Date of Analysis : Aug 12,2026 9:40am

PV Elite 28, January 2026



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FileName : 401-E-31006- RP

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Vessel Design Summary:

Step: 29 9:41am Aug 12,2026

Vessel Design Summary:

ASME Code, Section VIII Division 2, 2025

Diameter Spec : 1000.000 mm ID
Vessel Design Length, Tangent to Tangent 9570.00 mm
Specified Datum Line Distance 0.00 mm
Shell Side Design Temperature 80 °C
Channel Side Design Temperature 100 °C
Shell Side Design Pressure 16000.000 KPa.
Channel Side Design Pressure 16000.000 KPa.
Shop Shell Side Test Pressure 21991.135 KPa.
Shop Channel Side Test Pressure 21276.111 KPa.
Wind Design Code IS 875 (PART 3) Amd. 1&2 (2020)
Earthquake Design Code IS 1893 RSM Amd. 1&2 (2018)

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Flange	SA-350 LF2	1	...	K03011	+IT
Nozzle	SA-350 LF2	1	...	K03011	+IT
Tubes	SA-334 1	...	...	K03008	+IT
Tubesheet	SA-350 LF2	1	...	K03011	+IT
Flg Bolting	SA-193 B7	...	64 < t <= 100	G41400	
Hrz Bolting	SA-193 B7	...	<= 64	G41400	
Saddles	SA-516 70	...	...	K02700	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized

+Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (KPa. & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Chnl rear cover	16000.000	0.00	16733.000	1.5000	N/A	No
Chnl rear flange	16000.000	0.00	15577.000	1.5000	N/A	No
Chnl. rear	16005.890	0.00	16396.000	1.5000	N/A	No
Shell	16000.491	0.00	16405.000	1.5000	N/A	No
Chnl. front	16005.890	0.00	16396.000	1.5000	N/A	No
Chnl. front flange	16000.000	0.00	15577.000	1.5000	N/A	No
Chnl. front flange	16000.000	0.00	16733.000	1.5000	N/A	No

Liquid Level: 1000.00 mm Dens.: 0.001 kg./cm³ Sp. Gr.: 0.600

Element Types and Properties:

Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Body Flg	270.0	265.0	273.0	265.0	259.2	...	1.00	1.00
Body Flg	590.0	320.0	244.0	236.0	231.7	...	1.00	1.00



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Vessel Design Summary:

Step: 29 9:41am Aug 12,2026

Cylinder	1785.0	1195.0	56.0	56.0	54.6	...	1.00	1.00
Cylinder	7590.0	5610.0	55.0	55.0	53.6	...	1.00	1.00
Cylinder	8980.0	1195.0	56.0	56.0	54.6	...	1.00	1.00
Body Flg	9305.0	320.0	244.0	236.0	231.7	...	1.00	1.00
Body Flg	9570.0	265.0	273.0	265.0	259.2	...	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N1- Channel in	WRC-297	0.975	Shell	Passed
N4- Shell outl	WRC-297	0.976	Shell	Passed
N3- Shell inle	WRC-297	0.976	Shell	Passed
N2- Channel ou	WRC-297	0.975	Shell	Passed

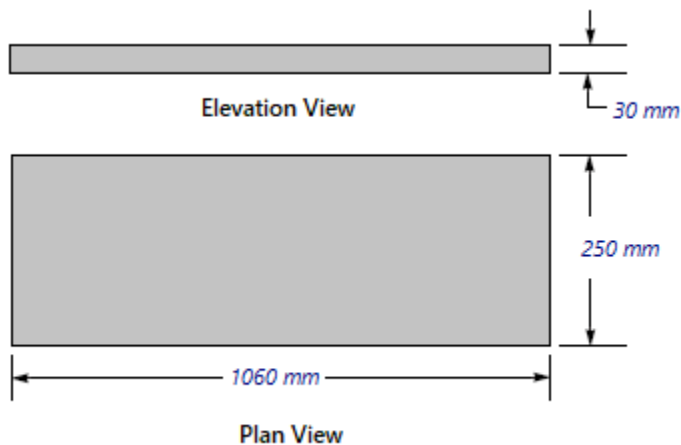
Foundation/Support Summary

Saddle Parameters:

Saddle Width	204.000	mm
Saddle Bearing Angle	120.000	deg.
Centerline Dimension	1050.000	mm
Wear Pad Width	250.000	mm
Wear Pad Thickness	30.000	mm
Wear Pad Bearing Angle	132.000	deg.
Distance from Saddle to Tangent	1122.000	mm

Baseplate Length	1060.000	mm
Baseplate Thickness	30.000	mm
Baseplate Width	250.000	mm
Number of Ribs (including outside ribs)	6	
Rib Thickness	15.000	mm
Web Thickness	15.000	mm
Height of Center Web	306.000	mm
Number of Bolts in Baseplate	4	

Baseplate Sketch



Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	25576.65	Kgf
Maximum Transverse Saddle Shear Load	1749.55	Kgf
Maximum Longitudinal Saddle Shear Load	3499.10	Kgf



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Vessel Design Summary:

Step: 29 9:41am Aug 12,2026

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	22878.13	Kgf
Maximum Transverse Saddle Shear Load	427.39	Kgf
Maximum Longitudinal Saddle Shear Load	98.33	Kgf

Weights:

Fabricated - Bare w/o removable internals	36748.6	kg.
Shop Test - Fabricated + water (full)	42869.7	kg.
Shipping - Fab. + rem. intls.+ shipping app.	36974.8	kg.
Erected - Fab. + rem. intls.+ insul. (etc)	36974.8	kg.
Empty - Fab. + intls. + details + wgths.	36974.8	kg.
Operating - Empty + operating liquid (no ca)	38287.0	kg.

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Nozzle Summary: Step: 27 9:40am Aug 12,2026

Nozzle Calculation Summary (KPa. & mm):

Description	MAWP	Ext	MAPNC	Tbl 4.5.2	Weld	Pass/Fail
N1- Channel inl	16402	...	...	OK	OK	Passed
N4- Shell outle	16405	...	...	OK	OK	Passed
N3- Shell inlet	16405	...	...	OK	OK	Passed
N2- Channel out	16396	...	...	OK	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 16405 Nozzle : N4- Shell outle [Shellside]
 Minimum MAWP Nozzles : 16396 Nozzle : N2- Channel out [Tubeside]

Note: MAWPs (Internal Case) shown above are at the High Point.

VIII-2 Nozzle to Gross Structural Discontinuity Check:

No Nozzles were too close to any gross structural discontinuities per 4.5.13.1

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Inside Radius mm
30	N1- Channel inle	1190.000	270.000	204.100
40	N4- Shell outlet	2680.000	270.000	180.700
40	N3- Shell inlet	6680.000	270.000	180.700
50	N2- Channel outl	8385.000	90.000	204.100

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Step: 26 9:40am Aug 12,2026

Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Diameter	Pad Thk mm	Cut Length mm	Flg Class
N4- Shell outlet	16.000 in	Actual	None	16.000	24.000	...	...	383.66	...
N3- Shell inlet	16.000 in	Actual	None	16.000	24.000	...	...	383.66	...
N1- Channel inle	18.000 in	Actual	None	18.000	26.000	...	...	407.96	...
N2- Channel outl	18.000 in	Actual	None	18.000	26.000	...	...	407.96	...

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
N4- Shell o	SA-350 LF2	55.000	10.000	...	...	...
N3- Shell i	SA-350 LF2	55.000	10.000	...	...	...
N1- Channel	SA-350 LF2	50.000	30.000	...	...	...
N2- Channel	SA-350 LF2	30.000	30.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
N4- Shell outlet	2680.000	270.0	250.00	0.00	Shell
N3- Shell inlet	6680.000	270.0	250.00	0.00	Shell
N1- Channel inle	1190.000	270.0	250.00	0.00	Chnl. rear
N2- Channel outl	8385.000	90.0	250.00	0.00	Chnl. front

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Bill of Material:

Step: 25 9:40am Aug 12,2026

Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	BLIND FLANGE: 273.0mm THK X 1520.0mm OD	SA-350 LF2
2	WELD NECK FLANGE: 244.0mm THK X 1520.0mm OD	SA-350 LF2
2	INSULATION: 1195mm X 40mm THK	
7	BAFFLES: 990.0mm DIA X 8.0mm THK	SA-516 70
2	TUBESHEET: 1110.0mm X 195.0mm THK	SA-350 LF2
1015	TUBES: 6000.0mm X 19.0mm DIA X 2.5mm THK	SA-334 1
2	SADDLE: 204.0mm X 120 DEG	SA-516 70
1	INSULATION: 5600mm X 115mm THK	Cold
2	GASKET: 1029.0mm OD X 1003.0mm ID	...
48	BODY FLANGE BOLTS: 82.6mm DIA	SA-193 B7
96	NUTS FOR BODY FLANGE BOLTS: 82.6mm DIA	...
2	CYLINDRICAL SEGMENT 56.0mm THK X 1247.9mm X 1195.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 56.0mm THK X 645.8mm X 1195.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 55.0mm THK X 2522.0mm X 5610.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 55.0mm THK X 619.6mm X 5610.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 56.0mm THK X 619.6mm X 1195.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 56.0mm THK X 2522.0mm X 1195.0mm	SA-516 70
1	NAMEPLATE	...

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MDMT Summary:

Step: 28 9:41am Aug 12,2026

Minimum Design Metal Temperature Results Summary (°C):

Description	Curve	Basic MDMT	Reduced MDMT	PWHT	Thickness ratio	Gov Thk mm	E*
Notes							
Shell [8]	B	11	7	Yes	0.921	55.000	1.00
Left Saddle [24]	B	-1	-1	Yes	1.000	30.000	1.00
Right saddle [24]	B	-1	-1	Yes	1.000	30.000	1.00
N4- Shell outl[1]	B	11	6	Yes	0.898	55.000	1.00
N3- Shell inle[1]	B	11	6	Yes	0.898	55.000	1.00
Tubesheet: SS[13]	B	11	-8	Yes	0.671	55.000	1.00
Warmest MDMT:		11	7				
Chnl rear cov[11]	!	-46		No			
Chnl rear fla[11]	!	-46		No			
Chnl. front f[11]	!	-46		Yes			
Chnl. front f[11]	!	-46		No			
Chnl. rear [8]	B	11	4	Yes	0.857	56.000	1.00
Chnl. front [8]	B	11	4	Yes	0.857	56.000	1.00
N1- Channel in[1]	B	11	5	Yes	0.881	56.000	1.00
N2- Channel ou[1]	B	11	5	Yes	0.881	56.000	1.00
Tubesheet: CS[14]	B	11	3	Yes	0.842	56.000	1.00
Bolting [21]		-40					
Bolting [21]		-40					
Warmest MDMT:		11	5				
Exchanger Side	Computed MDMT		Required MDMT		Pass/Fail		
Shell	7.0		-33.0		Fail		
Channel/Tube	5.0		-33.0		Fail		

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per equation 3.1.
- [5] - ASME Flange MDMT Calcs; Thickness ratio per equation 3.3.
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per 3.11.2.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld



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Warnings and Errors:

Step: 0 9:40am Aug 12,2026

Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.

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Input Echo:

Step: 1 9:40am Aug 12,2026

Units used in this Analysis (MM):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	6.8946	KPa.
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	16000	KPa.
Channel Side Design Pressure	16000	KPa.
Shell Side Design Temperature	80.0	°C
Channel Side Design Temperature	100.0	°C
Radiography, Shell Side	RT-1	
Radiography, Channel Side	RT-1	
Service Type, Shell Side	Sour	
Service Type, Channel Side	Sour	
MDMT (CET), Shell Side	-33.0	°C
MDMT (CET), Tube Side	-33.0	°C
User defined MAWP, Shell Side	0	KPa.
User defined MAWP, Channel Side	0	KPa.
User defined MAPnc, Shell Side	0	KPa.
User defined MAPnc, Channel Side	0	KPa.
User defined Test Pres., Shell Side	0	KPa.
User defined Test Pres., Channel Side	0	KPa.

Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	N	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070



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Input Echo: Step: 1 9:40am Aug 12,2026

0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Consider Vortex Shedding

N

Shop Pressure Test:

Type of Pressure Test	Hydrostatic
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	IS 875 (PART 3):2015-2020
Basic Wind Speed for IS-875	180 Km/hr
Wind Zone Number	5
Base Elevation	300 mm
Percent Wind for Hydrotest	33.0
Risk Factor [k1]	1.0
Terrain Category	2
Equipment Class	Class A
Topography Factor [k3]	1.0
Importance Factor [k4]	1.15
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000

Seismic Design Code	IS 1893-RSM
Importance Factor for IS-1893	2.000



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Input Echo: Step: 1 9:40am Aug 12,2026

Soil Factor	1.000
Zone Number	2.000
Percent Seismic for Hydrotest	0.000
Period of Vibration (User defined) Sec	0.000
Damping Factor	2.000

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Consider Vertical Acceleration	Yes
Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N
Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Using Metric Material Databases, ASME II D	Yes
Using ASME VIII-1 Appendix 46	No
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/7
Element From Node	10
Element To Node	20
Element Type	Flange
Description	Chnl rear cover
Distance "FROM" to "TO"	265 mm
Flange Outside Diameter	1520 mm
Element Thickness	265 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	273 mm
External Corrosion Allowance	0 mm



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Input Echo:

Step: 1 9:40am Aug 12,2026

Design Internal Pressure	16000	KPa.
Design Temperature Internal Pressure	100	°C
Design External Pressure	0	KPa.
Design Temperature External Pressure	40	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-350 LF2	+IT
Allowable Stress, Ambient	165	N./mm ²
Allowable Stress, Operating	151	N./mm ²
Allowable Stress, Hydrotest	237.5	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	0	mm
Yield Stress, Operating	227	N./mm ²
MDMT Curve	Impact Tested	
External Pressure Chart Name	CS-2	
UNS Number	K03011	
Class / Thickness / Grade	1::	
Product Form	Forgings	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	

Element#	2/7
Element From Node	20
Element To Node	30
Element Type	Flange
Description	Chnl rear flange
Distance "FROM" to "TO"	320 mm
Flange Inside Diameter	1000 mm
Element Thickness	236 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	244 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	16000 KPa.
Design Temperature Internal Pressure	100 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	40 °C
Effective Diameter Multiplier	1.2
Material Name	SA-350 LF2 +IT
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element#	3/7
Element From Node	30
Element To Node	40
Element Type	Cylinder
Description	Chnl. rear
Distance "FROM" to "TO"	1195 mm
Inside Diameter	1000 mm
Element Thickness	56 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	56 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	16000 KPa.
Design Temperature Internal Pressure	100 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	40 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	175 N./mm ²
Allowable Stress, Operating	159 N./mm ²
Allowable Stress, Hydrotest	247 N./mm ²
Material Density	0.00775 kg./cm ³



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Input Echo:

Step: 1 9:40am Aug 12,2026

P Number Thickness	0	mm
Yield Stress, Operating	239	N./mm ²
MDMT Curve	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	30	
Detail Type	Liquid	
Detail ID	Liquid: 30	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	1000	mm
Liquid Density	0.000599	kg./cm ³

Element From Node	30	
Detail Type	Insulation	
Detail ID	Cold insulation	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	1195	mm
Thickness of Insulation	40	mm
Density	0.80000E-04	kg./cm ³

Element From Node	30	
Detail Type	Nozzle	
Detail ID	N1- Channel inle	
Dist. from "FROM" Node / Offset dist	600	mm
Nozzle Diameter	18	in.
Nozzle Schedule	None	
Nozzle Class	None	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	351.11	Kgf
Grade of Attached Flange	None	
Nozzle Matl	SA-350 LF2 +IT	

Element#	4/7	
Element From Node	40	
Element To Node	50	
Element Type	Cylinder	
Description	Shell	
Distance "FROM" to "TO"	5610	mm
Inside Diameter	1000	mm
Element Thickness	55	mm
Internal Corrosion Allowance	1.5	mm
Nominal Thickness	55	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	16000	KPa.
Design Temperature Internal Pressure	80	°C
Design External Pressure	0	KPa.
Design Temperature External Pressure	40	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	40	
Detail Type	Saddle	
Detail ID	Left Saddle	
Dist. from "FROM" Node / Offset dist	1122	mm



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Input Echo:

Step: 1 9:40am Aug 12,2026

Width of Saddle	204	mm
Height of Saddle at Bottom	1050	mm
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	250	mm
Thickness of Wear Plate	30	mm
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.0	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm

Element From Node	40	
Detail Type	Saddle	
Detail ID	Right saddle	
Dist. from "FROM" Node / Offset dist	4488	mm
Width of Saddle	204	mm
Height of Saddle at Bottom	1050	mm
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	250	mm
Thickness of Wear Plate	30	mm
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.0	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm

Element From Node	40	
Detail Type	Liquid	
Detail ID	Liquid: 40	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	1000	mm
Liquid Density	0.49920E-04	kg./cm ³

Element From Node	40	
Detail Type	Insulation	
Detail ID	Cold insulation	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	5600	mm
Thickness of Insulation	115	mm
Density	0.80000E-04	kg./cm ³

Element From Node	40	
Detail Type	Nozzle	
Detail ID	N4- Shell outlet	
Dist. from "FROM" Node / Offset dist	700	mm
Nozzle Diameter	16	in.
Nozzle Schedule	None	
Nozzle Class	None	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	273.07	Kgf
Grade of Attached Flange	None	
Nozzle Matl	SA-350 LF2 +IT	

Element From Node	40	
Detail Type	Nozzle	
Detail ID	N3- Shell inlet	
Dist. from "FROM" Node / Offset dist	4700	mm
Nozzle Diameter	16	in.



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Input Echo:

Step: 1 9:40am Aug 12,2026

Nozzle Schedule	None
Nozzle Class	None
Layout Angle	270.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	273.07 Kgf
Grade of Attached Flange	None
Nozzle Matl	SA-350 LF2 +IT

Element#	5/7
Element From Node	50
Element To Node	60
Element Type	Cylinder
Description	Chnl. front
Distance "FROM" to "TO"	1195 mm
Inside Diameter	1000 mm
Element Thickness	56 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	56 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	16000 KPa.
Design Temperature Internal Pressure	100 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	40 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No
Element From Node	50
Detail Type	Liquid
Detail ID	Liquid: 50
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	1000 mm
Liquid Density	0.000599 kg./cm³
Element From Node	50
Detail Type	Insulation
Detail ID	Cold insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	1195 mm
Thickness of Insulation	40 mm
Density	0.80000E-04 kg./cm³
Element From Node	50
Detail Type	Nozzle
Detail ID	N2- Channel outl
Dist. from "FROM" Node / Offset dist	600 mm
Nozzle Diameter	18 in.
Nozzle Schedule	None
Nozzle Class	None
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	351.11 Kgf
Grade of Attached Flange	None
Nozzle Matl	SA-350 LF2 +IT

Element#	6/7
Element From Node	60
Element To Node	70
Element Type	Flange



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Input Echo:

Step: 1 9:40am Aug 12,2026

Description	Chnl. front flange
Distance "FROM" to "TO"	320 mm
Flange Inside Diameter	1000 mm
Element Thickness	236 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	244 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	16000 KPa.
Design Temperature Internal Pressure	100 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	40 °C
Effective Diameter Multiplier	1.2
Material Name	SA-350 LF2 +IT
Allowable Stress, Ambient	165 N./mm ²
Allowable Stress, Operating	151 N./mm ²
Allowable Stress, Hydrotest	237.5 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	0 mm
Yield Stress, Operating	227 N./mm ²
MDMT Curve	Impact Tested
External Pressure Chart Name	CS-2
UNS Number	K03011
Class / Thickness / Grade	1::
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element#	7/7
Element From Node	70
Element To Node	80
Element Type	Flange
Description	Chnl. front flange
Distance "FROM" to "TO"	265 mm
Flange Outside Diameter	1520 mm
Element Thickness	265 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	273 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	16000 KPa.
Design Temperature Internal Pressure	100 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	40 °C
Effective Diameter Multiplier	1.2
Material Name	SA-350 LF2 +IT
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No



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XY Coordinate Calculations: Step: 2 9:40am Aug 12,2026

XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
Chnl rear cover		270	...	265	...
Chnl rear flang		590	...	320	...
Chnl. rear		1785	...	1195	...
Shell		7590	...	5610	...
Chnl. front		8980	...	1195	...
Chnl. front fla		9305	...	320	...
Chnl. front fla		9570	...	265	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Flg Calc [Int P]: New Flange

Flng: 9 9:40am Aug 12,2026

Flange Input Data Values

Description: New Flange :

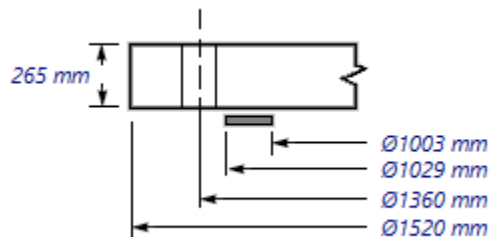
Chnl rear cover

Flange Type		Blind
Design Pressure	P	16000.00 KPa.
Design Temperature		100 °C
Internal Corrosion Allowance	ci	1.5000 mm
External Corrosion Allowance	ce	0.0000 mm
Use Corrosion Allowance in Thickness Calcs.		Yes
Flange Outside Diameter	A	1520.000 mm
Flange Thickness	t	265.0000 mm
Flange Material		SA-350 LF2 +IT
Flange Material UNS number		K03011
Flange Allowable Stress At Temperature	Sfo	151.00 N./mm ²
Flange Allowable Stress At Ambient	Sfa	165.00 N./mm ²
Bolt Material		SA-193 B7
Bolt Allowable Stress At Temperature	Sb	159.00 N./mm ²
Bolt Allowable Stress At Ambient	Sa	159.00 N./mm ²

Warning:

It seems as if the selected bolt material does not match the bolt diameter range allowed per the material specification. Please check the material selection for the given nominal bolt diameter.

Perimeter along the Center of the Bolts	L	4272.566 mm
Diameter of Bolt Circle	C	1360.000 mm
Nominal Bolt Diameter	a	82.5500 mm
Type of Threads	TEMA Thread Series	
Number of Bolts		24
Flange Face Outside Diameter	Fod	1032.000 mm
Flange Face Inside Diameter	Fid	1000.000 mm
Flange Facing Sketch		Sketch 1a
Gasket Outside Diameter	Go	1029.000 mm
Gasket Inside Diameter	Gi	1003.000 mm
Gasket Factor	m	4.2500
Gasket Design Seating Stress	y	69.64 N./mm ²
Gasket Thickness	tg	5.0000 mm
Raised face depth	Fd	0.0000 mm



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Gasket Contact Width,	$N = (Go - Gi)/2$	13.000 mm
Basic Gasket Width,	$bo = N/2$	6.500 mm



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Flg Calc [Int P]: New Flange Flng: 9 9:40am Aug 12, 2026

Effective Gasket Width, $b = Cb \cdot \sqrt{bo}$ 6.425 mm
Gasket Reaction Diameter, $G = Go - 2 \cdot b$ 1016.151 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 1016^2 \cdot 16000$$

$$= 1323191.625 \text{ Kg}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.42 \cdot \pi \cdot 1016 \cdot 4.25 \cdot 16000$$

$$= 284437.438 \text{ Kg}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(1323191 + 284437 + 0, 0)$$

$$= 1607629.125 \text{ Kg}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 69.6 \cdot 6.42 \cdot \pi \cdot 1016 + 0 \cdot 0 \cdot 0$$

$$= 145639.188 \text{ Kg}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg / Sa)$$

$$= \max((1607629 + 0 + 4 \cdot 0 / 1016) / 159, 145639 / 159)$$

$$= 991.574 \text{ cm}^2$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	991.574	1159.275	
Radial Distance between Bolts and Edge:	76.200	80.000	
Circ. Spacing between the Bolts:	171.450	177.516	...

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi (Go + Gi))$$

$$= 1159 \cdot 159 / (69.6 \cdot \pi (1029 + 1003))$$

$$= 41.462 \text{ mm} [\text{Note: Exceeds actual gasket width, 13.000}]$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa (Am + Ab) / 2$$

$$= 159 (992 + 1159) / 2$$

$$= 1743574.88 \text{ Kg}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo$$

$$= 1607629.12 \text{ Kg}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1360 - 1016) / 2$$

$$= 171.9246 \text{ mm}$$

Flange Stresses

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:

$$= 1.9 \cdot Wm1 \cdot hG \cdot Bsc / (t^2 \cdot G) + C \cdot Z \cdot Peq \cdot G^2 / t^2$$

$$= 1.9 \cdot 1607629 \cdot 172 \cdot 1 / (264^2 \cdot 1016) + 0.3 \cdot 1 \cdot 16000 \cdot 1016^2 / 264^2$$



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Flg Calc [Int P]: New Flange

Flng: 9 9:40am Aug 12, 2026

= 144.38 N./mm²

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:

= $1.9 \cdot W \cdot hG \cdot Bsc / (t^2 \cdot G)$

= $1.9 \cdot 1743574 \cdot 172 \cdot 1 / (264^2 \cdot 1016)$

= 79.17 N./mm²

Bolt Stress, Operating [BSo]:

= $Wm1 / Ab$

= 1607629/1159

= 136.00 N./mm²

Bolt Stress, Seating [BSa]:

= $(Wm2 / Ab)$

= (145639/1159)

= 12.32 N./mm²

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Tangential Flange	144.38	151.00	79.17	165.00
Bolting	136.00	159.00	12.32	159.00

Reqd. Blind Flange Thickness at Center 259.162 mm

Reqd. Blind Flange Thickness at Gasket 184.705 mm

Estimated M.A.W.P. (Operating) 16733 KPa.

Estimated Finished Weight of Flange at given Thk. 3726.9 kg.

Estimated Unfinished Weight of Forging at given Thk 3726.9 kg.

Minimum Design Metal Temperature Results:

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Tangential Flange	142.75	165.00	78.27	165.00
Bolting	136.00	159.00	12.32	159.00

Minimum Req'd Thickness, ambient, no Corrosion: 246.507 mm

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Flg Calc [Int P]: FLANGE

Flng: 10 9:40am Aug 12,2026

Flange Input Data Values

Description: FLANGE :

Chnl rear flange

Flange Type	Integral Weld Neck		
Design Pressure	P	16000.00	KPa.
Design Temperature		100	°C
Internal Corrosion Allowance	ci	1.5000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	1000.000	mm
Flange Outside Diameter	A	1520.000	mm
Flange Thickness	t	236.0000	mm
Thickness of Hub at Small End	go	56.0000	mm
Thickness of Hub at Large End	gl	83.0000	mm
Length of Hub	h	84.0000	mm
Flange Material	SA-350 LF2 +IT		
Flange Material UNS number	K03011		
Flange Allowable Stress At Temperature	Sfo	151.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	165.00	N./mm ²
Bolt Material	SA-193 B7		
Bolt Allowable Stress At Temperature	Sb	159.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	159.00	N./mm ²

Warning:

It seems as if the selected bolt material does not match the bolt diameter range allowed per the material specification. Please check the material selection for the given nominal bolt diameter.

Diameter of Bolt Circle	C	1360.000	mm
Nominal Bolt Diameter	a	82.5500	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		24	
Flange Face Outside Diameter	Fod	1032.000	mm
Flange Face Inside Diameter	Fid	1000.000	mm
Flange Facing Sketch	Sketch 1a		
Gasket Outside Diameter	Go	1029.000	mm
Gasket Inside Diameter	Gi	1003.000	mm
Gasket Factor	m	4.2500	
Gasket Design Seating Stress	y	69.64	N./mm ²
Gasket Thickness	tg	5.0000	mm
Raised face depth	Fd	0.0000	mm



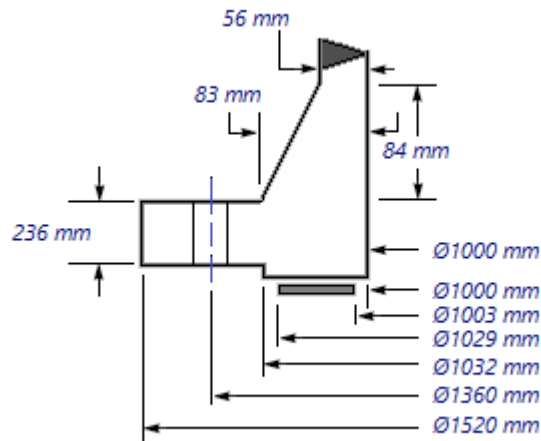
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Flg Calc [Int P]: FLANGE

Flng: 10 9:40am Aug 12, 2026



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Hub Small End Required Thickness due to Internal Pressure:

$$= (P(R + Ca) / (S - 0.5 \cdot P)) + (Ca + Ca_{ext})$$

$$= (16000(1000 + 1.5) / (151 - 0.5 \cdot 16000)) + (1.5)$$

$$= 57.6159 \text{ mm} \quad \text{Warning: Less than specified Small End Hub thickness}$$

Small End Hub MAWP:

$$= S \cdot \log((2 \cdot T + R) / R)$$

$$= 151 \cdot \log((2 \cdot 54.5 + 502) / 502)$$

$$= 15576.830 \text{ KPa.}$$

Small End Hub MAPnc:

$$= S_a \cdot \log((2.0 \cdot T + R) / R)$$

$$= 165 \cdot \log((2.0 \cdot 56 + 500) / 500)$$

$$= 17515.096 \text{ KPa.}$$

Corroded Flange Thickness, $t_c = T - c_i$	234.500 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	1003.000 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	81.500 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	54.500 mm
Code R Dimension, $R = ((C - B_{cor}) / 2) - g1_{cor}$	97.000 mm

Gasket Contact Width, $N = (G_o - G_i) / 2$	13.000 mm
Basic Gasket Width, $b_o = N / 2$	6.500 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{b_o}$	6.425 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	1016.151 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi / 4 \cdot G^2 \cdot P$$

$$= \pi / 4 \cdot 1016^2 \cdot 16000$$

$$= 1323191.625 \text{ Kg}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.42 \cdot \pi \cdot 1016 \cdot 4.25 \cdot 16000$$

$$= 284437.438 \text{ Kg}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi / 4 \cdot B_{cor}^2 \cdot P$$

$$= \pi / 4 \cdot 1003^2 \cdot 16000$$

$$= 1289164.250 \text{ Kg}$$



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Flg Calc [Int P]: FLANGE

Flng: 10 9:40am Aug 12, 2026

Pressure Force on Flange Face [Ht]:

$$\begin{aligned} &= H - H_d \\ &= 1323191 - 1289164 \\ &= 34027.371 \text{ Kgf} \end{aligned}$$

Design Bolt Load for the Operating Condition [Wo]:

$$\begin{aligned} &= \max(W_{o,1} + W_{o,2} + H'p, 0) \\ &= \max(1323191 + 284437 + 0, 0) \\ &= 1607629.125 \text{ Kgf} \end{aligned}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$\begin{aligned} &= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p \\ &= 69.6 \cdot 6.42 \cdot \pi \cdot 1016 + 0 \cdot 0 \cdot 0 \\ &= 145639.188 \text{ Kgf} \end{aligned}$$

Required Bolt Area [Am]:

$$\begin{aligned} &= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a) \\ &= \max((1607629 + 0 + 4 \cdot 0 / 1016) / 159, 145639 / 159) \\ &= 991.574 \text{ cm}^2 \end{aligned}$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$\begin{aligned} &= 2a + 6t / (m + 0.5) \\ &= 2 \cdot 82.5 + 6 \cdot 234 / (4.25 + 0.5) \\ &= 461.310 \text{ mm} \end{aligned}$$

Actual Circumferential Bolt Spacing [Bs]:

$$\begin{aligned} &= C \cdot \sin(\pi / n) \\ &= 1360 \cdot \sin(\pi / 24) \\ &= 177.516 \text{ mm} \end{aligned}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$\begin{aligned} &= \max(\sqrt{ B_s / (2a + t) }, 1) \\ &= \max(\sqrt{ 178 / (2 \cdot 82.5 + 234) }, 1) \\ &= 1.0000 \end{aligned}$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	991.574	1159.275	
Radial Distance between Hub and Bolts:	95.250	97.000	
Radial Distance between Bolts and Edge:	76.200	80.000	
Circ. Spacing between the Bolts:	171.450	177.516	461.310

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$\begin{aligned} &= A_b \cdot S_a / (y \cdot \pi (G_o + G_i)) \\ &= 1159 \cdot 159 / (69.6 \cdot \pi (1029 + 1003)) \\ &= 41.462 \text{ mm} [\text{Note: Exceeds actual gasket width, 13.000}] \end{aligned}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$\begin{aligned} &= S_a (A_m + A_b) / 2 \\ &= 159 (992 + 1159) / 2 \\ &= 1743574.88 \text{ Kgf} \end{aligned}$$

Gasket Load for the Operating Condition [HG]:

$$\begin{aligned} &= W_o - H \\ &= 1607629 - 1323191 \\ &= 284437.47 \text{ Kgf} \end{aligned}$$

Moment Arm Calculations:



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Flg Calc [Int P]: FLANGE

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Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1360 - 1016) / 2$$

$$= 171.9246 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5 ((C - B) / 2 + hg)$$

$$= 0.5 ((1360 - 1003) / 2 + 172)$$

$$= 175.2123 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (1360 - 1003 - 81.5) / 2$$

$$= 137.7500 \text{ mm}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	1289164.	137.7500	1.0000	177586.
Face Pressure, Mt	34027.	175.2123	1.0000	5962.
Gasket Load, Mg	284437.	171.9246	1.0000	48903.
Gasket Seating, Matm	1743575.	171.9246	1.0000	299769.

Total Moment for Operation, Mo 232451. Kg-m.
Total Moment for Gasket seating, Mg 299769. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 233.802 mm
Hub Ratio, h/h0 = HL / H0 0.359
Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.495

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.871
Factor V per 4.16.5 0.352
Factor f per 4.16.5 1.000

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 1520 / 1003$$

$$= 1.515$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169 (K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169 (1.52^2 \cdot \log_{10}(1.52))) / (1.52^2 - 1)) / (1.52 - 1)$$

$$= 4.844$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.52^2 (1 + 8.55246 \cdot \log_{10}(1.52)) - 1) / ((1.0472 + 1.9448 \cdot 1.52^2) \cdot (1.52 - 1))$$

$$= 1.704$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136 (K^2 - 1) \cdot (K - 1))$$

$$= (1.52^2 (1 + 8.55246 \cdot \log_{10}(1.52)) - 1) / (1.36136 (1.52^2 - 1) \cdot (1.52 - 1))$$

$$= 5.323$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.52^2 + 1) / (1.52^2 - 1)$$

$$= 2.542$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g0^2 \cdot h0 / V$$

$$= 5.32 \cdot 54.5^2 \cdot 234 / 0.35$$



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Flg Calc [Int P]: FLANGE

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$$= 10507500.000 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.87/234$$

$$= 0.004 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3/d$$

$$= (234 \cdot 0.0037 + 1) / 1.7 + 234^3/10507500$$

$$= 2.327$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot M_{op} / B_{cor}) / (L \cdot g_1^2)$$

$$= (1 \cdot 232451/1003) / (2.33 \cdot 81.5^2)$$

$$= 147.07 \text{ N./mm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot M_{atm} / B_{cor}) / (L \cdot g_1^2)$$

$$= (1 \cdot 299769/1003) / (2.33 \cdot 81.5^2)$$

$$= 189.66 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot M_{op} / B_{cor}) / (L \cdot t^2)$$

$$= (2.16 \cdot 232451/1003) / (2.33 \cdot 234^2)$$

$$= 38.45 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot M_{atm} / B_{cor}) / (L \cdot t^2)$$

$$= (2.16 \cdot 299769/1003) / (2.33 \cdot 234^2)$$

$$= 49.58 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot M_{op} / (t^2 \cdot B_{cor})) - Z \cdot S_{Ro}$$

$$= (4.84 \cdot 232451 / (234^2 \cdot 1003)) - 2.54 \cdot 38.4$$

$$= 102.43 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot M_{atm} / (t^2 \cdot B_{cor})) - Z \cdot S_{Ra}$$

$$= (4.84 \cdot 299769 / (234^2 \cdot 1003)) - 2.54 \cdot 49.6$$

$$= 132.10 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (S_{Ho} + \max(S_{Ro}, S_{To})) / 2$$

$$= (147 + \max(38.4, 102)) / 2$$

$$= 124.75 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (S_{Ha} + \max(S_{Ra}, S_{Ta})) / 2$$

$$= (190 + \max(49.6, 132)) / 2$$

$$= 160.88 \text{ N./mm}^2$$

Bolt Stress, Operating [BSo]:

$$= W_{m1} / A_b$$

$$= 1607629/1159$$

$$= 136.00 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (W_{m2} / A_b)$$

$$= (145639/1159)$$

$$= 12.32 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²



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Flg Calc [Int P]: FLANGE

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	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	147.07	226.50	189.66	247.50
Radial Flange	38.45	151.00	49.58	165.00
Tangential Flange	102.43	151.00	132.10	165.00
Maximum Average	124.75	151.00	160.88	165.00
Bolting	136.00	159.00	12.32	159.00

Minimum Required Flange Thickness 231.750 mm

Estimated M.A.W.P. (Operating) 15577 KPa.

Estimated Finished Weight of Flange at given Thk. 2034.5 kg.

Estimated Unfinished Weight of Forging at given Thk 2552.5 kg.

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 299769 / 1 \cdot 9807 \cdot 0.35 / (2.25 \cdot 201334 \cdot 54.5^2 \cdot 234 \cdot 0.3)$$

$$= 0.570 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 232451 / 1 \cdot 9807 \cdot 0.35 / (2.25 \cdot 196784 \cdot 54.5^2 \cdot 234 \cdot 0.3)$$

$$= 0.452 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 299769 / 1 \cdot 9807 \cdot 0.35 / (2.33 \cdot 201334 \cdot 54.5^2 \cdot 234 \cdot 0.3)$$

$$= 0.552 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 232451 / 1 \cdot 9807 \cdot 0.35 / (2.33 \cdot 196784 \cdot 54.5^2 \cdot 234 \cdot 0.3)$$

$$= 0.438 \text{ (should be } \leq 1)$$

Minimum Design Metal Temperature Results:

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	145.21	247.50	186.23	247.50
Radial Flange	38.79	165.00	49.74	165.00
Tangential Flange	99.98	165.00	128.23	165.00
Maximum Average	122.59	165.00	157.23	165.00
Bolting	136.00	159.00	12.32	159.00

Minimum Req'd Thickness, ambient, no Corrosion: 227.762 mm



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Flg Calc [Int P]: New Flange

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Flange Input Data Values

Description: New Flange :

Chnl. front flange

Flange Type	Integral Weld Neck		
Design Pressure	P	16000.00	KPa.
Design Temperature		100	°C
Internal Corrosion Allowance	ci	1.5000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	1000.000	mm
Flange Outside Diameter	A	1520.000	mm
Flange Thickness	t	236.0000	mm
Thickness of Hub at Small End	go	56.0000	mm
Thickness of Hub at Large End	gl	83.0000	mm
Length of Hub	h	84.0000	mm
Flange Material	SA-350 LF2 +IT		
Flange Material UNS number	K03011		
Flange Allowable Stress At Temperature	Sfo	151.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	165.00	N./mm ²
Bolt Material	SA-193 B7		
Bolt Allowable Stress At Temperature	Sb	159.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	159.00	N./mm ²

Warning:

It seems as if the selected bolt material does not match the bolt diameter range allowed per the material specification. Please check the material selection for the given nominal bolt diameter.

Diameter of Bolt Circle	C	1360.000	mm
Nominal Bolt Diameter	a	82.5500	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		24	
Flange Face Outside Diameter	Fod	1032.000	mm
Flange Face Inside Diameter	Fid	1000.000	mm
Flange Facing Sketch	Sketch 1a		
Gasket Outside Diameter	Go	1029.000	mm
Gasket Inside Diameter	Gi	1003.000	mm
Gasket Factor	m	4.2500	
Gasket Design Seating Stress	y	69.64	N./mm ²
Gasket Thickness	tg	5.0000	mm
Raised face depth	Fd	0.0000	mm



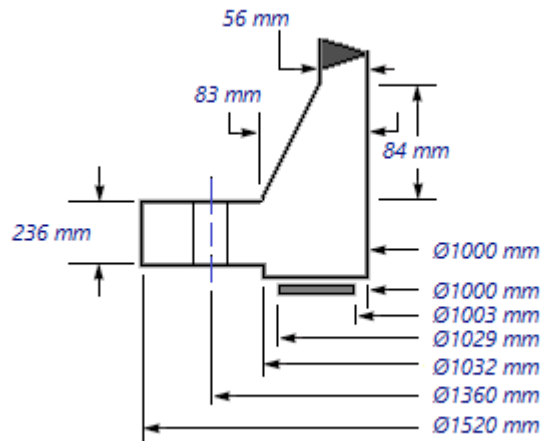
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Flg Calc [Int P]: New Flange

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Hub Small End Required Thickness due to Internal Pressure:

$$= (P(R + Ca) / (S - 0.5 \cdot P)) + (Ca + Ca_{ext})$$

$$= (16000(1000 + 1.5) / (151 - 0.5 \cdot 16000)) + (1.5)$$

$$= 57.6159 \text{ mm} \quad \text{Warning: Less than specified Small End Hub thickness}$$

Small End Hub MAWP:

$$= S \cdot \log((2 \cdot T + R) / R)$$

$$= 151 \cdot \log((2 \cdot 54.5 + 502) / 502)$$

$$= 15576.830 \text{ KPa.}$$

Small End Hub MAPnc:

$$= S_a \cdot \log((2.0 \cdot T + R) / R)$$

$$= 165 \cdot \log((2.0 \cdot 56 + 500) / 500)$$

$$= 17515.096 \text{ KPa.}$$

Corroded Flange Thickness, $t_c = T - c_i$	234.500 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	1003.000 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	81.500 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	54.500 mm
Code R Dimension, $R = ((C - B_{cor}) / 2) - g1_{cor}$	97.000 mm

Gasket Contact Width, $N = (G_o - G_i) / 2$	13.000 mm
Basic Gasket Width, $b_o = N / 2$	6.500 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{b_o}$	6.425 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	1016.151 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi / 4 \cdot G^2 \cdot P$$

$$= \pi / 4 \cdot 1016^2 \cdot 16000$$

$$= 1323191.625 \text{ Kg}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.42 \cdot \pi \cdot 1016 \cdot 4.25 \cdot 16000$$

$$= 284437.438 \text{ Kg}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi / 4 \cdot B_{cor}^2 \cdot P$$

$$= \pi / 4 \cdot 1003^2 \cdot 16000$$

$$= 1289164.250 \text{ Kg}$$



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Flg Calc [Int P]: New Flange

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Pressure Force on Flange Face [Ht]:

$$\begin{aligned}
 &= H - H_d \\
 &= 1323191 - 1289164 \\
 &= 34027.371 \text{ Kgf}
 \end{aligned}$$

Design Bolt Load for the Operating Condition [Wo]:

$$\begin{aligned}
 &= \max(W_{o,1} + W_{o,2} + H'p, 0) \\
 &= \max(1323191 + 284437 + 0, 0) \\
 &= 1607629.125 \text{ Kgf}
 \end{aligned}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$\begin{aligned}
 &= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p \\
 &= 69.6 \cdot 6.42 \cdot \pi \cdot 1016 + 0 \cdot 0 \cdot 0 \\
 &= 145639.188 \text{ Kgf}
 \end{aligned}$$

Required Bolt Area [Am]:

$$\begin{aligned}
 &= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a) \\
 &= \max((1607629 + 0 + 4 \cdot 0 / 1016) / 159, 145639 / 159) \\
 &= 991.574 \text{ cm}^2
 \end{aligned}$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$\begin{aligned}
 &= 2a + 6t / (m + 0.5) \\
 &= 2 \cdot 82.5 + 6 \cdot 234 / (4.25 + 0.5) \\
 &= 461.310 \text{ mm}
 \end{aligned}$$

Actual Circumferential Bolt Spacing [Bs]:

$$\begin{aligned}
 &= C \cdot \sin(\pi / n) \\
 &= 1360 \cdot \sin(\pi / 24) \\
 &= 177.516 \text{ mm}
 \end{aligned}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$\begin{aligned}
 &= \max(\sqrt{ B_s / (2a + t) }, 1) \\
 &= \max(\sqrt{ 178 / (2 \cdot 82.5 + 234) }, 1) \\
 &= 1.0000
 \end{aligned}$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	991.574	1159.275	
Radial Distance between Hub and Bolts:	95.250	97.000	
Radial Distance between Bolts and Edge:	76.200	80.000	
Circ. Spacing between the Bolts:	171.450	177.516	461.310

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$\begin{aligned}
 &= A_b \cdot S_a / (y \cdot \pi (G_o + G_i)) \\
 &= 1159 \cdot 159 / (69.6 \cdot \pi (1029 + 1003)) \\
 &= 41.462 \text{ mm} [\text{Note: Exceeds actual gasket width, 13.000}]
 \end{aligned}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$\begin{aligned}
 &= S_a (A_m + A_b) / 2 \\
 &= 159 (992 + 1159) / 2 \\
 &= 1743574.88 \text{ Kgf}
 \end{aligned}$$

Gasket Load for the Operating Condition [HG]:

$$\begin{aligned}
 &= W_o - H \\
 &= 1607629 - 1323191 \\
 &= 284437.47 \text{ Kgf}
 \end{aligned}$$

Moment Arm Calculations:



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Flg Calc [Int P]: New Flange

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Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1360 - 1016) / 2$$

$$= 171.9246 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5 ((C - B) / 2 + hg)$$

$$= 0.5 ((1360 - 1003) / 2 + 172)$$

$$= 175.2123 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (1360 - 1003 - 81.5) / 2$$

$$= 137.7500 \text{ mm}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	1289164.	137.7500	1.0000	177586.
Face Pressure, Mt	34027.	175.2123	1.0000	5962.
Gasket Load, Mg	284437.	171.9246	1.0000	48903.
Gasket Seating, Matm	1743575.	171.9246	1.0000	299769.

Total Moment for Operation, Mo 232451. Kg-m.
Total Moment for Gasket seating, Mg 299769. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 233.802 mm
Hub Ratio, h/h0 = HL / H0 0.359
Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.495

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.871
Factor V per 4.16.5 0.352
Factor f per 4.16.5 1.000

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 1520 / 1003$$

$$= 1.515$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169 (K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169 (1.52^2 \cdot \log_{10}(1.52))) / (1.52^2 - 1)) / (1.52 - 1)$$

$$= 4.844$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.52^2 (1 + 8.55246 \cdot \log_{10}(1.52)) - 1) / ((1.0472 + 1.9448 \cdot 1.52^2) \cdot (1.52 - 1))$$

$$= 1.704$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136 (K^2 - 1) \cdot (K - 1))$$

$$= (1.52^2 (1 + 8.55246 \cdot \log_{10}(1.52)) - 1) / (1.36136 (1.52^2 - 1) \cdot (1.52 - 1))$$

$$= 5.323$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.52^2 + 1) / (1.52^2 - 1)$$

$$= 2.542$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g0^2 \cdot h0 / V$$

$$= 5.32 \cdot 54.5^2 \cdot 234 / 0.35$$



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Flg Calc [Int P]: New Flange

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$$= 10507500.000 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.87/234$$

$$= 0.004 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3/d$$

$$= (234 \cdot 0.0037 + 1) / 1.7 + 234^3/10507500$$

$$= 2.327$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g_1^2)$$

$$= (1 \cdot 232451/1003) / (2.33 \cdot 81.5^2)$$

$$= 147.07 \text{ N./mm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g_1^2)$$

$$= (1 \cdot 299769/1003) / (2.33 \cdot 81.5^2)$$

$$= 189.66 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (2.16 \cdot 232451/1003) / (2.33 \cdot 234^2)$$

$$= 38.45 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (2.16 \cdot 299769/1003) / (2.33 \cdot 234^2)$$

$$= 49.58 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (4.84 \cdot 232451 / (234^2 \cdot 1003)) - 2.54 \cdot 38.4$$

$$= 102.43 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (4.84 \cdot 299769 / (234^2 \cdot 1003)) - 2.54 \cdot 49.6$$

$$= 132.10 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (147 + \max(38.4, 102)) / 2$$

$$= 124.75 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (SHa + \max(SRa, STa)) / 2$$

$$= (190 + \max(49.6, 132)) / 2$$

$$= 160.88 \text{ N./mm}^2$$

Bolt Stress, Operating [BSo]:

$$= Wm1 / Ab$$

$$= 1607629/1159$$

$$= 136.00 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (145639/1159)$$

$$= 12.32 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²



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Flg Calc [Int P]: New Flange

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	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	147.07	226.50	189.66	247.50
Radial Flange	38.45	151.00	49.58	165.00
Tangential Flange	102.43	151.00	132.10	165.00
Maximum Average	124.75	151.00	160.88	165.00
Bolting	136.00	159.00	12.32	159.00

Minimum Required Flange Thickness 231.750 mm

Estimated M.A.W.P. (Operating) 15577 KPa.

Estimated Finished Weight of Flange at given Thk. 2034.5 kg.

Estimated Unfinished Weight of Forging at given Thk 2552.5 kg.

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 299769 / 1 \cdot 9807 \cdot 0.35 / (2.25 \cdot 201334 \cdot 54.5^2 \cdot 234 \cdot 0.3)$$

$$= 0.570 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 232451 / 1 \cdot 9807 \cdot 0.35 / (2.25 \cdot 196784 \cdot 54.5^2 \cdot 234 \cdot 0.3)$$

$$= 0.452 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 299769 / 1 \cdot 9807 \cdot 0.35 / (2.33 \cdot 201334 \cdot 54.5^2 \cdot 234 \cdot 0.3)$$

$$= 0.552 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 232451 / 1 \cdot 9807 \cdot 0.35 / (2.33 \cdot 196784 \cdot 54.5^2 \cdot 234 \cdot 0.3)$$

$$= 0.438 \text{ (should be } \leq 1)$$

Minimum Design Metal Temperature Results:

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	145.21	247.50	186.23	247.50
Radial Flange	38.79	165.00	49.74	165.00
Tangential Flange	99.98	165.00	128.23	165.00
Maximum Average	122.59	165.00	157.23	165.00
Bolting	136.00	159.00	12.32	159.00

Minimum Req'd Thickness, ambient, no Corrosion: 227.762 mm



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Flg Calc [Int P]: New Flange

Flng: 12 9:40am Aug 12,2026

Flange Input Data Values

Description: New Flange :

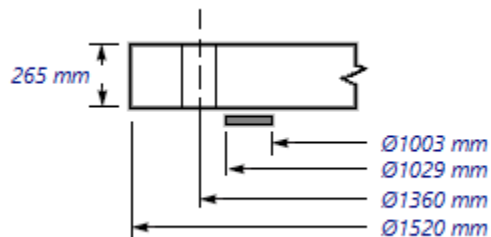
Chnl. front flange

Flange Type		Blind
Design Pressure	P	16000.00 KPa.
Design Temperature		100 °C
Internal Corrosion Allowance	ci	1.5000 mm
External Corrosion Allowance	ce	0.0000 mm
Use Corrosion Allowance in Thickness Calcs.		Yes
Flange Outside Diameter	A	1520.000 mm
Flange Thickness	t	265.0000 mm
Flange Material		SA-350 LF2 +IT
Flange Material UNS number		K03011
Flange Allowable Stress At Temperature	Sfo	151.00 N./mm ²
Flange Allowable Stress At Ambient	Sfa	165.00 N./mm ²
Bolt Material		SA-193 B7
Bolt Allowable Stress At Temperature	Sb	159.00 N./mm ²
Bolt Allowable Stress At Ambient	Sa	159.00 N./mm ²

Warning:

It seems as if the selected bolt material does not match the bolt diameter range allowed per the material specification. Please check the material selection for the given nominal bolt diameter.

Perimeter along the Center of the Bolts	L	4272.566 mm
Diameter of Bolt Circle	C	1360.000 mm
Nominal Bolt Diameter	a	82.5500 mm
Type of Threads	TEMA Thread Series	
Number of Bolts		24
Flange Face Outside Diameter	Fod	1032.000 mm
Flange Face Inside Diameter	Fid	1000.000 mm
Flange Facing Sketch		Sketch 1a
Gasket Outside Diameter	Go	1029.000 mm
Gasket Inside Diameter	Gi	1003.000 mm
Gasket Factor	m	4.2500
Gasket Design Seating Stress	y	69.64 N./mm ²
Gasket Thickness	tg	5.0000 mm
Raised face depth	Fd	0.0000 mm



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Gasket Contact Width,	$N = (Go - Gi)/2$	13.000 mm
Basic Gasket Width,	$bo = N/2$	6.500 mm



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Flg Calc [Int P]: New Flange Flng: 12 9:40am Aug 12,2026

Effective Gasket Width, $b = Cb \cdot \sqrt{bo}$ 6.425 mm
Gasket Reaction Diameter, $G = Go - 2 \cdot b$ 1016.151 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 1016^2 \cdot 16000$$

$$= 1323191.625 \text{ Kg}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.42 \cdot \pi \cdot 1016 \cdot 4.25 \cdot 16000$$

$$= 284437.438 \text{ Kg}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(1323191 + 284437 + 0, 0)$$

$$= 1607629.125 \text{ Kg}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 69.6 \cdot 6.42 \cdot \pi \cdot 1016 + 0 \cdot 0$$

$$= 145639.188 \text{ Kg}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G)/Sb, Wg/Sa)$$

$$= \max((1607629 + 0 + 4 \cdot 0/1016)/159, 145639/159)$$

$$= 991.574 \text{ cm}^2$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	991.574	1159.275	
Radial Distance between Bolts and Edge:	76.200	80.000	
Circ. Spacing between the Bolts:	171.450	177.516	...

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi (Go + Gi))$$

$$= 1159 \cdot 159 / (69.6 \cdot \pi (1029 + 1003))$$

$$= 41.462 \text{ mm} [\text{Note: Exceeds actual gasket width, 13.000 }]$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa (Am + Ab) / 2$$

$$= 159 (992 + 1159) / 2$$

$$= 1743574.88 \text{ Kg}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo$$

$$= 1607629.12 \text{ Kg}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1360 - 1016) / 2$$

$$= 171.9246 \text{ mm}$$

Flange Stresses

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:

$$= 1.9 \cdot Wm1 \cdot hG \cdot Bsc / (t^2 \cdot G) + C \cdot Z \cdot Peq \cdot G^2 / t^2$$

$$= 1.9 \cdot 1607629 \cdot 172 \cdot 1 / (264^2 \cdot 1016) + 0.3 \cdot 1 \cdot 16000 \cdot 1016^2 / 264^2$$



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Flg Calc [Int P]: New Flange

Flng: 12 9:40am Aug 12, 2026

= 144.38 N./mm²

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:

= $1.9 \cdot W \cdot hG \cdot Bsc / (t^2 \cdot G)$

= $1.9 \cdot 1743574 \cdot 172 \cdot 1 / (264^2 \cdot 1016)$

= 79.17 N./mm²

Bolt Stress, Operating [BSo]:

= $Wm1 / Ab$

= 1607629/1159

= 136.00 N./mm²

Bolt Stress, Seating [BSa]:

= $(Wm2 / Ab)$

= (145639/1159)

= 12.32 N./mm²

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Tangential Flange	144.38	151.00	79.17	165.00
Bolting	136.00	159.00	12.32	159.00

Reqd. Blind Flange Thickness at Center 259.162 mm

Reqd. Blind Flange Thickness at Gasket 184.705 mm

Estimated M.A.W.P. (Operating) 16733 KPa.

Estimated Finished Weight of Flange at given Thk. 3726.9 kg.

Estimated Unfinished Weight of Forging at given Thk 3726.9 kg.

Minimum Design Metal Temperature Results:

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Tangential Flange	142.75	165.00	78.27	165.00
Bolting	136.00	159.00	12.32	159.00

Minimum Req'd Thickness, ambient, no Corrosion: 246.507 mm

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Internal Pressure Calculations: Step: 7 9:40am Aug 12,2026

Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd KPa.	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) N./mm ²	Element#
Chnl rear cover		16000	273	1.5	1520	151	1
Chnl rear flange		16000	244	1.5	1000	151	2
Chnl. rear		16006	56	1.5	1000	159	3
Shell		16000	55	1.5	1000	161.86	4
Chnl. front		16006	56	1.5	1000	159	5
Chnl. front flan		16000	244	1.5	1000	151	6
Chnl. front flan		16000	273	1.5	1520	151	7

Element Required Thickness and MAWP :

From	To	Design Pressure KPa.	M.A.W.P. Corroded KPa.	M.A.P. New & Cold KPa.	Minimum Thickness mm	Required Thickness mm	Element#
Chnl rear cover		16000	16733	18493	265	259.162	1
Chnl rear flange		16000 >>>	15577	17515	236	231.75	2
Chnl. rear		16000	16396	18577	56	54.6162	3
Shell		16000	16405	18262	55	53.6131	4
Chnl. front		16000	16396	18577	56	54.6162	5
Chnl. front flan		16000 >>>	15577	17515	236	231.75	6
Chnl. front flan		16000	16733	18493	265	259.162	7

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

Shell Side MAWP = 16405 KPa.
 Shell Side MAPnc = 18262 KPa.
 Channel Side MAWP = 15577 KPa., < design pressure: 16000.000 KPa.
 Channel Side MAPnc = 17515 KPa.

Elements Suitable for Design Internal Pressure.

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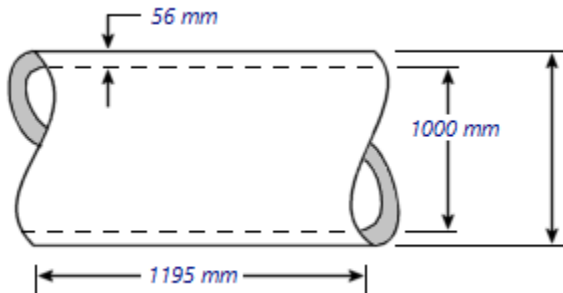
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Element: Chnl. rear

Step: 8 9:40am Aug 12,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Chnl. rear

Cylindrical Shell Design Information:			
Design Pressure P	16005.890 KPa.	Int. Design Temperature	100 °C
Material	SA-516 70	Ext. Design Temperature	40 °C
Length	1195.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	1.5000 mm	Allowable Stress (ope) S	159.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	175.0 N./mm ²
Inside Diameter Di	1000.000 mm	Actual Stress	147.3 N./mm ²
Spec. Min. Thk t,e, or sa	56.0000 mm	Spec. Nominal Thk.	56.000 mm
Surface Area	41746.7 cm ²	SG of Contents	0.600
Element Volume	938718.9 cm ³	Weight of Contents	562.9 Kg
Empty Weight	1720.7 Kg	Operating Weight	2283.6 Kg
		Comp. All. Ope. Stress	122.7 N./mm ²
Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 1003 \cdot (\exp(16006/(159 \cdot 1)) - 1) + 1.5 + 0$$

$$= 53.1 + 1.5 + 0 = 54.6 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 159 \cdot 1 \cdot \ln((2 \cdot 54.5 + 1003) / 1003) - 5.89$$

$$= 16402 - 5.89 = 16396 \text{ KPa.}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175 \cdot 1 \cdot \ln((2 \cdot 56 + 1000) / 1000)$$

$$= 18577 \text{ KPa.}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E(D_o - D))$$

$$= 16006 \cdot 1003 / (1(1112 - 1003))$$

$$= 147.2921 \text{ N./mm}^2$$



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Element: Chnl. rear

Step: 8 9:40am Aug 12,2026

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 16006 / (1 \cdot \ln((2 \cdot 54.5 + 1003) / 1003))$$

$$= 155.1578 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

$$\text{Cylinders formed from Plate} \quad (50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o) \quad 5.303 \%$$

Minimum Design Metal Temperature Results:

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature : 4 °C

User entered Minimum Design Temperature : -33 °C

The Curve for this Material is : B

From Figure 3.8M- with PWHT, the MDMT is : 11 °C

Temperature reduction per Graph 3.13 : 7 °C

The computed stress ratio is : 0.857

The computed thickness ratio [sr]:

$$= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$$

$$= (46.7 \cdot 1) / (56 - 1.5 - 0)$$

$$= 0.857$$

Warning: Impact Testing Is Mandatory!

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

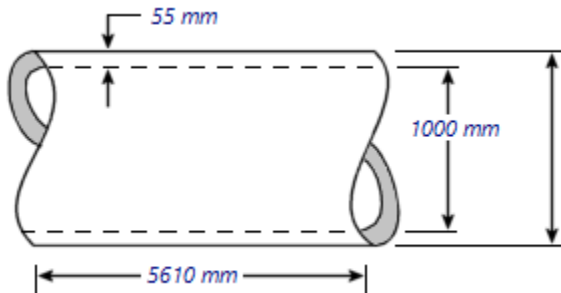
MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	15576.830 KPa.
		Min. Des. Mtl. Temp.	-33 °C
Governing Thk.	56.000 mm	Basic MDMT	11 °C
Stress/Thk. Ratio	0.857	Temperature Difference	7 °C
		Computed Min. Temp.	4 °C
		Joint Efficiency E*	100 %

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Shell

Cylindrical Shell Design Information:			
Design Pressure P	16000.491 KPa.	Int. Design Temperature	80 °C
Material	SA-516 70	Ext. Design Temperature	40 °C
Length	5610.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	1.5000 mm	Allowable Stress (ope) S	161.9 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	175.0 N./mm ²
Inside Diameter Di	1000.000 mm	Actual Stress	150.0 N./mm ²
Spec. Min. Thk t,e, or sa	55.0000 mm	Spec. Nominal Thk.	55.000 mm
Surface Area	195630.1 cm ²	SG of Contents	0.050
Element Volume	2792129.0 cm ³	Weight of Contents	139.5 Kg
Empty Weight	7926.1 Kg	Operating Weight	8065.6 Kg
		Comp. All. Ope. Stress	122.7 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 1003 \cdot (\exp(16000/(162 \cdot 1)) - 1) + 1.5 + 0$
 $= 52.1 + 1.5 + 0 = 53.6 \text{ mm}$ Equation 4.3.1

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 162 \cdot 1 \cdot \ln((2 \cdot 53.5 + 1003) / 1003) - 0.49$
 $= 16405 - 0.49 = 16405 \text{ KPa.}$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D)$
 $= 175 \cdot 1 \cdot \ln((2 \cdot 55 + 1000) / 1000)$
 $= 18262 \text{ KPa.}$

Computed Stress Intensity at Design Pressure [Si]:
 $= P \cdot D / (E(D_o - D))$
 $= 16000 \cdot 1003 / (1(1110 - 1003))$
 $= 149.9947 \text{ N./mm}^2$



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Element: Shell

Step: 9 9:40am Aug 12,2026

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 16000 / (1 \cdot \ln((2 \cdot 53.5 + 1003) / 1003))$$

$$= 157.8603 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

$$\text{Cylinders formed from Plate} \quad (50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o) \quad 5.213 \%$$

Minimum Design Metal Temperature Results:

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	7 °C
User entered Minimum Design Temperature	:	-33 °C
The Curve for this Material is	:	B
From Figure 3.8M- with PWHT, the MDMT is	:	11 °C
Temperature reduction per Graph 3.13	:	4 °C
The computed stress ratio is	:	0.921

The computed thickness ratio [sr]:

$$= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$$

$$= (49.3 \cdot 1) / (55 - 1.5 - 0)$$

$$= 0.921$$

Warning: Impact Testing Is Mandatory!

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	16404.936 KPa.
		Min. Des. Mtl. Temp.	-33 °C
Governing Thk.	55.000 mm	Basic MDMT	11 °C
Stress/Thk. Ratio	0.921	Temperature Difference	4 °C
		Computed Min. Temp.	7 °C
		Joint Efficiency E*	100 %

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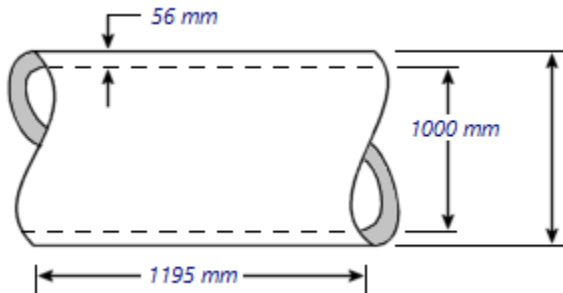
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Element: Chnl. front

Step: 10 9:40am Aug 12,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Chnl. front

Cylindrical Shell Design Information:			
Design Pressure P	16005.890 KPa.	Int. Design Temperature	100 °C
Material	SA-516 70	Ext. Design Temperature	40 °C
Length	1195.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	1.5000 mm	Allowable Stress (ope) S	159.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	175.0 N./mm ²
Inside Diameter Di	1000.000 mm	Actual Stress	147.3 N./mm ²
Spec. Min. Thk t,e, or sa	56.0000 mm	Spec. Nominal Thk.	56.000 mm
Surface Area	41746.7 cm ²	SG of Contents	0.600
Element Volume	938718.9 cm ³	Weight of Contents	562.9 Kg
Empty Weight	1720.7 Kg	Operating Weight	2283.6 Kg
		Comp. All. Ope. Stress	122.7 N./mm ²
Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 1003 \cdot (\exp(16006/(159 \cdot 1)) - 1) + 1.5 + 0$$

$$= 53.1 + 1.5 + 0 = 54.6 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 159 \cdot 1 \cdot \ln((2 \cdot 54.5 + 1003) / 1003) - 5.89$$

$$= 16402 - 5.89 = 16396 \text{ KPa.}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175 \cdot 1 \cdot \ln((2 \cdot 56 + 1000) / 1000)$$

$$= 18577 \text{ KPa.}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 16006 \cdot 1003 / (1 (1112 - 1003))$$

$$= 147.2921 \text{ N./mm}^2$$



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Element: Chnl. front

Step: 10 9:40am Aug 12,2026

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 16006 / (1 \cdot \ln((2 \cdot 54.5 + 1003) / 1003))$$

$$= 155.1578 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

$$\text{Cylinders formed from Plate} \quad (50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o) \quad 5.303 \%$$

Minimum Design Metal Temperature Results:

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature : 4 °C

User entered Minimum Design Temperature : -33 °C

The Curve for this Material is : B

From Figure 3.8M- with PWHT, the MDMT is : 11 °C

Temperature reduction per Graph 3.13 : 7 °C

The computed stress ratio is : 0.857

The computed thickness ratio [sr]:

$$= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$$

$$= (46.7 \cdot 1) / (56 - 1.5 - 0)$$

$$= 0.857$$

Warning: Impact Testing Is Mandatory!

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	15576.830 KPa.
		Min. Des. Mtl. Temp.	-33 °C
Governing Thk.	56.000 mm	Basic MDMT	11 °C
Stress/Thk. Ratio	0.857	Temperature Difference	7 °C
		Computed Min. Temp.	4 °C
		Joint Efficiency E*	100 %



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Shop Pressure Test Results:

Step: 11 9:40am Aug 12,2026

Hydrostatic Test Pressure Results:

Hydrotest Pressure Calculation [Shell Side]:

= Test Factor • ShellSide MAWP(St/S)

= 1.25 • 16405 (1.07)

= 21991.135 KPa.

Hydrotest Pressure Calculation [Channel Side]:

= Test Factor • Channel Side MAWP(St/S)

= 1.25 • 15577 (1.09)

= 21276.111 KPa.

Horizontal Test performed per: per VIII-2, Section 8.2

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
Chnl. rear	200.5	247.0	0.812	21285.91
Shell	210.8	247.0	0.854	22000.94
Chnl. front	200.5	247.0	0.812	21285.91

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N1- Channel inle	Nozzle	165.00	151.00	1.093
N4- Shell outlet	Nozzle	165.00	153.86	1.072
N3- Shell inlet	Nozzle	165.00	153.86	1.072
N2- Channel outl	Nozzle	165.00	151.00	1.093

Minimum

1.072

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
Chnl rear cover	165.00	151.00	1.093
Chnl rear flange	165.00	151.00	1.093
Chnl. rear	175.00	159.00	1.101
Shell	175.00	161.86	1.081
Chnl. front	175.00	159.00	1.101
Chnl. front flange	165.00	151.00	1.093
Chnl. front flange	165.00	151.00	1.093

Minimum

1.081

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	138.00	126.00	1.095
Tubesheet Material	165.00	151.00	1.093

Minimum

1.093

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
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Shop Pressure Test Results:

Step: 11 9:40am Aug 12,2026

N1- Channel inle	176.31	237.50	0.742
N4- Shell outlet	175.05	237.50	0.737
N3- Shell inlet	175.05	237.50	0.737
N2- Channel outl	176.31	237.50	0.742

Elements Suitable for Test Pressure.

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Ring MDMT Calcs:

Step: 12 9:40am Aug 12, 2026

MDMT Calculations for Stiffening Rings:

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ASME TS Calc: Pressure Test

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Input Echo, Tubesheet Number 1, Description: Fixed tubesheet

Shell Data:

Main Shell Description: Shell

Shell Test Pressure		21991.13 KPa.
Shell Thickness	ts	55.0000 mm
Shell Internal Corrosion Allowance	cas	0.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	1000.000 mm
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-516 70

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	175.00 N./mm ²
Shell Allowable Stress at Ambient		175.00 N./mm ²

Channel Data:

Channel Description: Chnl. rear

Channel Type:		Cylinder
Channel Test Pressure		21276.11 KPa.
Channel Thickness	tc	56.0000 mm
Channel Internal Corrosion Allowance	cac	0.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	1000.000 mm
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	175.00 N./mm ²
Channel Allowable Stress at Ambient		175.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	1015
Tube Wall Thickness	et	2.5000 mm
Tube Outside Diameter	D	19.0000 mm
Total Straight Tube Length	Lt	6000.00 mm
Straight Tube Length (bet. inner tubsht faces) L		5610.00 mm
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-334 1
Tube Material UNS Number		K03008
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		138.00 N./mm ²
Tube Allowable Stress At Ambient		138.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	205.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	25.0000 mm
Tube Layout Pattern		Triangular

Fillet Weld Leg	af	2.0000 mm
Groove Weld Leg	ag	2.0000 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		4.21.2
Is the Tube Inset		No

Radius to Outermost Tube Hole Center	ro	454.384 mm
Largest Center-to-Center Tube Distance	Ul	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	5190.7 cm ²



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Perimeter of Tube Layout Cp 2786.6 mm

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf A

Tubesheet Design Metal Temperature T 21.11 °C
Tubesheet Material SA-350 LF2

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Tubesheet Material UNS Number		K03011
Tubesheet Allowable Stress at Temperature	S	165.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	165.00 N./mm ²
Thickness of Tubesheet	h	195.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	0.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	0.0000 mm
Tubesheet Outside Diameter	A	1110.000 mm

Area of the Untubed Lanes	AL	0.0 cm ²
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Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	0.000 mm
Tube End condition corresponding to Span (l)	k	0.00

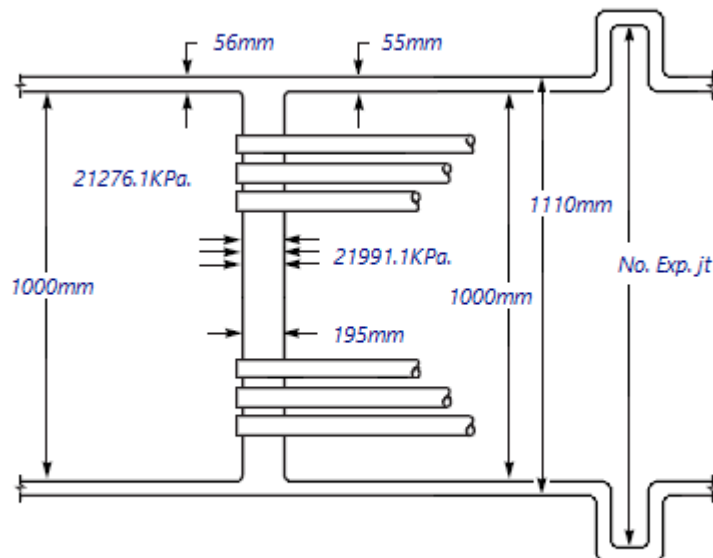
Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)	Yes
Perform Differential Pressure Design	No
Run Multiple Load Cases	Yes

Mean Shell Metal Temp. along Shell len.	Tsm	21.11 °C
Mean Tube Metal Temp. along Tube length	Ttm	21.11 °C
Junction Stress Reduction option	Perform Plastic Calculation	

Tubesheet Integral with	Both
Tubesheet Extended as Flange	No

Is Exchanger in Creep range (skip EP, Use 3S for Sps)	No
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Tubesheet Integral w/Shell and Channel



Configuration a:



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Messages regarding the Tubesheet & Exchanger Input Data:

Warning - Tube end condition k should be bet 0 - 1 (.00).

ASME Division 2 Tubesheet Results per Part 4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
Tubes - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.

Please insure these properties are consistent with the type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

$$\begin{aligned} &\text{Computed Minimum Required Thickness [t]:} \\ &= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i \quad \text{Equation 4.3.1} \\ &= 0.5 \cdot 14 \cdot (\exp(21276/(138 \cdot 1)) - 1) + 0 \\ &= 1.17 + 0 \\ &= 1.17 \text{ mm} \end{aligned}$$

Tube Required Thickness under External Pressure (Shellside pressure) :

Determine Parameter [Mx]:

$$\begin{aligned} &= L / \sqrt{(R_o \cdot t)} \\ &= 5610 / \sqrt{(9.5 \cdot 2.5)} \\ &= 1151.148 \end{aligned}$$

Compute the Factor [Ch]:

$$\begin{aligned} &\text{Since } M_x > 2 \cdot (D_o/t)^{0.94}, \text{ Ch} = 0.55 / (D_o/t) \\ &= 0.55 \cdot (19/2.5) \\ &= 0.0724 \end{aligned}$$



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Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot Ey(t / Do)$$

$$= 1.6 \cdot 0.072 \cdot 202701232 (2.5/19)$$

$$= 3088.425 \text{ N./mm}^2$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / Fy$$

$$= 3088.4/205.0$$

$$= 15.066$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 205/379$$

$$= 0.541$$

Ratio [AE]:

$$= Fhe / E$$

$$= 3088/202701232$$

$$= 0.015235$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.54) + (0.20000E-04 - 0.002)) / \ln(\ln(1 + 0.20000E-04) / \ln(1 + 0.002))$$

$$= 0.134$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys}(1 + \epsilon_{sys}) / \ln(1 + \epsilon_{psp})^{m1}$$

$$= 205 (1 + 0.0020) / \ln(1 + 0.0020)^{0.13403422}$$

$$= 472.527 \text{ N./mm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6 (1 - 0.54)$$

$$= 0.276$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (379 \cdot \exp(0.28)) / (0.28^{0.28})$$

$$= 712.658 \text{ N./mm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.541.5 - 0.5 \cdot 0.542.5 - 0.543.5$$

$$= 0.373$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	189	-2.474	0.216E-07	0.209E-07	0.785E-07	-0.738E-07	0.192E+08	0.536E-02
2	284	0.447	0.297E-06	-0.412E-06	0.228E-06	0.549E-06	0.150E+07	0.174E+00
3	237	-1.014	0.915E-07	0.334E-07	0.141E-06	0.858E-08	0.358E+07	0.510E-01
4	213	-1.744	0.463E-07	0.389E-07	0.107E-06	-0.783E-07	0.842E+07	0.101E-01
5	201	-2.109	0.320E-07	0.296E-07	0.920E-07	-0.804E-07	0.128E+08	0.518E-03
6	195	-2.291	0.264E-07	0.250E-07	0.851E-07	-0.777E-07	0.157E+08	0.277E-02
7	198	-2.200	0.291E-07	0.272E-07	0.885E-07	-0.792E-07	0.142E+08	0.123E-02
8	199	-2.154	0.305E-07	0.284E-07	0.903E-07	-0.798E-07	0.135E+08	0.382E-03
9	200	-2.132	0.313E-07	0.290E-07	0.912E-07	-0.801E-07	0.131E+08	0.612E-04
10	200	-2.143	0.309E-07	0.287E-07	0.907E-07	-0.800E-07	0.133E+08	0.162E-03
11	200	-2.137	0.311E-07	0.288E-07	0.909E-07	-0.801E-07	0.132E+08	0.508E-04
12	200	-2.135	0.312E-07	0.289E-07	0.910E-07	-0.801E-07	0.132E+08	0.507E-05



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13	200	-2.136	0.311E-07	0.289E-07	0.910E-07	-0.801E-07	0.132E+08	0.229E-04
14	200	-2.135	0.311E-07	0.289E-07	0.910E-07	-0.801E-07	0.132E+08	0.892E-05
15	200	-2.135	0.312E-07	0.289E-07	0.910E-07	-0.801E-07	0.132E+08	0.193E-05
16	200	-2.135	0.312E-07	0.289E-07	0.910E-07	-0.801E-07	0.132E+08	0.157E-05

Factor of Safety [FS]:

$$\begin{aligned} & \text{Since } F_{ic} > 0.55 F_y \text{ and } F_{ic} < F_y \\ & = 2.407 - 0.741 \cdot F_{ic} / F_y \\ & = 2.407 - 0.741 \cdot 201/205 \\ & = 1.682 \end{aligned}$$

Allowable Circumferential Compressive Stress [Fha]:

$$\begin{aligned} & = F_{ic} / FS \\ & = 201 / 1.68 \\ & = 119.291 \text{ N./mm}^2 \end{aligned}$$

Allowable External Pressure at the given Thickness [Pa]:

$$\begin{aligned} & = 2 \cdot F_{ha} (t / D_o) \\ & = 2 \cdot 119284 (2.5/19) \\ & = 31390.572 \text{ KPa.} \end{aligned}$$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	1.9280 mm
Allowable Internal Pressure at Corroded thickness	42139.45 KPa.
Required Internal Design Pressure	21276.11 KPa.
Allowable External Pressure at Corroded thickness	31390.57 KPa.
Required External Design Pressure	21991.13 KPa.
Required Thickness due to Shell Side pressure	1.9280 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME Fixed Tubesheet Calculations for Configuration a:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.8.4 Step 1:

Tube expansion depth ratio [ρ]:

$$\begin{aligned} & = l_{tx} / h \text{ (modified for corrosion if present)} \\ & = 0/195 \\ & = 0.0000 \text{ (must be } 0 \leq \rho \leq 1 \text{)} \end{aligned}$$

Effective tube hole diameter [d^*]:

$$\begin{aligned} & = \max(dt - 2 \cdot tt(E_t T / E) \cdot (S_t T / S) \cdot (\rho); dt - 2 \cdot tt) \\ & = \max(19 - 2 \cdot 2.5 (202701232/202701232) \cdot \\ & \quad (138/165) \cdot (0); 19 - 2 \cdot 2.5) \\ & = 19.0000 \text{ mm} \end{aligned}$$

Equivalent outer tube limit circle diameter [D_o]:

$$\begin{aligned} & = 2 \cdot r_o + dt \\ & = 2 \cdot 454 + 19 \\ & = 927.768 \text{ mm} \end{aligned}$$

Determine the basic ligament efficiency for shear [μ]:

$$\begin{aligned} & = (p - dt) / p \\ & = (25 - 19) / 25 \\ & = 0.240 \end{aligned}$$

Equivalent outer tube limit radius [a_o]:

$$\begin{aligned} & = D_o / 2 \\ & = 928/2 \\ & = 463.8840 \text{ mm} \end{aligned}$$



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Effective tube pitch [p*]:

$$= p / \sqrt{(1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot Do \cdot p) / (\pi \cdot Do^2))}$$

$$= 25 / \sqrt{(1 - 4 \cdot \min(0 \cdot 100 ; 4 \cdot 928 \cdot 25) / (\pi \cdot 928^2))}$$

$$= 25.0000 \text{ mm}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (25 - 19) / 25$$

$$= 0.2400$$

Ratio [ps]:

$$= a_s / a_o$$

$$= 500 / 464$$

$$= 1.0779$$

Ratio [pc]:

$$= a_c / a_o$$

$$= 500 / 464$$

$$= 1.0779$$

Compute parameter [xt]:

$$= 1 - Nt((dt - 2 \cdot tt) / (2 \cdot a_o))^2$$

$$= 1 - 1015 ((19 - 2 \cdot 2.5) / (2 \cdot 464))^2$$

$$= 0.7689$$

Determine parameter [xs]:

$$= 1 - Nt(dt / (2 \cdot a_o))^2$$

$$= 1 - 1015 (19 / (2 \cdot 464))^2$$

$$= 0.5743$$

Determine the value [h'g]:

$$= \max((h_g - CATC); 0) \text{ (For pressure only cases)}$$

$$= \max((0 - 0); 0)$$

$$= 0.000 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 2:

Axial Shell Stiffness [Ks]:

$$= \pi \cdot t_s(D_s + t_s) E_s / L$$

$$= \pi \cdot 55 (1000 + 55) 202701232 / 5610$$

$$= 671668.6875 \text{ Kg/mm}$$

Axial Tube Stiffness [Kt]:

$$= \pi \cdot t_t(D_t - t_t) E_t / L$$

$$= \pi \cdot 2.5 (19 - 2.5) 202701232 / 5610$$

$$= 477.4895 \text{ Kg/mm}$$

Stiffness Factor [Ks,t]:

$$= K_s / (Nt \cdot K_t)$$

$$= 671669 / (1015 \cdot 477)$$

$$= 1.38588$$

Rigidity Ratio [J]:

$$= 1 / (1 + K_s / K_j)$$

$$= 1 / (1 + 671669 / 0)$$

$$= 1.0000 (= 1 \text{ if No Exp. Jt.})$$

Shell coefficient [β_s]:

$$= ((12(1 - v_s^2))^{1/4}) / ((D_s + t_s) t_s)$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((1000 + 55) 55)$$

$$= 0.0075 \text{ 1/mm}$$

Shell coefficient [ks]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - v_s^2))$$



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$$= 0.0075 \cdot 202701232 \cdot 55^3 / (6(1 - 0.3^2))$$

$$= 46612226048.0000 \text{ KPa} \cdot \text{mm}^2$$

Shell coefficient [λ_s]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(1000 + 55) \cdot 46.61\text{E}+09 / 195^3 \cdot (1 + 195 \cdot 0.0075 + (195 \cdot 0.0075)^2 / 2)$$

$$= 141435744.0000 \text{ KPa}$$

Shell coefficient [δ_s]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot v_s / (2(D_s + t_s)))$$

$$= 1000^2 / (4 \cdot 202701232 \cdot 55) \cdot (1 - 1000 \cdot 0.3 / (2(1000 + 55)))$$

$$= 0.19235\text{E}-01 \text{ mm/N./mm}^2$$

Channel coefficient [β_c]:

$$= ((12(1 - v_c^2))^{1/4}) / \sqrt{(D_c + t_c)t_c}$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((1000 + 56)56)$$

$$= 0.0075 \text{ 1/mm}$$

Channel coefficient [k_c]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - v_c^2))$$

$$= 0.0075 \cdot 202701232 \cdot 56^3 / (6(1 - 0.3^2))$$

$$= 48736858112.0000 \text{ KPa} \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (Cm\Delta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot Cq\Delta)$$

$$= 6(1000 + 56 / \cos(0)) \cdot 48.74\text{E}+09 / 195^3 \cdot (1 + 195 \cdot 0.0075 \cdot (1 + 1) / 2 + (195 \cdot 0.0075)^2 / 2 \cdot 1)$$

$$= 146597104.0 \text{ KPa}$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot v_c / (2(D_c + t_c)))$$

$$= 1000^2 / (4 \cdot 202701232 \cdot 56) \cdot (1 - 1000 \cdot 0.3 / (2(1000 + 56)))$$

$$= 0.18894\text{E}-01 \text{ mm/N./mm}^2$$

Channel coefficient [zetac]:

$$\text{zetac} = 0$$

ASME Section VIII-2 Part 4.18.8.4 Step 3:

$$h/p = 7.800000 ; \mu^* = 0.240000 ; E^* = 41356984 \text{ KPa}$$

$$E^*/E = 0.204029 ; \nu^* = 0.407183$$

Note: As h/p (7.800) is > 2, data values for $h/p = 2$ were used.

Tube Bundle Stiffness Factor [X_a]:

$$= ((24(1 - \nu^{*2})Nt \cdot Et \cdot t(d_t - t_t) a o^2) / (E^* \cdot L \cdot H^3))^{1/4}$$

$$= ((24(1 - 0.41^2)1015 \cdot 202701232 \cdot 2.5(19 - 2.5)464^2) / (41356984 \cdot 5610 \cdot 195^3))^{1/4}$$

$$= 2.1471$$

Values from Table 4.18.3

$$Z_d = 0.138660 ; Z_v = 0.147370 ; Z_m = 0.620842$$

$$Z_a = 0.961373\text{E}+00 ; Z_w = 0.147370$$

ASME Section VIII-2 Part 4.18.8.4 Step 4:

Diameter ratio [K]:

$$= A / D_o$$

$$= 1110/928$$

$$= 1.1964$$



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Compute coefficient [F]:

$$= (1 - \nu) / E \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.41) / 41356984 \cdot (141435744 + 146597104 + 202701232 \cdot \ln(1.2))$$

$$= 4.6498$$

Compute parameter [ϕ]:

$$= (1 + \nu) F$$

$$= (1 + 0.41) 4.65$$

$$= 6.5431$$

Compute parameter [Q1]:

$$= (\rho_s - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m)$$

$$= (1.08 - 1 - 6.54 \cdot 0.15) / (1 + 6.54 \cdot 0.62)$$

$$= -0.175101$$

Compute Parameter [Qz1]:

$$= (Z_d + Q1 \cdot Z_w) / 2 \cdot X_a^4$$

$$= (0.14 + -0.18 \cdot 0.15) / 2 \cdot 2.15^4$$

$$= 1.1993$$

Compute parameter [Qz2]:

$$= (Z_v + Q1 \cdot Z_m) / 2 \cdot X_a^4$$

$$= (0.15 + -0.18 \cdot 0.62) / 2 \cdot 2.15^4$$

$$= 0.4108$$

Compute parameter [U]:

$$= (Z_w + (\rho_s - 1) Z_m) X_a^4 / (1 + \phi \cdot Z_m)$$

$$= (0.15 + (1.08 - 1) 0.62) 2.15^4 / (1 + 6.54 \cdot 0.62)$$

$$= 0.8217$$

ASME Section VIII-2 Part 4.18.8.4 Step 5:

Determine factor [γ_b]:

$$= 0 \text{ (config a)}$$

Compute Parameter [γ]:

$$= 0.000 \text{ mm (For Pressure only cases)}$$

Calculate parameter [ω_s]:

$$= \rho_s \cdot k_s \cdot \beta_s \cdot \delta S (1 + h \cdot \beta_s)$$

$$= 1.08 \cdot 46.61 \text{E}+09 \cdot 0.0075 \cdot 0.019234987 (1 + 195 \cdot 0.0075)$$

$$= 18026.0078 \text{ mm}^2$$

Calculate parameter [ω_s]:

$$= a o^2 (\rho_s^2 - 1) \cdot (\rho_s - 1) / 4 - \omega_s$$

$$= 464^2 (1.08^2 - 1) \cdot (1.08 - 1) / 4 - 18026$$

$$= -17348.4355 \text{ mm}^2$$

Parameter [ω_c]:

$$= \rho_c \cdot k_c [\delta C (C_m \Theta + \beta C + C_q \Delta \cdot h \cdot \beta C^2) - \zeta_c (C_m \Theta + h \cdot C_q \Theta \cdot h \cdot \beta C / 2)]$$

$$= 1.08 \cdot 48.74 \text{E}+09 [0.018894468 (1 + 0.0075 \cdot 1 \cdot 195) - 0.000 (1/195 + 1 \cdot 0.0075/2)]$$

$$= 18236.1953 \text{ mm}^2$$

Parameter [ω_c]:

$$= a o^2 [(\rho_c^2 + 1) \cdot (\rho_c - 1) / 4 - (\rho_s - 1) / 2] - \omega_c$$

$$= 464^2 [(1.08^2 + 1) \cdot (1.08 - 1) / 4 - (1.08 - 1) / 2] - 18236$$

$$= -17558.6250 \text{ mm}^2$$

Pressure [P^*S]:

$$= 0 \text{ For Pressure only cases or Configurations d,e,f,A,B,C,D}$$

Pressure [P^*C]:



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ASME TS Calc: Pressure Test

Case: 1 9:40a Aug 12, 2026

= 0 For Pressure only cases or Configurations b,c,d,B,C,D

ASME Section VIII-2 Part 4.18.8.4 Step 6:

Pressure [P's]:

$$= Ps \{ xs + 2(1 - xs) vt + [2/Kst \cdot (Ds/Do)^2] vs - [(ps^2 - 1)/(J \cdot Kst)] - [(1 - J)/(2J \cdot Kst)] [(Dj^2 - (Ds)^2)/Do^2] \}$$

$$= 21991 \{ 0.57 + 2(1 - 0.57) 0.3 + [2/1.39 \cdot (1000/928)^2] 0.3 - [(1.08^2 - 1)/(1 \cdot 1.39)] - [(1 - 1)/(2 \cdot 1 \cdot 1.39)] [(0^2 - (1000)^2)/928^2] \}$$

$$= 26740.5625 \text{ KPa.}$$

Pressure [P't]:

$$= [xt + 2(1 - xt) vt + 1/(J \cdot Kst)] Pt$$

$$= [0.77 + 2(1 - 0.77) 0.3 + 1/(1 \cdot 1.39)] 21276$$

$$= 34661.2227 \text{ KPa.}$$

Pressure [P_y]:

$$= Nt \cdot Kt \cdot \gamma / (\pi \cdot ao^2)$$

$$= 1015 \cdot 477 \cdot 0 / (3.14 \cdot 464^2)$$

$$= 0.0000 \text{ KPa.}$$

Pressure [P_w]:

$$= -\gamma b \cdot U \cdot W^* / (2 \cdot \pi \cdot ao^2)$$

$$= -0 \cdot 0.82 \cdot 0 / (2 \cdot \pi \cdot 464^2)$$

$$= -0.0000 \text{ KPa.}$$

Calculate the pressure [Prim]:

$$= -(U/ao^2) \cdot (\omega^*s \cdot Ps - \omega^*c \cdot Pt)$$

$$= -(0.82/464^2) \cdot (-17348 \cdot 21991 - -17559 \cdot 21276)$$

$$= 30.2894 \text{ KPa.}$$

Calculate the pressure [P_w]:

$$= (U/ao^2) \cdot (\omega^*s \cdot P^*s - \omega^*c \cdot P^*c)$$

$$= (0.82/464^2) \cdot (18026 \cdot 0 - 18236 \cdot 0)$$

$$= 0.0000 \text{ KPa.}$$

Determine the effective pressure [Pe]:

$$= J \cdot Kst / (1 + J \cdot Kst (Qz1 + (ps - 1) Qz2)) \cdot (P's - P't + P_y + P_w + P_w + Prim)$$

$$= 0.1000E+01 \cdot 1.39 / (1 + 1 \cdot 1.39 (1.2 + (1.08 - 1) 0.41)) \cdot (26741 - 34661 + 0 + 0 + -0 + 30.3)$$

$$= -4040.4678 \text{ KPa.}$$

ASME Section VIII-2 Part 4.18.8.4 Step 7:

Determine factor [Q2]:

$$= [((\omega^*s \cdot Ps - \omega^*c \cdot Pt) - (\omega^*s \cdot P^*s - \omega^*c \cdot P^*c)) + W^* \cdot \gamma b / (2 \cdot \pi)] / (1 + \phi \cdot Zm)$$

$$= [((-17348 \cdot 21991 - -17559 \cdot 21276) - (18026 \cdot 0 - 18236 \cdot 0)) \cdot 0.0001 + 0 \cdot 0 / (2 \cdot \pi)] / (1 + 6.54 \cdot 0.62)$$

$$= -159.7972 \text{ Kg}$$

Calculate factor [Q3]:

$$= Q1 + 2 \cdot Q2 / (Pe \cdot ao^2)$$

$$= -0.18 + 2 \cdot -160 / (-4040 \cdot 464^2)$$

$$= -0.17150$$

Fm Value from Table 4.18.3 = 0.100667

Tube sheet Bending Stress - original thickness [σ]:

$$= (1.5 \cdot Fm / \mu^*) \cdot (2 \cdot ao / (H - h'g))^2 \cdot Pe$$

$$= (1.5 \cdot 0.1/0.24) \cdot (2 \cdot 464 / (195 - 0))^2 \cdot -4040$$

$$= -57.5485 \text{ N./mm}^2$$



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Allowable Tubesheet Bending Stress [σ allowed]:
 $= 1.5 \cdot \beta T \cdot S_y = 1.5 \cdot 0.95 \cdot 250 = 356 \text{ N./mm}^2, 4.18.233$

Tubesheet Bending Stress - final thickness [σ_f]:
 $= (1.5 \cdot F_m / \mu \cdot) \cdot (2 \cdot a_o / (h - h'g))^2 \cdot P_e$
 $= (1.5 \cdot 0.05 / 0.24) \cdot (2 \cdot 464 / (41.2 - 0))^2 \cdot -2245$
 $= -356.2352 \text{ N./mm}^2$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:
 $= h + \text{Cats} + \text{Catc}$
 $= 41.2 + 0 + 0$
 $= 41.1756 \text{ mm}$

ASME Section VIII-2 Part 4.18.8.4 Step 8:

Shear Stress check [τ_{limit}]:
 $= (2 \cdot \mu \cdot h / a_o) \cdot (0.8 \cdot \beta T \cdot S_y)$
 $= (2 \cdot 0.24 \cdot 195 / 464) \cdot 190$
 $= 38.3365 \text{ N./mm}^2$

Shear Stress is not required to be computed; [P_e] $\leq$ τ_{limit}

Tubesheet Shear Stress is probably low, use the following req. thk:

Tubesheet thickness (Incl. Corr.) = 3.8100 mm
 Tubesheet Shear Stress = 112.2932 N./mm²

Reqd Tubesheet Thickness for Given Loadings (including CA) [Hreqd]:
 $= \max(\text{HreqB}; \text{HreqS})$
 $= \max(41.2; 3.81)$
 $= 41.1756 \text{ mm}$

ASME Section VIII-2 Part 4.18.8.4 Step 9:

The F_{tmin} and F_{tmax} Coefficients from Table 4.18.4:
 $F_{\text{tmin}} = 0.7, F_{\text{tmax}} = 1.2$

First Extreme Tube Axial Stress from among all the tubes [σ_{t1}]:
 $= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{\text{tmin}}) / (X_t - X_s)$
 $= ((21991 \cdot 0.57 - 21276 \cdot 0.77) - (-4040) \cdot 0.7) / (0.77 - 0.57)$
 $= -4.5372 \text{ N./mm}^2$

Second Extreme value of Tube Axial Stress from among all the tubes [σ_{t2}]:
 $= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{\text{tmax}}) / (X_t - X_s)$
 $= ((21991 \cdot 0.57 - 21276 \cdot 0.77) - (-4040) \cdot 1.2) / (0.77 - 0.57)$
 $= 5.8568 \text{ N./mm}^2$

Maximum Tube Axial Stress [$\sigma_{t,\text{max}}$]:
 $= \max(|\sigma_{t,1}|; |\sigma_{t,2}|) = 5.86 \text{ N./mm}^2$

Allowable Tube Stress [$\sigma_{t,A}$]:
 $= \beta T \cdot S_{y,t} = 194.7465 \text{ N./mm}^2$

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [Fs]:
 $= \max((3.25 - 0.25(Z_d + Q_3 \cdot Z_w))X_a^4; 1.25)$
 $= \max((3.25 - 0.25(0.14 + -0.17 \cdot 0.15)2.15^4); 1.25)$
 $= 2.0000 \text{ (Should be } \leq 2)$

Determine the factor [rt]:
 $= (\sqrt{dt^2 + (dt - 2 \cdot tt)^2})/4$



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ASME TS Calc: Pressure Test

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$$= (\sqrt{(19^2 + (19 - 2 \cdot 2.5)^2)}) / 4$$

$$= 5.9002 \text{ mm}$$

Determine the factor [Ct]:

$$= \sqrt{(2 \cdot \pi^2 \cdot Et / Syt)}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 202701232 / 205)}$$

$$= 143.8624$$

Determine the factor [Ft]:

$$= k \cdot L / r$$

$$= 0 \cdot 0 / 5.9$$

$$= 0.0000$$

Buckling Allowable Stress [Stb]:

$$= (Sy_t / Fs) \cdot (1 - Ft / (2 \cdot Ct))$$

$$= (205 / 2) \cdot (1 - 0 / (2 \cdot 144))$$

$$= 194.746 \text{ N./mm}^2 \quad (\text{Never greater than } Sot)$$

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load [Wt]:

$$= \sigma_{t,max} \cdot \text{Tube Area}$$

$$= 5.86 \cdot 1.3$$

$$= 77.39 \text{ Kg}$$

Tube Metal Area [At]:

$$= \pi (do - t) t$$

$$= \pi (19 - 2.5) 2.5$$

$$= 1.296 \text{ cm}^2$$

Tube Weld Size Results per VIII-2 4.21.2:

Tube Strength [Ft]:

$$= \pi \cdot t (do - t) S$$

$$= \pi \cdot 2.5 (19 - 2.5) 195$$

$$= 2573.444 \text{ Kg}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot ag (do + 0.67 \cdot ag) Sw \quad (\text{but not } > Ft)$$

$$= 0.85 \cdot \pi \cdot 2 (19 + 0.67 \cdot 2) 195$$

$$= 2157.2014 \text{ Kg}$$

Weld Strength Factor [fw]:

$$= S / Sw \quad \text{where } Sw = \min(S \text{ and } St)$$

$$= 195 / 195$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot af (do + 0.67 \cdot af) Sw \quad (\text{but not } > Ft)$$

$$= 0.55 \cdot \pi \cdot 2 (19 + 0.67 \cdot 2) 195$$

$$= 1395.8363 \text{ Kg}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{((0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t) \cdot fw \cdot fd)} - 0.75 \cdot do \right)$$

$$= 2 \left(\sqrt{((0.75 \cdot 19)^2 + 1.07 \cdot 2.5 \cdot (19 - 2.5) \cdot 1 \cdot 1)} - 0.75 \cdot 19 \right)$$

$$= 2.945 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.5000 mm

Minimum Required Groove Weld Leg [agr]: 1.5000 mm

Tube-Tubesheet Jt allowable, 2573 is $\geq$ tube strength 2573 Kg



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Note: This tube-tubesheet joint is a Full Strength joint

ASME Section VIII-2 Part 4.18.8.4 Step 10:

Shell Axial Membrane Allowable Stress:

$$= \beta T \cdot S_{y,s} = 0.95 \cdot 260 = 370 \text{ N./mm}^2$$

Axial Membrane Stress in Shell $[\sigma_{s,m}]$:

$$= \frac{a o^2}{(D_s + t_s) t_s} \cdot [P_e + (p_s^2 - 1)(P_s - P_t)] + \frac{a s^2 \cdot P_t}{(D_s + t_s) t_s}$$

$$= \frac{464^2}{(1000 + 55) 55} \cdot [-4040 + (1.08^2 - 1)(21991 - 21276)] + \frac{500^2 \cdot 21276}{(1000 + 55) 55}$$

$$= 77.1171 \text{ N./mm}^2$$

ASME Section VIII-2 Part 4.18.8.4 Step 11:

Note:

For a given Shell thickness of 55 mm, the minimum Shell length adjacent to the tubesheet should be 422.137 mm

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	57.5 <=	356.2 N./mm ²	Pass
Tubesheet Shear Stress	16.1 <=	190.0 N./mm ²	Pass
Maximum Tube Stress	5.9 <=	194.7 N./mm ²	Pass
Minimum Tube Stress (Buckling)	-4.5 <=	-194.7 N./mm ²	Pass
Maximum Force on any one Tube	77.4 <=	2573.4 Kgf	Pass
Axial Membrane Stress in Shell	77.1 <=	370.5 N./mm ²	Pass

Thickness Results for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	41.176	195.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.500	2.000	Ok
Tube-Tubesheet Groove Weld Leg :	1.500	2.000	Ok

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 3 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Stresses Bend Allwd Shear Allwd	Case Type	Pass/Fail
D1uc	168.259 ...	277	356 74 190	Ps+Pt-Th D1	Ok
D2uc	147.470 ...	228	356 57 190	Ps+Pt-Th D2	Ok
D3uc	41.176 ...	58	356 16 190	Ps+Pt-Th D3	Ok
Max:	168.2589 ... mm	0.779	0.387 (Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[(+)-Th]:

With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:

With or Without Corrosion Allowance



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[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:

Operating Load Cases using the Maximum and Minimum Operating Pressures and Operating Temperatures

Shell Axial Membrane Stress Summary:

---- Shell Stresses ----					:	---- Shell Band Stress ----					:	Pass
Case#	Ten	Allwd	Cmp	Allwd	:	Ten	Allwd	Cmp	Allwd	:		Fail
D1uc	10	370	...	...	:	...	...	...	...	:		Ok
D2uc	67	370	...	...	:	...	...	...	...	:		Ok
D3uc	77	370	...	...	:	...	...	...	...	:		Ok
Max RATIO	0.208			...	:		...		...	:		

Tube, Shell and Channel Stress Summary:

---- Tube Stresses							Tube Loads		Shell Stress		Channel Stress		Pass
Case#	Ten	Allwd	Cmp	Allwd	Ld	Allwd			Stress	Allwd	Stress	Allwd	Fail
D1uc	24	194	-12	-194	321	2573	...	370	...	370	...	370	Ok
D2uc	18	194	-18	-194	244	2573	...	370	...	370	...	370	Ok
D3uc	6	194	-5	-194	77	2573	...	370	...	370	...	370	Ok
Max RATIO	0.125			0.095		0.125		...		...		...	

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	168.259	195.000	Ok
Tube thickness :	1.928	2.500	Ok
Tube-tubesheet fillet weld leg :	1.500	2.000	Ok
Tube-tubesheet groove weld leg :	1.500	2.000	Ok

Min Shell length of thk, (55.000) adj. to tubesheet: 422.137 mm

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Extends by : +.000 mm
 Due to Pressure Shell Compresses by : -0.638 mm
 Due to Pressure + Thermal Shell Compresses by : -0.638 mm

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.1165	...	...	...	...	...
2	-0.6378	...	...	...	...	...
3	-0.5213	...	...	...	...	...
4	...	...	...	...	...	...
5	...	...	...	...	...	...
6	...	...	...	...	...	...
7	...	...	...	...	...	...
8	...	...	...	...	...	...



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ASME TS Calc: Pressure Test

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External Pressure Calculations: Step: 13 9:40am Aug 12,2026

External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 2, 2025

External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B N./mm ²
10	20	No Calc	...	263.5	No Calc	No Calc
20	30	No Calc	...	234.5	No Calc	No Calc
30	Tsht	1514.97	1112	54.5	No Calc	No Calc
Tsht	40	No Calc	...	54.5	No Calc	No Calc
40	Tsht	5610	1110	53.5	No Calc	No Calc
Tsht	50	No Calc	...	53.5	No Calc	No Calc
50	60	1515.03	1112	54.5	No Calc	No Calc
60	70	No Calc	...	234.5	No Calc	No Calc
70	80	No Calc	...	263.5	No Calc	No Calc

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure KPa.	External M.A.W.P. KPa.
10	20	265	140.594	...	No Calc
20	30	236	146.888	...	No Calc
30	Tsht	56	No Calc	...	13442
Tsht	40	56	No Calc	...	No Calc
40	Tsht	55	No Calc	...	9656
Tsht	50	55	No Calc	...	No Calc
50	60	56	No Calc	...	13442
60	70	236	146.888	...	No Calc
70	80	265	140.594	...	No Calc

Minimum

9656

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	No Calc	No Calc	No Calc	No Calc
30	Tsht	1514.97	No Calc	No Calc	No Calc
Tsht	40	No Calc	No Calc	No Calc	No Calc
40	Tsht	5610	No Calc	No Calc	No Calc
Tsht	50	No Calc	No Calc	No Calc	No Calc
50	60	1515.03	No Calc	No Calc	No Calc
60	70	No Calc	No Calc	No Calc	No Calc
70	80	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

Step: 14

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Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm ³	Corroded Metal Wgt. kg.	Corroded ID Volume cm ³	Extra due Misc % kg.
10	20	3839.46	...	3839.46	...	383.946
20	30	2098.35	258809	2086.35	259205	209.835
30	40	1720.68	938719	1676.97	944360	172.068
40	50	7926.07	2792129	7720.87	2818609	792.607
50	60	1720.68	938719	1676.97	944360	172.068
60	70	2098.35	258809	2086.35	259205	209.835
70	80	3839.46	...	3839.46	...	383.946
Total		23243	5187184.00	22926	5225739.50	2324



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset Dtl. Cent. mm	Description
30	Liqd	562.888	597.5	...	...	Liquid: 30
30	Insl	13.8557	597.5	...	...	Cold insulation
30	Nozl	386.221	600	728.6	...	N1- Channel inle
40	Sadl	328.551	1122	732.5	...	Left Saddle
40	Sadl	328.551	4488	732.5	...	Right saddle
40	Liqd	139.521	2805	...	...	Liquid: 40
40	Insl	198.504	2800	...	...	Cold insulation
40	Nozl	300.381	700	703.2	...	N4- Shell outlet
40	Nozl	300.381	4700	703.2	...	N3- Shell inlet
50	Liqd	562.888	597.5	...	...	Liquid: 50
50	Insl	13.8557	597.5	...	...	Cold insulation
50	Nozl	386.221	600	-728.6	...	N2- Channel outl
30	FTsh	1130.33	1292.5	...	...	Fixed tubesheet
30	Tube	6728.43	4195	...	...	Tubes
30	RTsh	1130.33	7097.5	...	...	Tubesheet
40	Baf1	23.1256	788	...	...	Baffles
40	Baf1	23.1256	1508	...	...	Baffles
40	Baf1	23.1256	2228	...	...	Baffles
40	Baf1	23.1256	2948	...	...	Baffles
40	Baf1	23.1256	3668	...	...	Baffles
40	Baf1	23.1256	4388	...	...	Baffles
40	Baf1	23.1256	5108	...	...	Baffles

Total Weight of Each Detail Type:

Saddles	657.1
Liquid	1265.3
Insulation	226.2
Nozzles	1373.2
Exchanger Components	9151.0
Liquid in Tubes	46.9

Sum of the Detail Weights 12719.7 kg.

Weight Summation Results: (kg.)



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Element and Detail Weights:

Step: 14 9:40am Aug 12,2026

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	25567.3	25567.3	25567.3	25567.3	25567.3	25567.3
Saddles	657.1	657.1	657.1	657.1	657.1	657.1
Nozzles	1373.2	1373.2	1373.2	1373.2	1373.2	1373.2
Exchanger	9151.0	9151.0	9151.0	9151.0	9151.0	9151.0
Insulation	...	...	226.2	226.2	226.2	226.2
Ope. Liquid	...	...	...	...	...	1265.3
Tube Ope Lqd	...	...	...	...	...	46.9
Test Liquid	...	5184.0	...	...	...	...
Tube Tst Lqd	...	937.1	...	...	...	...
Totals	36748.6	42869.7	36974.8	36974.8	36974.8	38287.0

Field Installation Options:

Miscellaneous Weight Percent: 10.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	36748.6 kg.
Shop Test Wt.	- Fabricated weight + water (full)	42869.7 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	36974.8 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	36974.8 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	36974.8 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	38287.0 kg.
Oper. Wt. + CA	- Corr wt. + operating liquid	37938.7 kg.

Exchanger Tube Data

Volume of Exchanger tubes :	937650.7 cm ³
Weight of Ope Liq in tubes :	46.9 kg.
Weight of Water in tubes :	937.1 kg.

The Corroded Weight and thickness are used in the Horizontal Vessel Analysis (Ope Case) and Earthquake Load Calculations.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	31182.2	...
20	30	25020.4	10304.4
30	40	41746.7	37542
40	50	195630	176243
50	60	41746.7	37542
60	70	25020.4	10304.4
70	80	31182.2	...

Total	391528.8	271936.3 cm ²
Total	421.4	292.7 ft ²



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Nozzle Flange MAWP:

Step: 15 9:40am Aug 12,2026

Nozzle Flange MAWP Results: (KPa. & °C)

Nozzle	Flange Rating		Des.			Equiv.	- - - -	Max Pressure	
Description	Ope.	Ambient	Temp	Class	Grade/Group	Press	UG-44	50%	DNV

Shellside Flange Rating**Channelside Flange Rating**

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Wind Load Calculation:

Step: 16 9:40am Aug 12,2026

Wind Load Calculation Results:

Wind Load Calculations per India Std. IS 875 (PART 3):2015, Amd. 1&2 (2020):

Input Values:

Wind Design Code	IS 875 (PART 3):2015-2020	
Basic Wind Speed for IS-875	180	Km/hr
Wind Zone Number	5	
Base Elevation	300	mm
Percent Wind for Hydrotest	33.0	
Risk Factor	[k1] 1.0	
Terrain Category	2	
Equipment Class	Class A	
Topography Factor	[k3] 1.0	
Importance Factor	[k4] 1.15	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	
Use Gust Response Factor (Dynamic Analysis)	No	

Calculated Values:

Vessel Length to Diameter ratio: 7.755

Design Wind Speed [Vz]:

= Basic Wind Speed • k1 • k2,1 • k3 • k4 (for the 1st element)

= 50 • 1 • 1 • 1 • 1.15

= 57.496 meters/s

Design Wind Pressure per 4.5.2 [pd]:

= max(0.7 • pz, Kd • Ka • Kc • 0.6 • Vz²)

= max(0.7 • 1984, 1 • 0.96 • 1 • 0.6 • 57.5²)

= 1905.5 N/m²

Element Wind Load (operating condition):

= Wind Area • Cf • pd

= 0.48 • 0.86 • 1905

= 787.6 N

From	Height mm	k1	k2	k3	Vz m/sec	Cf
10	1350.00	1.0000	1.0000	1.0000	57.50	0.8551
20	1350.00	1.0000	1.0000	1.0000	57.50	0.8551
30	1350.00	1.0000	1.0000	1.0000	57.50	0.8551
40	1350.00	1.0000	1.0000	1.0000	57.50	0.8551
50	1350.00	1.0000	1.0000	1.0000	57.50	0.8551
60	1350.00	1.0000	1.0000	1.0000	57.50	0.8551
70	1350.00	1.0000	1.0000	1.0000	57.50	0.8551

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area cm²	Wind Pressure Kgs/m²	Element Wind Load Kgf
10	20	1350	1824	4833.6	194.387	88.3472
20	30	1350	1200	3840	194.387	70.1865
30	40	1350	1430.4	17093.3	194.387	312.426
40	50	1350	1607.51	90181.2	194.387	1648.31
50	60	1350	1430.4	17093.3	194.387	312.426
60	70	1350	1200	3840	194.387	70.1865
70	80	1350	1824	4833.6	194.387	88.3472



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Wind Load Calculation:

Step: 16 9:40am Aug 12, 2026

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation: Step: 17 9:40am Aug 12,2026

Earthquake Load Calculation Results:

Seismic Analysis Results per IS 1893 (Part 4): 2024

Input Data:

Importance Factor as Entered by User	I	2.0000
Zone Number		2
Soil Type	Rocky or Hard Soil	
Response reduction factor (Force factor)	R	2.0000
Time Period of Vibration (program or user)		0.0303 Seconds
Value of Sa/g from user Spectrum Data		0.0939
Percent Damping		2.0 %
Considering MCE for load determination		No

Calculated results:

Horizontal Seismic Coefficient per 7.3.1 [Ah]:

$$= (Sa/g)_{ss} \cdot DampVal \cdot (I/R)$$

$$= (0.094 \cdot 1) \cdot 1$$

$$= 0.094$$

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Actual Time Period = 0.03 secs

Computed Sa/g = 0.094

Note:

It was indicated to ignore the damping factor, assuming that the Site Spectrum specified is per the actual damping value of this vessel.

The value of "DampVal" has been set to 1 (not used).

Earthquake Element Load, for the first Element:

$$= \text{Earthquake Weight} \cdot Ah \cdot \text{Scalar}$$

$$= 4142 \cdot 0.094 \cdot 1$$

$$= 388.8 \text{ Kgf}$$

Vertical load per 8.4:

$$= 2/3 \cdot Ah \cdot W \cdot \text{Scalar}$$

$$= 2/3 \cdot 0.094 \cdot 37282 \cdot 1$$



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Earthquake Load Calculation:

Step: 17

9:40am Aug 12, 2026

= 2332.7 Kgf

Earthquake Load Calculation:

From	To	Earthquake Height mm	Earthquake Weight Kgf	Element Ope Load Kgf
10	20	760	4142.4	388.789
20	30	500	4142.4	388.789
30	40	500	4142.4	388.789
40	Sad1	500	4142.4	388.789
Sad1	50	500	4142.4	388.789
40	50	500	4142.4	388.789
50	60	500	4142.4	388.789
60	70	500	4142.4	388.789
70	80	760	4142.4	388.789

The Earthquake Weight does not include the weight of the saddles.

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Center of Gravity Calculation: Step: 18 9:40am Aug 12,2026

Shop/Field Installation Options :

Insulation is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	4785.000 mm
Center of Gravity of the Liquid	4785.000 mm
Center of Gravity of the Insulation	4780.613 mm
Center of Gravity of the Nozzles	4740.470 mm
Center of Gravity of the Tubesheet(s)	4785.000 mm
Center of Gravity of the Tubes	4785.000 mm
Center of Gravity of the Baffles	4928.000 mm
Center of Gravity of Bare Shell New and Cold	4786.277 mm
Center of Gravity of Bare Shell Corroded	4786.292 mm
Vessel CG in the Operating Condition	4784.831 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	4784.829 mm
Vessel CG in the Test Condition	4784.883 mm

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Saddle Calcs: Operating Case: Step: 19 9:40am Aug 12,2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Note:

Wear Pad Width (250.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

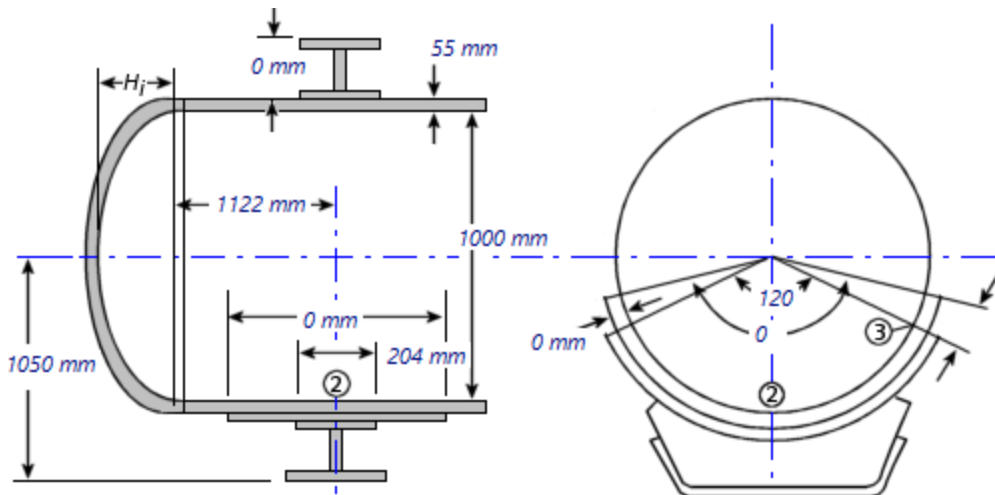
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(204 + 1.56 \cdot \sqrt{528 \cdot 53.5}, 2 \cdot 1122)$$

$$= 466.2534 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	528.25	mm
Shell Thickness used in this Case	t	55.000	mm
Stiffened Vessel Length per 4.15.6	L	5610.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1122.00	mm
Saddle Width	b1 or b	204.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		161.86	N./mm ²
Head Allowable Stress used in Calculation		151.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1050.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case	25248.10	Kgf
Pressure used in Analysis	P	16000.245 KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	77.38	161.86
Long. Stress at Bottom of Midspan	80.61	161.86
Long. Stress at Top of Saddles	88.88	161.86
Long. Stress at Bottom of Saddles	73.52	161.86



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Saddle Calcs: Operating Case: Step: 19 9:40am Aug 12,2026

Tangential Shear in Shell	6.15	129.48
Circ. Stress at Horn of Saddle	9.34	202.32
Circ. Compressive Stress in Shell	0.75	161.86

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (2590/2 + 0) \cdot 1050/1013$$

$$= 4026.3 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(298, 0) \cdot 1050/3366$$

$$= 92.9 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(3499, 0) \cdot 1050/3366$$

$$= 1091.5 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (3499/2 + 0) \cdot 1050/1013$$

$$= 5439.0 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 19809 + \max(92.9, 4026, 1092, 5439)$$

$$= 25248.1 \text{ Kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1906 (0 + \pi/4(1.11 \cdot 1.2)^2)$$

$$= 2921.953 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	25576.65	Kgf
Transverse Shear Load	1749.55	Kgf
Longitudinal Shear Load	3499.10	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot \text{Rm})$$

$$= \min(2(1000/2 + 55 + 30)\sin(120/2), 2 \cdot 528)$$

$$= 1013.250 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (\text{Rm}^2 - \text{hm}^2)/(2a \cdot L)) / (1 + (4\text{hm})/3L)]$$

$$= -25248 \cdot 1122 [1 - (1 - 1122/5610 + (528^2 - 0^2)/(2 \cdot 1122 \cdot 5610)) / (1 + (4 \cdot 0)/(3 \cdot 5610))]$$

$$= -5037.8 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(\text{Rm}^2 - \text{hm}^2)/(L^2)) / (1 + (4\text{hm}^2)/(3L)) - 4a/L)$$



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Saddle Calcs: Operating Case: Step: 19 9:40am Aug 12, 2026

$$= 25248 \cdot 5610 / 4 \cdot ((1 + 2(528^2 - 0^2)) / (5610^2)) / (1 + (4 \cdot 0) / (3 \cdot 5610)) - 4 \cdot 1122 / 5610$$

$$= 7710.2 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 16000 \cdot 528 / (2 \cdot 53.5) - 7710 / (\pi \cdot 528^2 \cdot 53.5)$$

$$= 77.38 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 16000 \cdot 528 / (2 \cdot 53.5) + 7710 / (\pi \cdot 528^2 \cdot 53.5)$$

$$= 80.61 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 16000 \cdot 528 / (2 \cdot 53.5) - 5038 / (0.11 \cdot \pi \cdot 528^2 \cdot 53.5)$$

$$= 88.88 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 16000 \cdot 528 / (2 \cdot 53.5) + 5038 / (0.19 \cdot \pi \cdot 528^2 \cdot 53.5)$$

$$= 73.52 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3))$$

$$= 25248 (5610 - 2 \cdot 1122) / (5610 + (4 \cdot 0 / 3))$$

$$= 15148.9 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 15149 / (528 \cdot 53.5)$$

$$= 6.15 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(528 \cdot 53.5)}$$

$$= 131.127 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.76 \cdot 25248 \cdot 0.1 / (53.5(204 + 131 + 131))$$

$$= -0.75 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 3 \cdot K_7 \cdot Q / (2 \cdot t^2)$$

$$= -25248 / (4 \cdot 53.5(204 + 131 + 131)) - 3 \cdot 0.053 \cdot 25248 / (2 \cdot 53.5^2)$$

$$= -9.34 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(204 + 1.56 \cdot \sqrt{(528 \cdot 53.5)}, 2 \cdot 1122)$$

$$= 466.25 \text{ mm}$$

Distance between Saddle Supports [Ls]:

$$= 3366.0 \text{ mm}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot L_s(\text{Design Temperature} - \text{Ambient Temperature})$$

$$= 0.11973 \text{E-}04 \cdot 3366(80 - 21.1)$$

$$= 2.373 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length Bplen 1060.0000 mm



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Saddle Calcs: Operating Case: Step: 19 9:40am Aug 12,2026

Baseplate Thickness	Bpthk	30.0000	mm
Baseplate Width	Bpwid	250.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	6	
Rib Thickness	Ribtk	15.0000	mm
Web Thickness	Webtk	15.0000	mm
Web Location	Webloc	Side	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	262.0	N./mm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(502 \cdot 53.5)} + 250)^3) / 12$$

$$= 645.096 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 645 + 270 (148 - 26.7)^2$$

$$= 40596.184 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	505.5	53.5	26.7	270.5	723472.4	0.406E+05
Wearplate	250.0	30.0	68.5	75.0	513749.9	0.483E+04
Web	15.0	435.0	301.0	65.2	1964024.8	0.255E+05
BasePlate	250.0	30.0	533.5	75.0	4001249.8	0.111E+06
Totals	...	...	...	485.7	7202497.0	0.182E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 7202497 / 486$$

$$= 148.289 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 25248$$

$$= 5138.5347 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 2.3411 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 157.1972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1050 - 502 \cdot \sin(120/2) / 1.05$$

$$= 635.263 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 3264.3850 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 2.6043 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 174.6635 \text{ N./mm}^2$$



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Saddle Calcs: Operating Case: Step: 19 9:40am Aug 12,2026

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{\frac{3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}}{2 \cdot \text{BasePlateLength} \cdot \text{AllStress}}}$$

$$= \sqrt{\frac{3(25248 + 329) \cdot 250}{2 \cdot 1060 \cdot 175}}$$

$$= 22.540 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (500 + 55 + 30)$$

$$= 1013.250 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 1013 / (6 - 1)$$

$$= 202.650 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 203 \cdot 250 / 2$$

$$= 253.312 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 25577 / (1060 \cdot 250)$$

$$= 9.652 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= \text{Ap} \cdot \text{Bp}$$

$$= 253 \cdot 9.65$$

$$= 2444.861 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 28.4 + 15.2$$

$$= 43.549 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / \text{Ar}$$

$$= 2445 / 43.5$$

$$= 5.506 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	15.0	189.0	109.5	28.4	310432.5	0.120E+04
Web	101.3	15.0	7.5	15.2	11399.1	673.
Totals	...	...	...	43.5	321831.6	0.188E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 204 - 15$$

$$= 189.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= \text{AY} / \text{A}$$

$$= 321832 / 43.5$$

$$= 73.901 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(\text{ytot}, \text{Saddle Width} - \text{ytot})$$

$$= \max(73.9, 204 - 73.9)$$



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Saddle Calcs: Operating Case: Step: 19 9:40am Aug 12,2026

= 130.099 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{1876/43.5}$

= 65.637 mm

Length of Outer Rib [L]:

= Saddle Height - $\cos(\theta/2)$ (radius + shlthk + wpdthk) - bpthk

= 1050 - $\cos(120/2)$ (500 + 55 + 30) - 30

= 727.500 mm

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$

= $\sqrt{2 \cdot \pi^2 \cdot 199943008/262}$

= 122.739

Slenderness ratio [KL/r]:

= KL/r

= 1 • 728/65.6

= 11.084

Bending Moment [Rm]:

= $Fl / (2 \cdot B_{pln}) \cdot e \cdot L / 2$

= 3499 / (2 • 1060) • 203 • 728/2

= 121.669 Kg-m.

Compressive Allowable, $KL/r < Cc$ (11.1 < 123) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$

= $(1 - (11.1)^2 / (2 \cdot 123^2)) 262 /$

$(5/3 + 3 \cdot (11.1) / (8 \cdot 123) - (11.1)^3 / (8 \cdot 123^3))$

= 153.4 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= $S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$

= 5.51/153 + (122 • 130/18761614)/175

= 0.083

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	15.0	189.0	109.5	28.4	310432.5	0.163E+04
Web	202.6	15.0	7.5	30.4	22798.1	742.
Totals	...	...	...	58.7	333230.6	0.238E+04

Distance to Centroid from Datum [ytot]:

= AY / A

= 333231/58.7

= 56.723 mm

Distance to Centroid [C1]:

= max(ytot, abs(Saddle Width - ytot))

= max(56.7, abs(204 - 56.7))

= 147.277 mm

Length of Inner Rib [L]:

= Saddle Height - $\sqrt{(Ro + Wpdthk)^2 - (Pitch/2)^2}$ - Bpthk

= 1050 - $\sqrt{(585 + 30)^2 - (203/2)^2}$ - 30

= 443.842 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{2376/58.7}$



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Saddle Calcs: Operating Case: Step: 19 9:40am Aug 12,2026

= 63.593 mm

Slenderness ratio [KL/r]:

= KL/r

= 1 • 444/63.6

= 6.979

Unit Force [Force,u]:

= FI / (2 • Baseplate Length)

= 3499/(2 • 1060)

= 1.651 Kg/mm

Moment at base of inner Rib [Mbase,c]:

= Unit Force • e • L

= 1.65 • 203 • 444

= 148.458 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

= Bending Moment / Section Modulus

= 148/161313

= 9.025 N./mm²

Compressive Allowable, KL/r < Cc (6.98 < 123) per AISC E2-1 [Sca]:

= (1-(KL/r)²/(2•Cc²))Fy/(5/3+3•(KL/r)/(8•Cc)-(KL/r)³/(8•Cc³))

= (1-(6.98)²/(2 • 123²))262/

(5/3+3•(6.98)/(8• 123)-(6.98³)/(8•123³))

= 155 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= Sc/Sca + (Mbase,c • C1/I)/Sba

= 7.75/155 + (148 • 147/2376)/175

= 0.102

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	137.29 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
Nominal Bolt Diameter	Bnd	30.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	175.00 N./mm ²
Area Available in a Single Bolt	BlArea	3.9077 cm ²
Saddle Load Q0 (Weight)	Q0	20137.6 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	1091.5 Kg
Maximum Transverse Force	Ft	1749.6 Kg
Maximum Longitudinal Force	F1	3499.1 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

= FI / (Bolt Area • Number of Bolts)

= 3499/(3.91 • 4)

= 22.0 N./mm². Must be less than 107.9 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

= Ft / (Bolt Area • Number of Bolts)

= 1750/(3.91 • 4)

= 11.0 N./mm². Must be less than 107.9 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:



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= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:

= $F_l / (\text{BoltShearAllowable} \cdot N_{\text{bolts}})$

= $3499 / (108 \cdot 4)$

= 0.7953 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

= $B \cdot F_t + \text{Sum of X Moments}$

= $1050 \cdot 1750 + 0$

= 1837.07 Kg-m.

Eccentricity (e):

= R_{mom} / QO

= $1837 / 20138$

= 91.22 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:

= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:

= $\max[Bltarearl, Bltarears, Bltareart]$

= $\max[0, 0.8, 0]$

= 0.7953 cm²

MDMT Calculations for: Left Saddle, Saddle to Wearplate Junction, Curve: B

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature : -1 °C

User entered Minimum Design Temperature : -33 °C

The Curve for this Material is : B

From Figure 3.8M- with PWHT, the MDMT is : -1 °C

Temperature reduction per Graph 3.13 : 0 °C

The computed stress ratio is : 1.000

The computed thickness ratio [sr]:

= $(t_r \cdot E^*) / (t_{\text{nom}} - c_i - c_{\text{ext}})$

= $(49.3 \cdot 1) / (30 - 1.5 - 0)$

= 1.000

Warning: Impact Testing Is Mandatory!

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:

Wear Pad Width (250.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$

and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

= $\min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$

= $\min(204 + 1.56 \cdot \sqrt{528 \cdot 53.5}, 2 \cdot 1122)$

= 466.2534 mm

Input and Calculated Values:

Vessel Mean Radius	Rm	528.25	mm
Shell Thickness used in this Case	t	55.000	mm
Stiffened Vessel Length per 4.15.6	L	5610.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1122.00	mm
Saddle Width	b1 or b	204.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees



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Shell Allowable Stress used in Calculation	161.86	N./mm ²
Head Allowable Stress used in Calculation	151.00	N./mm ²
Circumferential Efficiency in Plane of Saddle	1.00	
Circumferential Efficiency at Mid-Span	1.00	
Distance from Saddle Base to Centerline	B	1050.00 mm
Coefficient of Friction	mu	0.00

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case	25244.29	Kgf
Pressure used in Analysis	P	16000.245 KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	77.38	161.86
Long. Stress at Bottom of Midspan	80.61	161.86
Long. Stress at Top of Saddles	88.88	161.86
Long. Stress at Bottom of Saddles	73.52	161.86
Tangential Shear in Shell	6.15	129.48
Circ. Stress at Horn of Saddle	9.34	202.32
Circ. Compressive Stress in Shell	0.75	161.86

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= Ftr(Ft/Num \text{ of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (2590/2 + 0) \cdot 1050/1013$
 $= 4026.3 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(Fl,wind; \text{Sum of X Forces}) \cdot B / Ls$
 $= \max(298, 0) \cdot 1050/3366$
 $= 92.9 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(Fl,eq; \text{Sum of X Forces}) \cdot B / Ls$
 $= \max(3499, 0) \cdot 1050/3366$
 $= 1091.5 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= Ftr(Ft/Num \text{ of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (3499/2 + 0) \cdot 1050/1013$
 $= 5439.0 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(Fwl, Fwt, Fsl, Fst)$
 $= 19805 + \max(92.9, 4026, 1092, 5439)$
 $= 25244.3 \text{ Kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4 (OD \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1906 (0 + \pi/4 (1.11 \cdot 1.2)^2)$
 $= 2921.953 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	25572.84	Kgf
Transverse Shear Load	1749.55	Kgf
Longitudinal Shear Load	3499.10	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1



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Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(1000/2 + 55 + 30) \sin(120/2), 2 \cdot 528)$$

$$= 1013.250 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$$

$$= -25244 \cdot 1122 [1 - (1 - 1122/5610 + (528^2 - 0^2)/(2 \cdot 1122 \cdot 5610)) / (1 + (4 \cdot 0)/(3 \cdot 5610))]$$

$$= -5037.1 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 25244 \cdot 5610/4 ((1 + 2(528^2 - 0^2)/(5610^2)) / (1 + (4 \cdot 0)/(3 \cdot 5610)) - 4 \cdot 1122/5610)$$

$$= 7709.0 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t)$$

$$= 16000 \cdot 528/(2 \cdot 53.5) - 7709/(\pi \cdot 528^2 \cdot 53.5)$$

$$= 77.38 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$$

$$= 16000 \cdot 528/(2 \cdot 53.5) + 7709/(\pi \cdot 528^2 \cdot 53.5)$$

$$= 80.61 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m/(2t) - M1/(K1 \cdot \pi \cdot R_m^2 t)$$

$$= 16000 \cdot 528/(2 \cdot 53.5) - 5037/(0.11 \cdot \pi \cdot 528^2 \cdot 53.5)$$

$$= 88.88 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m/(2t) + M1/(K1^* \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 16000 \cdot 528/(2 \cdot 53.5) + 5037/(0.19 \cdot \pi \cdot 528^2 \cdot 53.5)$$

$$= 73.52 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot h_m/3))$$

$$= 25244 (5610 - 2 \cdot 1122)/(5610 + (4 \cdot 0/3))$$

$$= 15146.6 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 15147/(528 \cdot 53.5)$$

$$= 6.15 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(528 \cdot 53.5)}$$

$$= 131.127 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.76 \cdot 25244 \cdot 0.1/(53.5(204 + 131 + 131))$$

$$= -0.75 \text{ N./mm}^2$$



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Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t \cdot (b + X_1 + X_2)) - 3 \cdot K_7 \cdot Q / (2 \cdot t^2)$$

$$= -25244 / (4 \cdot 53.5 \cdot (204 + 131 + 131)) -$$

$$3 \cdot 0.053 \cdot 25244 / (2 \cdot 53.5^2)$$

$$= -9.34 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(204 + 1.56 \cdot \sqrt{528 \cdot 53.5}, 2 \cdot 1122)$$

$$= 466.25 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1060.0000	mm
Baseplate Thickness	Bpthk	30.0000	mm
Baseplate Width	Bpwid	250.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	6	
Rib Thickness	Ribtk	15.0000	mm
Web Thickness	Webtk	15.0000	mm
Web Location	Webloc	Side	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	262.0	N./mm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{R \cdot t} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{502 \cdot 53.5}) + 250)53.5^3 / 12$$

$$= 645.096 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 645 + 270(148 - 26.7)^2$$

$$= 40596.184 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	505.5	53.5	26.7	270.5	723472.4	0.406E+05
Wearplate	250.0	30.0	68.5	75.0	513749.9	0.483E+04
Web	15.0	435.0	301.0	65.2	1964024.8	0.255E+05
BasePlate	250.0	30.0	533.5	75.0	4001249.8	0.111E+06
Totals	...	...	...	485.7	7202497.0	0.182E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 7202497 / 486$$

$$= 148.289 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K_1 \cdot Q$$

$$= 0.2 \cdot 25244$$

$$= 5137.7603 \text{ Kgf}$$



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Tension Stress, $St = (Fh/As) = 2.3408 \text{ N./mm}^2$
 Allowed Stress, $Sa = 0.6 * \text{Yield Str} = 157.1972 \text{ N./mm}^2$

Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$
 $= 1050 - 502 \cdot \sin(120/2) / 1.05$
 $= 635.263 \text{ mm}$

Bending Moment, $M = Fh * d = 3263.8928 \text{ Kg-m.}$

Bending Stress, $Sb = (M * C1 / I) = 2.6039 \text{ N./mm}^2$
 Allowed Stress, $Sa = 2/3 * \text{Yield Str} = 174.6635 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (2 * \text{BasePlateLength} * \text{AllStress}))}$
 $= \sqrt{(3(25244 + 329)250 / (2 * 1060 * 175))}$
 $= 22.538 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 * \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 * \cos(90 - 120/2) (500 + 55 + 30)$
 $= 1013.250 \text{ mm}$

Distance between Ribs [e]:
 $= \text{Web Length} / (\text{Nr ribs} - 1)$
 $= 1013 / (6 - 1)$
 $= 202.650 \text{ mm}$

Baseplate Pressure Area [Ap]:
 $= e * \text{Bpwid} / 2$
 $= 203 * 250 / 2$
 $= 253.312 \text{ cm}^2$

Bearing Pressure [Bp]:
 $= (Q + \text{SaddleWt}) / (\text{BasePlateLength} * \text{BasePlateWidth})$
 $= 25573 / (1060 * 250)$
 $= 9.650 \text{ Kg/cm}^2$

Axial Load [P]:
 $= Ap * Bp$
 $= 253 * 9.65$
 $= 2444.498 \text{ Kg}$

Area of the Rib and Web [Ar]:
 $= \text{Rib Area} + \text{Web Area}$
 $= 28.4 + 15.2$
 $= 43.549 \text{ cm}^2$

Compressive Stress [Sc]:
 $= P/Ar$
 $= 2444/43.5$
 $= 5.505 \text{ N./mm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	15.0	189.0	109.5	28.4	310432.5	0.120E+04
Web	101.3	15.0	7.5	15.2	11399.1	673.
Totals	...	...	...	43.5	321831.6	0.188E+04



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Rib dimension [D]:

= Saddle Width - Web Thickness
= 204 - 15
= 189.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A
= 321832/43.5
= 73.901 mm

Distance to Centroid [C1]:

= max(ytot, Saddle Width - ytot)
= max(73.9, 204 - 73.9)
= 130.099 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
= $\sqrt{1876/43.5}$
= 65.637 mm

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
= 1050 - cos(120/2)(500 + 55 + 30) - 30
= 727.500 mm

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
= $\sqrt{2 \cdot \pi^2 \cdot 199943008/262}$
= 122.739

Slenderness ratio [KL/r]:

= KL/r
= 1 • 728/65.6
= 11.084

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2
= 3499 / (2 • 1060) • 203 • 728/2
= 121.669 Kg-m.

Compressive Allowable, KL/r < Cc (11.1 < 123) per AISC E2-1 [Sca]:

= (1 - (KL/r)² / (2 • Cc²)) Fy / (5/3 + 3 • (KL/r) / (8 • Cc) - (KL/r)³ / (8 • Cc³))
= (1 - (11.1)² / (2 • 123²)) 262 /
(5/3 + 3 • (11.1) / (8 • 123) - (11.1³) / (8 • 123³))
= 153.4 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I) / Sba
= 5.5/153 + (122 • 130/18761614) / 175
= 0.083

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	15.0	189.0	109.5	28.4	310432.5	0.163E+04
Web	202.6	15.0	7.5	30.4	22798.1	742.
Totals	...	...	...	58.7	333230.6	0.238E+04

Distance to Centroid from Datum [ytot]:

= AY / A
= 333231/58.7
= 56.723 mm



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Distance to Centroid [C1]:

$$= \max(y_{tot}, \text{abs}(\text{Saddle Width} - y_{tot}))$$

$$= \max(56.7, \text{abs}(204 - 56.7))$$

$$= 147.277 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{ (Ro + Wpdthk)^2 - (Pitch/2)^2 } - Bpthk$$

$$= 1050 - \sqrt{ (585 + 30)^2 - (203/2)^2 } - 30$$

$$= 443.842 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ \text{Total Inertia} / \text{Total Area} }$$

$$= \sqrt{ 2376/58.7 }$$

$$= 63.593 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 444/63.6$$

$$= 6.979$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 3499 / (2 \cdot 1060)$$

$$= 1.651 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.65 \cdot 203 \cdot 444$$

$$= 148.458 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 148/161313$$

$$= 9.025 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($6.98 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (6.98)^2 / (2 \cdot 123^2)) 262 / (5/3 + 3 \cdot (6.98) / (8 \cdot 123) - (6.98^3) / (8 \cdot 123^3))$$

$$= 155 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c/S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$$

$$= 7.75/155 + (148 \cdot 147/2376) / 175$$

$$= 0.102$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification	SA-193 B7	
Bolt Allowable Stress	Stba	137.29 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
Nominal Bolt Diameter	Bnd	30.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	175.00 N./mm ²
Area Available in a Single Bolt	BltArea	3.9077 cm ²
Saddle Load Q0 (Weight)	Q0	20133.8 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	1091.5 Kg
Maximum Transverse Force	Ft	1749.6 Kg
Maximum Longitudinal Force	F1	3499.1 Kg
Saddle Bolted to Steel Foundation	No	



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Shear Stress in a Single Bolt, Longitudinal Direction [τ_{b,l}]:

$$= F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 3499 / (3.91 \cdot 4)$$

$$= 22.0 \text{ N./mm}^2. \text{ Must be less than } 107.9 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_{b,t}]:

$$= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 1750 / (3.91 \cdot 4)$$

$$= 11.0 \text{ N./mm}^2. \text{ Must be less than } 107.9 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (QO > QL --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F_l / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 3499 / (108 \cdot 4)$$

$$= 0.7953 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot F_t + \text{Sum of X Moments}$$

$$= 1050 \cdot 1750 + 0$$

$$= 1837.07 \text{ Kg-m.}$$

Eccentricity (e):

$$= R_{mom} / QO$$

$$= 1837 / 20134$$

$$= 91.24 \text{ mm} < B_{plen} / 6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0, 0.8, 0]$$

$$= 0.7953 \text{ cm}^2$$

MDMT Calculations for: Right saddle, Saddle to Wearplate Junction, Curve: B

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	-1 °C
User entered Minimum Design Temperature	:	-33 °C
The Curve for this Material is	:	B
From Figure 3.8M- with PWHT, the MDMT is	:	-1 °C
Temperature reduction per Graph 3.13	:	0 °C
The computed stress ratio is	:	1.000

The computed thickness ratio [sr]:

$$= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$$

$$= (49.3 \cdot 1) / (30 - 1.5 - 0)$$

$$= 1.000$$

Warning: Impact Testing Is Mandatory!

Review notes about nozzle loadings and supports in the Warnings report.

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Saddle Calcs: Test Case:

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Note:

Wear Pad Width (250.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

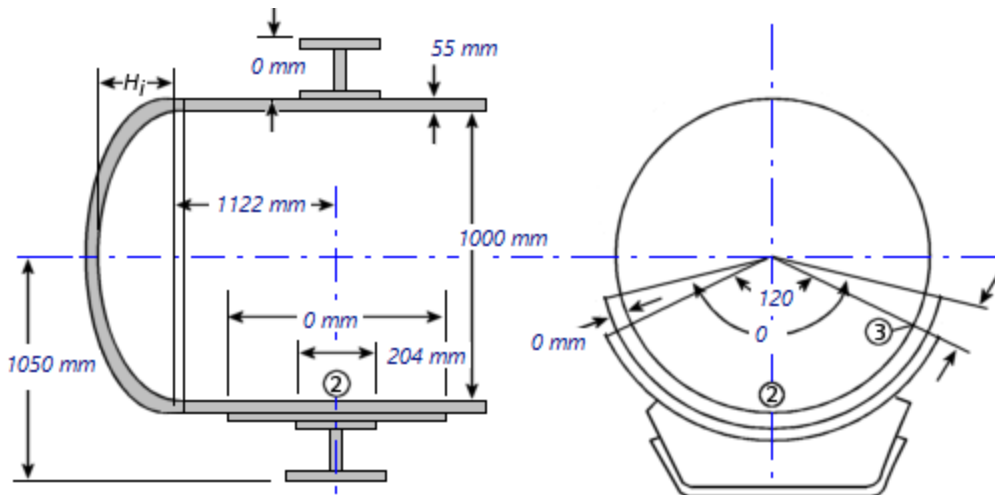
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(204 + 1.56 \cdot \sqrt{527 \cdot 55}, 2 \cdot 1122)$$

$$= 469.7157 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	527.50	mm
Shell Thickness used in this Case	t	55.000	mm
Stiffened Vessel Length per 4.15.6	L	5610.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1122.00	mm
Saddle Width	b1 or b	204.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		247.00	N./mm ²
Head Allowable Stress used in Calculation		247.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1050.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces	22549.58	Kgf
Pressure used in Analysis	P	21996.305 KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	104.08	247.00
Long. Stress at Bottom of Midspan	106.89	247.00
Long. Stress at Top of Saddles	114.10	247.00
Long. Stress at Bottom of Saddles	100.72	247.00



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Saddle Calcs: Test Case: Step: 20 9:40am Aug 12,2026

Tangential Shear in Shell	5.35	138.58
Circ. Stress at Horn of Saddle	7.94	370.49
Circ. Compressive Stress in Shell	0.65	247.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr}(F_t/Num \text{ of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (855/2 + 0) \cdot 1050/1013$
 $= 1328.7 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(98.3, 0) \cdot 1050/3366$
 $= 30.7 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 21221 + \max(30.7, 1329, 0, 0)$
 $= 22549.6 \text{ Kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4 (OD \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1906 (0 + \pi/4 (1.11 \cdot 1.2)^2)$
 $= 2921.953 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	22878.13	Kgf
Transverse Shear Load	427.39	Kgf
Longitudinal Shear Load	98.33	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$
 $= \min(2(1000/2 + 55 + 30)\sin(120/2), 2 \cdot 527)$
 $= 1013.250 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K11* = 0.1923	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$
 $= -22550 \cdot 1122 [1 - (1 - 1122/5610 + (527^2 - 0^2)/(2 \cdot 1122 \cdot 5610))/(1 + (4 \cdot 0)/(3 \cdot 5610))]$
 $= -4501.0 \text{ Kg-m.}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$
 $= 22550 \cdot 5610/4 ((1 + 2(527^2 - 0^2)/(5610^2))/(1 + (4 \cdot 0)/(3 \cdot 5610)) - 4 \cdot 1122/5610)$
 $= 6884.5 \text{ Kg-m.}$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 $= P \cdot R_m/(2t) - M_2/(\pi \cdot R_m^2 t)$
 $= 21996 \cdot 527/(2 \cdot 55) - 6885/(\pi \cdot 527^2 \cdot 55)$
 $= 104.08 \text{ N./mm}^2$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:



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Saddle Calcs: Test Case:

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$$\begin{aligned}
 &= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t) \\
 &= 21996 \cdot 527 / (2 \cdot 55) + 6885 / (\pi \cdot 527^2 \cdot 55) \\
 &= 106.89 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t) \\
 &= 21996 \cdot 527 / (2 \cdot 55) - 4501 / (0.11 \cdot \pi \cdot 527^2 \cdot 55) \\
 &= 114.10 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t) \\
 &= 21996 \cdot 527 / (2 \cdot 55) + 4501 / (0.19 \cdot \pi \cdot 527^2 \cdot 55) \\
 &= 100.72 \text{ N./mm}^2
 \end{aligned}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$\begin{aligned}
 &= Q(L-2a) / (L + (4 \cdot hm / 3)) \\
 &= 22550 (5610 - 2 \cdot 1122) / (5610 + (4 \cdot 0 / 3)) \\
 &= 13529.7 \text{ Kgf}
 \end{aligned}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$\begin{aligned}
 &= K2 \cdot T / (Rm \cdot t) \\
 &= 1.17 \cdot 13530 / (527 \cdot 55) \\
 &= 5.35 \text{ N./mm}^2
 \end{aligned}$$

Decay Length (4.15.20) [x_1, x_2]:

$$\begin{aligned}
 &= 0.78 \cdot \sqrt{(Rm \cdot t)} \\
 &= 0.78 \cdot \sqrt{(527 \cdot 55)} \\
 &= 132.858 \text{ mm}
 \end{aligned}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$\begin{aligned}
 &= -K5 \cdot Q \cdot k / (t(b + X1 + X2)) \\
 &= -0.76 \cdot 22550 \cdot 0.1 / (55(204 + 133 + 133)) \\
 &= -0.65 \text{ N./mm}^2
 \end{aligned}$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8Rm$ (4.15.22) [σ_7]:

$$\begin{aligned}
 &= -Q / (4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q / (2 \cdot t^2) \\
 &= -22550 / (4 \cdot 55(204 + 133 + 133)) - \\
 &\quad 3 \cdot 0.053 \cdot 22550 / (2 \cdot 55^2) \\
 &= -7.94 \text{ N./mm}^2
 \end{aligned}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$\begin{aligned}
 &= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a) \\
 &= \min(204 + 1.56 \cdot \sqrt{(527 \cdot 55)}, 2 \cdot 1122) \\
 &= 469.72 \text{ mm}
 \end{aligned}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1060.0000	mm
Baseplate Thickness	Bpthk	30.0000	mm
Baseplate Width	Bpwid	250.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	6	
Rib Thickness	Ribtk	15.0000	mm
Web Thickness	Webtk	15.0000	mm
Web Location	Webloc	Side	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw, c	306.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$\begin{aligned}
 &= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12 \\
 &= ((1.56 \cdot \sqrt{(500 \cdot 55)} + 250)^3) / 12 \\
 &= 705.286 \text{ cm}^4
 \end{aligned}$$

Resolved Inertia for the Shell [I,shell]:



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$$\begin{aligned}
 &= \text{ShellInertia} + A\text{Shell}(C1 - Y)^2 \\
 &= 705 + 280 (147 - 27.5)^2 \\
 &= 40709.359 \text{ cm}^4
 \end{aligned}$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	508.7	55.0	27.5	279.8	769403.8	0.407E+05
Wearplate	250.0	30.0	70.0	75.0	524999.9	0.451E+04
Web	15.0	435.0	302.5	65.2	1973812.4	0.261E+05
BasePlate	250.0	30.0	535.0	75.0	4012500.0	0.113E+06
Totals	...	...	...	495.0	7280715.5	0.184E+06

Distance to Centroid [C1]:

$$\begin{aligned}
 &= AY / A \\
 &= 7280715/495 \\
 &= 147.075 \text{ mm}
 \end{aligned}$$

Angle [β]:

$$\begin{aligned}
 &= 180 - \text{Saddle Angle}/2 \\
 &= 180 - 120/2 \\
 &= 120.0
 \end{aligned}$$

Saddle Splitting Coefficient [K1]:

$$\begin{aligned}
 &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta)) \\
 &= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120)) \\
 &= 0.2035
 \end{aligned}$$

Saddle Splitting Force [Fh]:

$$\begin{aligned}
 &= K1 \cdot Q \\
 &= 0.2 \cdot 22550 \\
 &= 4589.3276 \text{ Kgf}
 \end{aligned}$$

$$\begin{aligned}
 \text{Tension Stress, } St &= (Fh/As) = 2.0909 \text{ N./mm}^2 \\
 \text{Allowed Stress, } Sa &= 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2
 \end{aligned}$$

Saddle Splitting Dimension [d]:

$$\begin{aligned}
 &= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians}) \\
 &= 1050 - 500 \cdot \sin(120/2) / 1.05 \\
 &= 636.503 \text{ mm}
 \end{aligned}$$

$$\text{Bending Moment, } M = Fh \cdot d = 2921.1804 \text{ Kg-m.}$$

$$\begin{aligned}
 \text{Bending Stress, } Sb &= (M \cdot C1 / I) = 2.2874 \text{ N./mm}^2 \\
 \text{Allowed Stress, } Sa &= 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2
 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned}
 &= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})) \\
 &= \sqrt{(3(22550 + 329) \cdot 250) / (2 \cdot 1060 \cdot 173)} \\
 &= 21.399 \text{ mm}
 \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$\begin{aligned}
 &= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\
 &= 2 \cdot \cos(90 - 120/2) (500 + 55 + 30) \\
 &= 1013.250 \text{ mm}
 \end{aligned}$$

Distance between Ribs [e]:

$$\begin{aligned}
 &= \text{Web Length} / (N_{\text{ribs}} - 1) \\
 &= 1013 / (6 - 1) \\
 &= 202.650 \text{ mm}
 \end{aligned}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$



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Saddle Calcs: Test Case: Step: 20 9:40am Aug 12,2026

$$= 203 \cdot 250/2$$

$$= 253.312 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 22878 / (1060 \cdot 250)$$

$$= 8.633 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 253 \cdot 8.63$$

$$= 2186.911 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 28.4 + 15.2$$

$$= 43.549 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 2187 / 43.5$$

$$= 4.925 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	15.0	189.0	109.5	28.4	310432.5	0.120E+04
Web	101.3	15.0	7.5	15.2	11399.1	673.
Totals	...	...	...	43.5	321831.6	0.188E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 204 - 15$$

$$= 189.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 321832 / 43.5$$

$$= 73.901 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(\text{ytot}, \text{Saddle Width} - \text{ytot})$$

$$= \max(73.9, 204 - 73.9)$$

$$= 130.099 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(1876 / 43.5)}$$

$$= 65.637 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1050 - \cos(120/2) (500 + 55 + 30) - 30$$

$$= 727.500 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199943008 / 260)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL / r$$



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Saddle Calcs: Test Case: Step: 20 9:40am Aug 12,2026

$$= 1 \cdot 728/65.6$$

$$= 11.084$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 98.3 / (2 \cdot 1060) \cdot 203 \cdot 728/2$$

$$= 3.419 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($11.1 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (Klr)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (Klr) / (8 \cdot Cc) - (Klr)^3 / (8 \cdot Cc^3))$$

$$= (1 - (11.1)^2 / (2 \cdot 123^2)) 260 /$$

$$(5/3 + 3 \cdot (11.1) / (8 \cdot 123) - (11.1^3) / (8 \cdot 123^3))$$

$$= 152.3 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 4.92/152 + (3.42 \cdot 130/18761614) / 173$$

$$= 0.034$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	15.0	189.0	109.5	28.4	310432.5	0.163E+04
Web	202.6	15.0	7.5	30.4	22798.1	742.
Totals	...	...	...	58.7	333230.6	0.238E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 333231/58.7$$

$$= 56.723 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(ytot, \text{abs}(\text{Saddle Width} - ytot))$$

$$= \max(56.7, \text{abs}(204 - 56.7))$$

$$= 147.277 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{ (Ro + Wpdthk)^2 - (Pitch/2)^2 } - Bpthk$$

$$= 1050 - \sqrt{ (585 + 30)^2 - (203/2)^2 } - 30$$

$$= 443.842 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (2376/58.7) }$$

$$= 63.593 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 444/63.6$$

$$= 6.979$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 98.3 / (2 \cdot 1060)$$

$$= 0.046 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.046 \cdot 203 \cdot 444$$

$$= 4.172 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$



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Saddle Calcs: Test Case:

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$$= 4.17/161313$$

$$= 0.254 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($6.98 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (6.98)^2 / (2 \cdot 123^2)) 260 / (5/3 + 3 \cdot (6.98) / (8 \cdot 123) - (6.98^3) / (8 \cdot 123^3))$$

$$= 153.8 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$$

$$= 6.94/154 + (4.17 \cdot 147/2376) / 173$$

$$= 0.047$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	172.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
Nominal Bolt Diameter	Bnd	30.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	175.00 N./mm ²
Area Available in a Single Bolt	BltArea	3.9077 cm ²
Saddle Load QO (Weight)	QO	21549.5 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	30.7 Kg
Maximum Transverse Force	Ft	427.4 Kg
Maximum Longitudinal Force	Fl	98.3 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 98.3 / (3.91 \cdot 4)$$

$$= 0.6 \text{ N./mm}^2. \text{ Must be less than } 103.2 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 427 / (3.91 \cdot 4)$$

$$= 2.7 \text{ N./mm}^2. \text{ Must be less than } 103.2 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} QO > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= Fl / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 98.3 / (103 \cdot 4)$$

$$= 0.0234 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 1050 \cdot 427 + 0$$

$$= 448.77 \text{ Kg-m.}$$

Eccentricity (e):

$$= Rmom / QO$$

$$= 449/21549$$

$$= 20.82 \text{ mm} < B_{plen}/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:



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Saddle Calcs: Test Case: Step: 20 9:40am Aug 12,2026

= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:

= max[Bltarearl, Bltarears, Bltareart]

= max[0, 0.023, 0]

= 0.0234 cm²

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:

Wear Pad Width (250.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$

and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

= min($b + 1.56 \cdot \sqrt{R_m \cdot t}$, 2a)

= min(204 + $1.56 \cdot \sqrt{527 \cdot 55}$, 2 • 1122)

= 469.7157 mm

Input and Calculated Values:

Vessel Mean Radius	Rm	527.50	mm
Shell Thickness used in this Case	t	55.000	mm
Stiffened Vessel Length per 4.15.6	L	5610.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1122.00	mm
Saddle Width	b1 or b	204.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		247.00	N./mm ²
Head Allowable Stress used in Calculation		247.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1050.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		22546.58	Kgf
Pressure used in Analysis	P	21996.305	KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	104.08	247.00
Long. Stress at Bottom of Midspan	106.89	247.00
Long. Stress at Top of Saddles	114.10	247.00
Long. Stress at Bottom of Saddles	100.72	247.00
Tangential Shear in Shell	5.35	138.58
Circ. Stress at Horn of Saddle	7.93	370.49
Circ. Compressive Stress in Shell	0.65	247.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

= Ftr(Ft/Num of Saddles + Z Force Load) • B / E

= 3 (855/2 + 0) • 1050/1013

= 1328.7 Kgf

Longitudinal Saddle Reaction Force [Fwl]:

= max(Fl,wind; Sum of X Forces) • B / Ls

= max(98.3, 0) • 1050/3366



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Saddle Calcs: Test Case:

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= 30.7 Kgf

Load Combination Results for Q + Wind or Seismic [Q]:

= Saddle Load + max(Fwl, Fwt, Fsl, Fst)

= 21218 + max(30.7, 1329, 0, 0)

= 22546.6 Kgf

Longitudinal Wind Force [F]:

= WindScalar • WindPress(Platform Area + ($\pi/4$ (OD • WindDiaMult)²))

= 1.1 • 1906 (0 + $\pi/4$ (1.11 • 1.2)²)

= 2921.953 N

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	22875.14	Kgf
Transverse Shear Load	427.39	Kgf
Longitudinal Shear Load	98.33	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

= min(2(ShellID/2 + t + WearPadThickness)sin($\theta/2$), 2•Rm)

= min(2(1000/2 + 55 + 30)sin(120/2), 2•527)

= 1013.250 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

= -Q•a [1 - (1 - a/L + (Rm²-hm²)/(2a•L))/(1+(4hm)/3L)]

= -22547•1122[1-(1-1122/5610+(527²-0²)/

(2•1122•5610))/(1+(4•0)/(3•5610))]

= -4500.4 Kg-m.

Moment per Equation 4.15.2 [M2]:

= Q•L/4((1+2(Rm²-hm²)/(L²))/(1+(4hm²)/(3L))-4a/L)

= 22547•5610/4((1+2(527²-0²)/(5610²))/(1+(4•0)/

(3•5610))-4•1122/5610)

= 6883.6 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

= P • Rm/(2t) - M2/(π •Rm²t)

= 21996 • 527/(2•55) - 6884/(π •527²•55)

= 104.08 N./mm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

= P • Rm/(2t) + M2/(π • Rm² • t)

= 21996 • 527/(2 • 55) + 6884/(π • 527² • 55)

= 106.89 N./mm²

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_3]:

= P • Rm/(2t) - M1/(K1• π •Rm²t)

= 21996•527/(2•55)--4500/(0.11• π •527²•55)

= 114.10 N./mm²

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_4]:

= P • Rm/(2t) + M1/(K1* • π • Rm² • t)

= 21996•527/(2•55)+4500/(0.19• π •527²•55)

= 100.72 N./mm²

Maximum Shear Force in the Saddle (4.15.3) [T]:

= Q(L-2a)/(L+(4•hm/3))



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Saddle Calcs: Test Case: Step: 20 9:40am Aug 12,2026

$$= 22547 (5610 - 2 \cdot 1122) / (5610 + (4 \cdot 0/3))$$

$$= 13528.0 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 13528 / (527 \cdot 55)$$

$$= 5.35 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{ (R_m \cdot t) }$$

$$= 0.78 \cdot \sqrt{ (527 \cdot 55) }$$

$$= 132.858 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t (b + X_1 + X_2))$$

$$= -0.76 \cdot 22547 \cdot 0.1 / (55 (204 + 133 + 133))$$

$$= -0.65 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t (b + X_1 + X_2)) - 3 \cdot K_7 \cdot Q / (2 \cdot t^2)$$

$$= -22547 / (4 \cdot 55 (204 + 133 + 133)) -$$

$$3 \cdot 0.053 \cdot 22547 / (2 \cdot 55^2)$$

$$= -7.93 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{ (R_m \cdot t) }, 2a)$$

$$= \min(204 + 1.56 \cdot \sqrt{ (527 \cdot 55) }, 2 \cdot 1122)$$

$$= 469.72 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1060.0000	mm
Baseplate Thickness	Bpthk	30.0000	mm
Baseplate Width	Bpwid	250.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	6	
Rib Thickness	Ribtk	15.0000	mm
Web Thickness	Webtk	15.0000	mm
Web Location	Webloc	Side	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw, c	306.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{ (R \cdot t) } + \text{WearPlateWidth}) t^3) / 12$$

$$= ((1.56 \cdot \sqrt{ (500 \cdot 55) } + 250) 55^3) / 12$$

$$= 705.286 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C_1 - Y)^2$$

$$= 705 + 280 (147 - 27.5)^2$$

$$= 40709.359 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	508.7	55.0	27.5	279.8	769403.8	0.407E+05
Wearplate	250.0	30.0	70.0	75.0	524999.9	0.451E+04
Web	15.0	435.0	302.5	65.2	1973812.4	0.261E+05
BasePlate	250.0	30.0	535.0	75.0	4012500.0	0.113E+06
Totals	...	...	...	495.0	7280715.5	0.184E+06

Distance to Centroid [C1]:

$$= AY / A$$



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Saddle Calcs: Test Case: Step: 20 9:40am Aug 12, 2026

$$= 7280715/495$$

$$= 147.075 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 22547$$

$$= 4588.7188 \text{ Kgf}$$

Tension Stress, $St = (Fh/As) = 2.0906 \text{ N./mm}^2$

Allowed Stress, $Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1050 - 500 \cdot \sin(120/2) / 1.05$$

$$= 636.503 \text{ mm}$$

Bending Moment, $M = Fh \cdot d = 2920.7930 \text{ Kg-m.}$

Bending Stress, $Sb = (M \cdot C1 / I) = 2.2871 \text{ N./mm}^2$

Allowed Stress, $Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(22547 + 329)250) / (2 \cdot 1060 \cdot 173)}$$

$$= 21.398 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (500 + 55 + 30)$$

$$= 1013.250 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 1013 / (6 - 1)$$

$$= 202.650 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 203 \cdot 250/2$$

$$= 253.312 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 22875 / (1060 \cdot 250)$$

$$= 8.632 \text{ Kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 253 \cdot 8.63$$

$$= 2186.625 \text{ Kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$



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Saddle Calcs: Test Case: Step: 20 9:40am Aug 12,2026

$$= 28.4 + 15.2$$

$$= 43.549 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/Ar$$

$$= 2187/43.5$$

$$= 4.924 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	15.0	189.0	109.5	28.4	310432.5	0.120E+04
Web	101.3	15.0	7.5	15.2	11399.1	673.
Totals	...	...	...	43.5	321831.6	0.188E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 204 - 15$$

$$= 189.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 321832/43.5$$

$$= 73.901 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(ytot, \text{Saddle Width} - ytot)$$

$$= \max(73.9, 204 - 73.9)$$

$$= 130.099 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ \text{Total Inertia} / \text{Total Area} }$$

$$= \sqrt{ 1876/43.5 }$$

$$= 65.637 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1050 - \cos(120/2) (500 + 55 + 30) - 30$$

$$= 727.500 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{ 2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress} }$$

$$= \sqrt{ 2 \cdot \pi^2 \cdot 199943008/260 }$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 728/65.6$$

$$= 11.084$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 98.3 / (2 \cdot 1060) \cdot 203 \cdot 728/2$$

$$= 3.419 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($11.1 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (11.1)^2 / (2 \cdot 123^2)) 260 / (5/3 + 3 \cdot (11.1) / (8 \cdot 123) - (11.1^3) / (8 \cdot 123^3))$$

$$= 152.3 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)



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Saddle Calcs: Test Case: Step: 20 9:40am Aug 12,2026

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 4.92/152 + (3.42 \cdot 130/18761614) / 173$$

$$= 0.034$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	15.0	189.0	109.5	28.4	310432.5	0.163E+04
Web	202.6	15.0	7.5	30.4	22798.1	742.
Totals	...	...	...	58.7	333230.6	0.238E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 333231/58.7$$

$$= 56.723 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(ytot, \text{abs}(\text{Saddle Width} - ytot))$$

$$= \max(56.7, \text{abs}(204 - 56.7))$$

$$= 147.277 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{ (Ro + Wpdthk)^2 - (Pitch/2)^2 } - Bpthk$$

$$= 1050 - \sqrt{ (585 + 30)^2 - (203/2)^2 } - 30$$

$$= 443.842 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ \text{Total Inertia} / \text{Total Area} }$$

$$= \sqrt{ 2376/58.7 }$$

$$= 63.593 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 444/63.6$$

$$= 6.979$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 98.3 / (2 \cdot 1060)$$

$$= 0.046 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.046 \cdot 203 \cdot 444$$

$$= 4.172 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 4.17/161313$$

$$= 0.254 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < Cc$ ($6.98 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (6.98)^2 / (2 \cdot 123^2)) 260 / (5/3 + 3 \cdot (6.98) / (8 \cdot 123) - (6.98^3) / (8 \cdot 123^3))$$

$$= 153.8 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I) / Sba$$

$$= 6.94/154 + (4.17 \cdot 147/2376) / 173$$

$$= 0.047$$

Input Data for Base Plate Bolting Calculations:



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Saddle Calcs: Test Case:

Step: 20 9:40am Aug 12,2026

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	172.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
Nominal Bolt Diameter	Bnd	30.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	175.00 N./mm ²
Area Available in a Single Bolt	BltArea	3.9077 cm ²
Saddle Load Q0 (Weight)	Q0	21546.5 Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	30.7 Kgf
Maximum Transverse Force	Ft	427.4 Kgf
Maximum Longitudinal Force	Fl	98.3 Kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

= Fl / (Bolt Area • Number of Bolts)

= 98.3/(3.91 • 4)

= 0.6 N./mm². Must be less than 103.2 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

= Ft / (Bolt Area • Number of Bolts)

= 427/(3.91 • 4)

= 2.7 N./mm². Must be less than 103.2 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

= 0.0 (Q0 > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:

= Fl / (BoltShearAllowable • Nbolts)

= 98.3/(103 • 4)

= 0.0234 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

= B • Ft + Sum of X Moments

= 1050 • 427 + 0

= 448.77 Kg-m.

Eccentricity (e):

= Rmom / Q0

= 449/21546

= 20.83 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:

= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:

= max[Bltarearl, Bltarears, Bltareart]

= max[0, 0.023, 0]

= 0.0234 cm²

Review notes about nozzle loadings and supports in the Warnings report.

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Nozzle Calcs.: N1- Channel inl Nozl: 9 9:40am Aug 12,2026

Input, Nozzle Desc: N1- Channel inlet

From: 30

Pressure for Reinforcement Calculations	P	16000.000	KPa.
Temperature for Internal Pressure	Temp	100	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	159.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	175.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1000.00	mm
Shell Finished (Minimum) Thickness	t	56.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1190.00	mm
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	151.00	N./mm ²
Allowable Stress At Ambient	Sna	165.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		15.9528	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	26.0000	mm
Hub Height of Integral Nozzle	h	190.0000	mm
Height of Beveled Transition	L'	74.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		100.0000	mm
Corrosion Allowance	can	1.5000	mm
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	30.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		50.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N1- Channel inlet

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	15.953	in.
Actual Thickness Used in Calculation	1.024	in.



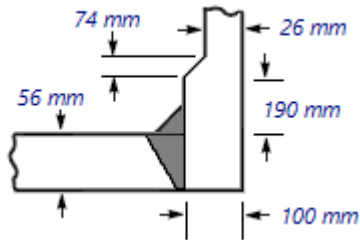
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Nozzle Calcs.: N1- Channel in1

Nozl: 9 9:40am Aug 12, 2026



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 408 \cdot (\exp(16000/(151 \cdot 1)) - 1) + 1.5 + 0$$

$$= 22.8 + 1.5 + 0 = 24.3 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 151 \cdot 1 \cdot \ln((2 \cdot 24.5 + 408) / 408) - 0$$

$$= 17117 - 0 = 17117 \text{ KPa.}$$

Nozzle/Tubesheet Proximity Warning:

This nozzle opening seems to be too close to the tubesheet (see 4.18.3(h) and Figure 4.18.19).

Nozzle Dimension d, 392.400 mm, is < 1.8 * sqrt(D * t): 420.844 mm.

This nozzle or its diameter limit intersects weld seam # [3] on this element. The distance between the nozzle and seam is, 140.04 mm, while the distance from the center of the nozzle to the edge of the diameter limit is: 467.92 mm

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 9.840 mm

Min. required shell thickness, internal p + c : 54.643 mm

Nozzle neck minimum required thickness : 9.840 mm

Minimum nozzle neck thickness 26.010 mm, is greater than required thickness 9.840 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 86.3,	Allowable	: 151.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 308.7 N./mm ²	Passed
Occasional	: 62.9,	Allowable	: 200.8 N./mm ²	Passed
Shear	: 15.1,	Allowable	: 105.7 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[151/159, 1]$$

$$= 0.950$$



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Nozzle Calcs.: N1- Channel in1 Nozl: 9 9:40am Aug 12, 2026

Thickness of Nozzle at Shell [tn]:

= hub thickness - corrosion allowance
= 100 - 1.5
= 98.500 mm

Thickness of Nozzle at Top [tn2]:

= thickness - corrosion allowance
= 26 - 1.5
= 24.500 mm

Shell Diameter to Thickness ratio [D/t]:

= Di/t
= 1003/54.5
= 18.404 must be less than 400.

Effective Pressure Radius [Reff]:

= Di/2 + corrosion allowance
= 1000/2 + 1.5
= 501.500 mm

Effective Length of Vessel Wall [LR]:

= $\min(\sqrt{Reff \cdot t}, 2 \cdot Rn)$
= $\min(\sqrt{502 \cdot 54.5}, 2 \cdot 204)$
= 165.323 mm

Thickness Limit Candidate [LH1]:

= $\min[1.5t, te] + \sqrt{Rn \cdot tn}$
= $\min[1.5 \cdot 54.5, 0] + \sqrt{204 \cdot 98.5}$
= 141.788 mm

Thickness Limit Candidate [LH2]:

= Lpr1
= 250
= 250.000 mm

Thickness Limit Candidate [LH3]:

= $8(t + te)$
= $8(54.5 + 0)$
= 436.000 mm

Effective Nozzle Wall Length Outside the Vessel [LH]:

= $\min[LH1, LH2, LH3] + t$ for set-in nozzles
= $\min[142, 250, 436] + 54.5$
= 196.288 mm

Effective Vessel Thickness [teff]:

= $t + (A5 \cdot frp) / LR$
= $54.5 + ((0 \cdot 1) / 165)$
= 54.500 mm

Parameter [λ]:

= $\min(12, (2 \cdot Rn + tn) / (\sqrt{(Di + teff) \cdot teff}))$
= $\min(12, (2 \cdot 204 + 98.5) / (\sqrt{(1003 + 54.5) \cdot 54.5}))$
= 2.111

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

= $t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$
= $54.5 \cdot 165 \cdot \max((2.11/5)^{0.85}, 1)$
= 90.101 cm²

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

= $tn \cdot LH$
= $98.5 \cdot 196$



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Nozzle Calcs.: N1- Channel in1 Nozl: 9 9:40am Aug 12,2026

$$= 193.344 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$

$$= 0.5 \cdot 30^2$$

$$= 4.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$

$$= 90.1 + 0.95(193 + 0) + 4.5 + 0 + 0$$

$$= 278.217 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 98.5 / \ln[1 + 98.5/204]$$

$$= 250.126 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{\text{eff}} / \ln[1 + t_{\text{eff}}/R_{\text{eff}}]$$

$$= 54.5 / \ln[1 + 54.5/502]$$

$$= 528.282 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 159 \cdot 1$$

$$= 238.5 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{\text{xn}} \cdot L_{\text{H}}$$

$$= 16000 \cdot 250 \cdot 196$$

$$= 80106.7 \text{ Kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{\text{xs}}(L_{\text{R}} + t_n)$$

$$= 16000 \cdot 528(165 + 98.5)$$

$$= 227402.4 \text{ Kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{\text{xs}} \cdot R_{\text{nc}}$$

$$= 16000 \cdot 528 \cdot 204$$

$$= 175924.0 \text{ Kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (227402 + 175924 + 80107) / 16000$$

$$= 2962.9 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{\text{allow}} / (2 \cdot A_p / AT - R_{\text{xs}} / t_{\text{eff}})$$

$$= 238 / (2 \cdot 2963 / 278 - 528 / 54.5)$$

$$= 20548.1 \text{ KPa.}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t / R_{\text{xs}}]$$

$$= 159 [54.5 / 528]$$

$$= 16402.1 \text{ KPa.}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{\text{max1}}, P_{\text{max2}})$$

$$= \min(20548, 16402)$$

$$= 16402.068 \text{ KPa.}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / AT$$

$$= (80107 + 227402 + 175924) / 278$$



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 Nozzle Calcs.: N1- Channel inl Nozl: 9 9:40am Aug 12,2026

$$= 170.405 \text{ N./mm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot Rxs / t_{eff}$$

$$= 16000 \cdot 528 / 54.5$$

$$= 155.1 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 170 - 155, 155)$$

$$= 185.7 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	238.50	N./mm ²
Local Primary Membrane Stress	PL	185.71	N./mm ²
Maximum Allowable Working Pressure	Pmax	16402.07	KPa.

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (204 + 98.5) / 204$$

$$= 1.483 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lr]:

$$= \pi / 2 \cdot (R_n + t_n)$$

$$= \pi / 2 \cdot (204 + 98.5)$$

$$= 475.323 \text{ mm}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 21.213, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$t_c, L41T = 21.213, \text{ must be } \geq \min(0.7 \cdot 98.500, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(f_y \cdot ky, 1.5 \cdot S_n(A_2 + A_3), \pi / 4 \cdot P \cdot R_n^2 \cdot ky^2)$$

$$= \min(175924 \cdot 1.48, 1.5 \cdot 151(193+0), \pi / 4 \cdot 16000 \cdot 204^2 \cdot 1.48^2)$$

$$= 117339.492 \text{ Kgf}$$

Weld Stress Value [T]:

$$= fwelds / (L_r(0.49 \cdot L41T + 0.6 \cdot t_{w1} + 0.49 \cdot L43T))$$

$$= 117339 / (475(0.49 \cdot 21.2 + 0.6 \cdot 50 + 0.49 \cdot 0))$$

$$= 59.933 < \text{ or } = \text{ to } 158.998 \text{ Weld Stress Passed}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	5 °C
User entered Minimum Design Temperature	:	-33 °C
The Curve for this Material is	:	B
From Figure 3.8M- with PWHT, the MDMT is	:	11 °C
Temperature reduction per Graph 3.13	:	6 °C
The computed stress ratio is	:	0.881

The computed thickness ratio [sr]:

$$= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$$

$$= (48 \cdot 1) / (56 - 1.5 - 0)$$

$$= 0.881$$

Warning: Impact Testing Is Mandatory!

Gov. MDMT of the nozzle to shell joint welded assembly : 5 °C



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Nozzle Calcs.: N1- Channel inl Nozl: 9 9:40am Aug 12,2026

The Drop for this Nozzle is : 101.9632 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 407.9633 mm

Input Echo, WRC297 Item 1, Description: N1- Channel in:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1000.000	mm
Shell Thickness	56.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 175.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 159.00	N./mm ²

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm
Nozzle Diameter	405.200	mm
Nozzle Thickness	100.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-350 LF2	
Nozzle Cold S.I. Allowable	SNmc 165.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 151.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	16000.00	KPa.
Radial Load	P	-5736.27	Kgf
Circumferential Shear	Vc	4589.03	Kgf
Longitudinal Shear	Vl	4589.03	Kgf
Circumferential Moment	Mc	6192.80	Kg-m.
Longitudinal Moment	Ml	6192.80	Kg-m.
Torsional Moment	Mt	7224.93	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, D_v/T_v (19.857) is < 20
check the Limitations of WRC bulletin 297.

Warning - The ratio, D_n/D_v (.544) is > 0.5
check the Limitations of WRC bulletin 297.

Warning - The ratio, D_n/T_n (6.052) is < 10
check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 605.2 + 2 \cdot 1.65 \cdot \sqrt{528.75(56.0 - 1.5)}$$

$$= 1165.393 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1057.500	mm
Nozzle Outside Diameter	(d) =	605.200	mm
Vessel Thickness used	(T) =	54.500	mm
Nozzle Thickness used	(t) =	98.500	mm
T / t	=	0.553	



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Nozzle Calcs.: N1- Channel inl Nozl: 9 9:40am Aug 12,2026

$$d / t = 10.000$$

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.521$$

Nr/P	=	0.066
Mr/P	=	0.119
MØ/P	=	0.037
NØ/P	=	0.091
MrD/Mc	=	0.270
NrDL/Mc	=	0.106
MØd/Mc	=	0.084
NØDL/Mc	=	0.077
MrD/Ml	=	0.103
NrDL/Ml	=	0.056
MØD/Ml	=	0.031
NØDL/Ml	=	0.110

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-3	-3	3	3
Outplane Bending (ML)	-6	6	6	-6
Normal Pressure Stress	139	155	139	155
Outplane Stress Summary	135	155	155	150

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	13	-13	13	-13
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-20	20	20	-20
Inplane Pressure Stress	69	69	69	69
Inplane Stress Summary	61	76	107	38

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	3	3	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	135	155	155	150



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Nozzle Calcs.: N1- Channel inl Nozl: 9 9:40am Aug 12,2026

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	-2	-2	2	2
Outplane Bending (Mc)	-16	16	16	-16
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	69	69	69	69
Outplane Stress Summary	56	81	95	52

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	13	-13	13	-13
Inplane Membrane (Mc)	-3	-3	3	3
Inplane Bending (Mc)	-54	54	54	-54
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	139	155	139	155
Inplane Stress Summary	96	194	212	92

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	2	2	2	2
Shear Stress Summary	1	1	3	3

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	96	194	212	92

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-3	-3	3	3
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	32	51	32	51
Hoop Stress Summary	30	49	37	56

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
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Nozzle Calcs.: N1- Channel inl Nozl: 9 9:40am Aug 12,2026

(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	2	-2	2	-2
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-3	-3	3	3
Axial Bending (ML)	-4	4	4	-4
Axial Pressure Stress	16	16	16	16
Axial Stress Summary	11	14	27	12

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1
Shear Stress Summary	1	1	0	0

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	30	49	37	56

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-2	-2	2	2
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	32	51	32	51
Hoop Stress Summary	31	50	36	55

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	2	-2	2	-2
Axial Membrane (Mc)	-3	-3	3	3
Axial Bending (Mc)	-13	13	13	-13
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	16	16	16	16
Axial Stress Summary	2	23	36	3

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0



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Nozzle Calcs.: N1- Channel inl Nozl: 9 9:40am Aug 12,2026

Shear due to Torsion	1	1	1	1
Shear Stress Summary	0	0	1	1

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	31	50	38	55

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	139	155	139	155	139	155	139	155
Circ. Pl (SUS)	-2	-2	4	4	-2	-2	4	4
Circ. Q (SUS)	-2	2	10	-10	-41	41	67	-67
Long. Pm (SUS)	69	69	69	69	69	69	69	69
Long. Pl (SUS)	0	0	2	2	-1	-1	3	3
Long. Q (SUS)	-7	7	33	-33	-12	12	20	-20
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	139.0	155.0	139.0	155.0	139.0	155.0	139.0	155.0
Pm+Pl (SUS)	137.0	153.0	143.0	159.0	137.0	153.0	143.0	159.0
Pm+Pl+Q (Total)	135.1	155.1	153.1	149.0	96.1	194.0	210.0	92.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	155.00	159.00	Passed
Pm+Pl (SUS)	159.00	238.50	Passed
Pm+Pl+Q (TOTAL)	210.03	477.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	32	51	32	51	32	51	32	51
Circ. Pl (SUS)	-2	-2	4	4	-1	-1	3	3
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	16	16	16	16	16	16	16	16
Long. Pl (SUS)	-3	-3	3	3	-3	-3	3	3
Long. Q (SUS)	-2	2	6	-6	-11	11	15	-15
Shear Pm (SUS)	0	0	0	0	0	0	0	0



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Nozzle Calcs.: N1- Channel in1 Nozl: 9 9:40am Aug 12,2026

Shear P1 (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	32.0	51.0	32.0	51.0	32.0	51.0	32.0	51.0
Pm+P1 (SUS)	30.0	49.0	36.0	55.0	31.0	50.0	35.0	54.0
Pm+P1+Q (Total)	30.1	49.0	36.1	55.0	31.0	50.0	35.6	54.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	51.00	151.00	Passed
Pm+P1 (SUS)	55.00	226.50	Passed
Pm+P1+Q (TOTAL)	55.02	453.00	Passed

*Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.*

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Nozzle Calcs.: N4- Shell outlet

Nozl: 10 9:40am Aug 12,2026

Input, Nozzle Desc: N4- Shell outlet

From: 40

Pressure for Reinforcement Calculations	P	16000.000	KPa.
Temperature for Internal Pressure	Temp	80	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	161.86	N./mm ²
Parent Allowable Stress At Ambient	Sva	175.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1000.00	mm
Shell Finished (Minimum) Thickness	t	55.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		2680.00	mm
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	153.86	N./mm ²
Allowable Stress At Ambient	Sna	165.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		14.1102	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	24.0000	mm
Hub Height of Integral Nozzle	h	185.0000	mm
Height of Beveled Transition	L'	66.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		90.0000	mm
Corrosion Allowance	can	1.5000	mm
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		55.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N4- Shell outlet

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	14.110	in.
Actual Thickness Used in Calculation	0.945	in.



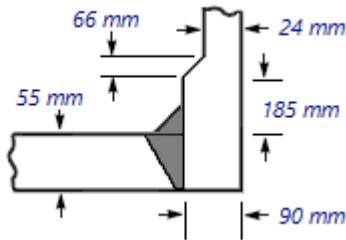
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Nozzle Calcs.: N4- Shell outle

Nozl: 10 9:40am Aug 12,2026



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 361 \cdot (\exp(16000/(154 \cdot 1)) - 1) + 1.5 + 0$$

$$= 19.8 + 1.5 + 0 = 21.3 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 154 \cdot 1 \cdot \ln((2 \cdot 22.5 + 361) / 361) - 0$$

$$= 18054 - 0 = 18054 \text{ KPa.}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 9.840 mm

Min. required shell thickness, internal p + c : 53.656 mm

Nozzle neck minimum required thickness : 9.840 mm

Minimum nozzle neck thickness 24.003 mm , is greater than required thickness 9.840 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 86.1,	Allowable	: 153.9 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 312.4 N./mm ²	Passed
Occasional	: 60.5,	Allowable	: 204.6 N./mm ²	Passed
Shear	: 16.6,	Allowable	: 107.7 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[154/162, 1]$$

$$= 0.951$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 90 - 1.5$$

$$= 88.500 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 24 - 1.5$$

$$= 22.500 \text{ mm}$$



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Nozzle Calcs.: N4- Shell outle

Noz1: 10

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Shell Diameter to Thickness ratio [D/t]:

$$\begin{aligned}
 &= D_i/t \\
 &= 1003/53.5 \\
 &= 18.748 \text{ must be less than 400.}
 \end{aligned}$$

Effective Pressure Radius [Reff]:

$$\begin{aligned}
 &= D_i/2 + \text{corrosion allowance} \\
 &= 1000/2 + 1.5 \\
 &= 501.500 \text{ mm}
 \end{aligned}$$

Effective Length of Vessel Wall [LR]:

$$\begin{aligned}
 &= \min(\sqrt{(R_{eff} \cdot t)}, 2 \cdot R_n) \\
 &= \min(\sqrt{(502 \cdot 53.5)}, 2 \cdot 181) \\
 &= 163.799 \text{ mm}
 \end{aligned}$$

Thickness Limit Candidate [LH1]:

$$\begin{aligned}
 &= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)} \\
 &= \min[1.5 \cdot 53.5, 0] + \sqrt{(181 \cdot 88.5)} \\
 &= 126.459 \text{ mm}
 \end{aligned}$$

Thickness Limit Candidate [LH2]:

$$\begin{aligned}
 &= L_{pr1} \\
 &= 250 \\
 &= 250.000 \text{ mm}
 \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned}
 &= 8(t + t_e) \\
 &= 8(53.5 + 0) \\
 &= 428.000 \text{ mm}
 \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned}
 &= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles} \\
 &= \min[126, 250, 428] + 53.5 \\
 &= 179.959 \text{ mm}
 \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned}
 &= t + ((A5 \cdot frp) / LR) \\
 &= 53.5 + ((0 \cdot 1) / 164) \\
 &= 53.500 \text{ mm}
 \end{aligned}$$

Parameter [λ]:

$$\begin{aligned}
 &= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}})) \\
 &= \min(12, (2 \cdot 181 + 88.5) / (\sqrt{(1003 + 53.5) \cdot 53.5})) \\
 &= 1.892
 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned}
 &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\
 &= 53.5 \cdot 164 \cdot \max((1.89/5)^{0.85}, 1) \\
 &= 87.633 \text{ cm}^2
 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned}
 &= t_n \cdot LH \\
 &= 88.5 \cdot 180 \\
 &= 159.264 \text{ cm}^2
 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned}
 &= 0.5 \cdot \text{Leg}^2 \\
 &= 0.5 \cdot 10^2 \\
 &= 0.500 \text{ cm}^2
 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$



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Nozzle Calcs.: N4- Shell outle Nozl: 10 9:40am Aug 12, 2026

$$= 87.6 + 0.95(159 + 0) + 0.5 + 0 + 0$$

$$= 239.525 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 88.5 / \ln[1 + 88.5/181]$$

$$= 222.018 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 53.5 / \ln[1 + 53.5/502]$$

$$= 527.798 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 162 \cdot 1$$

$$= 242.8 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 16000 \cdot 222 \cdot 180$$

$$= 65189.7 \text{ Kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (L_R + t_n)$$

$$= 16000 \cdot 528 (164 + 88.5)$$

$$= 217270.5 \text{ Kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 16000 \cdot 528 \cdot 181$$

$$= 155611.9 \text{ Kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (217271 + 155612 + 65190) / 16000$$

$$= 2684.9 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$$

$$= 243 / (2 \cdot 2685 / 240 - 528 / 53.5)$$

$$= 19339.2 \text{ KPa.}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S / [t / R_{xs}]$$

$$= 162 [53.5 / 528]$$

$$= 16405.4 \text{ KPa.}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(19339, 16405)$$

$$= 16405.428 \text{ KPa.}$$

Average Primary Membrane Stress [σAvg]:

$$= (f_N + f_S + f_Y) / A_T$$

$$= (65190 + 217271 + 155612) / 240$$

$$= 179.359 \text{ N./mm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 16000 \cdot 528 / 53.5$$

$$= 157.9 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$



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Nozzle Calcs.: N4- Shell outle Nozl: 10 9:40am Aug 12,2026

$$= \max(2 \cdot 179 - 158, 158)$$

$$= 200.9 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	242.78	N./mm ²
Local Primary Membrane Stress	PL	200.86	N./mm ²
Maximum Allowable Working Pressure	Pmax	16405.43	KPa.

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (181 + 88.5) / 181$$

$$= 1.490 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [L_T]:

$$= \pi/2 \cdot (R_n + t_n)$$

$$= \pi/2 \cdot (181 + 88.5)$$

$$= 422.858 \text{ mm}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$t_c, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 88.500, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$$

$$= \min(155612 \cdot 1.49, 1.5 \cdot 154(159+0), \pi/4 \cdot 16000 \cdot 181^2 \cdot 1.49^2)$$

$$= 92865.953 \text{ Kg}$$

Weld Stress Value [T]:

$$= fwelds / (L_T(0.49 \cdot L41T + 0.6 \cdot t_{w1} + 0.49 \cdot L43T))$$

$$= 92866 / (423(0.49 \cdot 7.07 + 0.6 \cdot 53.5 + 0.49 \cdot 0))$$

$$= 60.558 < \text{or } = \text{ to } 161.855 \text{ Weld Stress Passed}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	6 °C
User entered Minimum Design Temperature	:	-33 °C
The Curve for this Material is	:	B
From Figure 3.8M- with PWHT, the MDMT is	:	11 °C
Temperature reduction per Graph 3.13	:	5 °C
The computed stress ratio is	:	0.898

The computed thickness ratio [sr]:

$$= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$$

$$= (48 \cdot 1) / (55 - 1.5 - 0)$$

$$= 0.898$$

Warning: Impact Testing Is Mandatory!

Gov. MDMT of the nozzle to shell joint welded assembly : 6 °C

The Drop for this Nozzle is : 78.6553 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 383.6553 mm

Input Echo, WRC297 Item 1, Description: N4- Shell outl:

Diameter Basis for Cylindrical Shell	ID
Shell Internal Corrosion Allowance	1.5000 mm
Shell External Corrosion Allowance	0.0000 mm



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Nozzle Calcs.: N4- Shell outle Nozl: 10 9:40am Aug 12,2026

Shell Diameter		1000.000	mm
Shell Thickness		55.0000	mm
Shell Stress Concentration Factor		1.000	
Vessel Material		SA-516 70	
Vessel Cold S.I. Allowable	Smc	175.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	161.86	N./mm ²
Diameter Basis for Nozzle		ID	
Nozzle Corrosion Allowance		1.5000	mm
Nozzle Diameter		358.400	mm
Nozzle Thickness		90.0000	mm
Nozzle Stress Concentration Factor		1.000	
Nozzle Material		SA-350 LF2	
Nozzle Cold S.I. Allowable	SNmc	165.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	153.86	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	16000.00	KPa.
Radial Load	P	-5098.92	Kgf
Circumferential Shear	Vc	4079.14	Kgf
Longitudinal Shear	Vl	4079.14	Kgf
Circumferential Moment	Mc	4893.08	Kg-m.
Longitudinal Moment	Ml	4893.08	Kg-m.
Torsional Moment	Mt	5708.59	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tn (5.982) is < 10
check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 538.4 + 2 \cdot 1.65 \cdot \sqrt{528.25(55.0 - 1.5)}$$

$$= 1093.167 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1056.500 mm
Nozzle Outside Diameter	(d) =	538.400 mm
Vessel Thickness used	(T) =	53.500 mm
Nozzle Thickness used	(t) =	88.500 mm
T / t	=	0.605
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.265$$

Nr/P	=	0.070
Mr/P	=	0.127
M0/P	=	0.039
N0/P	=	0.096
MrD/Mc	=	0.271
NrDL/Mc	=	0.110
M0d/Mc	=	0.086
N0DL/Mc	=	0.071
MrD/Ml	=	0.108



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Nozzle Calcs.: N4- Shell outle

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NrDL/M1 = 0.062

MØD/M1 = 0.034

NØDL/M1 = 0.118

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-3	-3	3	3
Outplane Bending (ML)	-6	6	6	-6
Normal Pressure Stress	142	158	142	158
Outplane Stress Summary	138	158	158	153

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	13	-13	13	-13
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-20	20	20	-20
Inplane Pressure Stress	71	71	71	71
Inplane Stress Summary	63	77	107	40

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	3	3	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	138	158	158	153

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	-2	-2	2	2
Outplane Bending (Mc)	-16	16	16	-16
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	71	71	71	71



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Nozzle Calcs.: N4- Shell outle Nozl: 10 9:40am Aug 12,2026

Outplane Stress Summary	58	82	95	54
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	13	-13	13	-13
Inplane Membrane (Mc)	-3	-3	3	3
Inplane Bending (Mc)	-50	50	50	-50
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	142	158	142	158

Inplane Stress Summary	102	193	210	99
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	2	2	2	2

Shear Stress Summary	1	1	3	3
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	102	193	211	99
------------------------	-----	-----	-----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-3	-3	3	3
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	31	50	31	50

Hoop Stress Summary	29	48	36	56
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	3	-3	3	-3
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-3	-3	3	3
Axial Bending (ML)	-5	5	5	-5
Axial Pressure Stress	15	15	15	15

Axial Stress Summary	10	13	28	11
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Nozzle Calcs.: N4- Shell outle

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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1
Shear Stress Summary	1	1	0	0

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	29	48	36	56

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-2	-2	2	2
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	31	50	31	50
Hoop Stress Summary	30	50	35	54

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	3	-3	3	-3
Axial Membrane (Mc)	-3	-3	3	3
Axial Bending (Mc)	-14	14	14	-14
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	15	15	15	15
Axial Stress Summary	0	23	38	1

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1
Shear Stress Summary	0	0	1	1

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	30	50	39	54

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):



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Nozzle Calcs.: N4- Shell outle

Nozl: 10 9:40am Aug 12,2026

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	142	158	142	158	142	158	142	158
Circ. Pl (SUS)	-2	-2	4	4	-2	-2	4	4
Circ. Q (SUS)	-2	2	10	-10	-37	37	63	-63
Long. Pm (SUS)	71	71	71	71	71	71	71	71
Long. Pl (SUS)	0	0	2	2	-1	-1	3	3
Long. Q (SUS)	-7	7	33	-33	-12	12	20	-20
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	142.0	158.0	142.0	158.0	142.0	158.0	142.0	158.0
Pm+Pl (SUS)	140.0	156.0	146.0	162.0	140.0	156.0	146.0	162.0
Pm+Pl+Q (Total)	138.1	158.0	156.1	152.0	103.1	193.0	209.0	99.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	158.00	161.86	Passed
Pm+Pl (SUS)	162.00	242.78	Passed
Pm+Pl+Q (TOTAL)	209.03	485.57	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	31	50	31	50	31	50	31	50
Circ. Pl (SUS)	-2	-2	4	4	-1	-1	3	3
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	15	15	15	15	15	15	15	15
Long. Pl (SUS)	-3	-3	3	3	-3	-3	3	3
Long. Q (SUS)	-2	2	8	-8	-11	11	17	-17
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	31.0	50.0	31.0	50.0	31.0	50.0	31.0	50.0
Pm+Pl (SUS)	29.0	48.0	35.0	54.0	30.0	49.0	34.0	53.0
Pm+Pl+Q (Total)	29.1	48.0	35.1	54.0	30.0	49.0	35.6	53.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
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Nozzle Calcs.: N4- Shell outle

Nozl: 10

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Pm (SUS)	50.00	153.86	Passed
Pm+Pl (SUS)	54.00	230.78	Passed
Pm+Pl+Q (TOTAL)	54.02	461.57	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Nozzle Calcs.: N3- Shell inlet

Nozl: 11 9:40am Aug 12,2026

Input, Nozzle Desc: N3- Shell inlet

From: 40

Pressure for Reinforcement Calculations	P	16000.000	KPa.
Temperature for Internal Pressure	Temp	80	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	161.86	N./mm ²
Parent Allowable Stress At Ambient	Sva	175.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1000.00	mm
Shell Finished (Minimum) Thickness	t	55.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		6680.00	mm
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	153.86	N./mm ²
Allowable Stress At Ambient	Sna	165.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		14.1102	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	24.0000	mm
Hub Height of Integral Nozzle	h	185.0000	mm
Height of Beveled Transition	L'	66.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		90.0000	mm
Corrosion Allowance	can	1.5000	mm
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		55.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N3- Shell inlet

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	14.110	in.
Actual Thickness Used in Calculation	0.945	in.



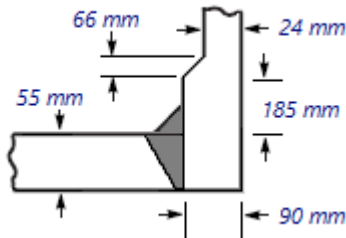
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Nozzle Calcs.: N3- Shell inlet

Noz1: 11 9:40am Aug 12,2026



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 361 \cdot (\exp(16000/(154 \cdot 1)) - 1) + 1.5 + 0$$

$$= 19.8 + 1.5 + 0 = 21.3 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 154 \cdot 1 \cdot \ln((2 \cdot 22.5 + 361) / 361) - 0$$

$$= 18054 - 0 = 18054 \text{ KPa.}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 9.840 mm

Min. required shell thickness, internal p + c : 53.656 mm

Nozzle neck minimum required thickness : 9.840 mm

Minimum nozzle neck thickness 24.003 mm , is greater than required thickness 9.840 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 86.1,	Allowable	: 153.9 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 312.4 N./mm ²	Passed
Occasional	: 60.5,	Allowable	: 204.6 N./mm ²	Passed
Shear	: 16.6,	Allowable	: 107.7 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[154/162, 1]$$

$$= 0.951$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 90 - 1.5$$

$$= 88.500 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 24 - 1.5$$

$$= 22.500 \text{ mm}$$



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Nozzle Calcs.: N3- Shell inlet

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Shell Diameter to Thickness ratio [D/t]:

$$\begin{aligned}
 &= D_i/t \\
 &= 1003/53.5 \\
 &= 18.748 \text{ must be less than 400.}
 \end{aligned}$$

Effective Pressure Radius [Reff]:

$$\begin{aligned}
 &= D_i/2 + \text{corrosion allowance} \\
 &= 1000/2 + 1.5 \\
 &= 501.500 \text{ mm}
 \end{aligned}$$

Effective Length of Vessel Wall [LR]:

$$\begin{aligned}
 &= \min(\sqrt{(R_{eff} \cdot t)}, 2 \cdot R_n) \\
 &= \min(\sqrt{(502 \cdot 53.5)}, 2 \cdot 181) \\
 &= 163.799 \text{ mm}
 \end{aligned}$$

Thickness Limit Candidate [LH1]:

$$\begin{aligned}
 &= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)} \\
 &= \min[1.5 \cdot 53.5, 0] + \sqrt{(181 \cdot 88.5)} \\
 &= 126.459 \text{ mm}
 \end{aligned}$$

Thickness Limit Candidate [LH2]:

$$\begin{aligned}
 &= L_{pr1} \\
 &= 250 \\
 &= 250.000 \text{ mm}
 \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned}
 &= 8(t + t_e) \\
 &= 8(53.5 + 0) \\
 &= 428.000 \text{ mm}
 \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned}
 &= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles} \\
 &= \min[126, 250, 428] + 53.5 \\
 &= 179.959 \text{ mm}
 \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned}
 &= t + ((A5 \cdot frp) / LR) \\
 &= 53.5 + ((0 \cdot 1) / 164) \\
 &= 53.500 \text{ mm}
 \end{aligned}$$

Parameter [λ]:

$$\begin{aligned}
 &= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}})) \\
 &= \min(12, (2 \cdot 181 + 88.5) / (\sqrt{(1003 + 53.5) \cdot 53.5})) \\
 &= 1.892
 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned}
 &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\
 &= 53.5 \cdot 164 \cdot \max((1.89/5)^{0.85}, 1) \\
 &= 87.633 \text{ cm}^2
 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned}
 &= t_n \cdot LH \\
 &= 88.5 \cdot 180 \\
 &= 159.264 \text{ cm}^2
 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned}
 &= 0.5 \cdot \text{Leg}^2 \\
 &= 0.5 \cdot 10^2 \\
 &= 0.500 \text{ cm}^2
 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$



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Nozzle Calcs.: N3- Shell inlet Nozl: 11 9:40am Aug 12, 2026

$$= 87.6 + 0.95(159 + 0) + 0.5 + 0 + 0$$

$$= 239.525 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 88.5 / \ln[1 + 88.5/181]$$

$$= 222.018 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 53.5 / \ln[1 + 53.5/502]$$

$$= 527.798 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 162 \cdot 1$$

$$= 242.8 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 16000 \cdot 222 \cdot 180$$

$$= 65189.7 \text{ Kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (L_R + t_n)$$

$$= 16000 \cdot 528 (164 + 88.5)$$

$$= 217270.5 \text{ Kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 16000 \cdot 528 \cdot 181$$

$$= 155611.9 \text{ Kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (217271 + 155612 + 65190) / 16000$$

$$= 2684.9 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$$

$$= 243 / (2 \cdot 2685 / 240 - 528 / 53.5)$$

$$= 19339.2 \text{ KPa.}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S / [t / R_{xs}]$$

$$= 162 [53.5 / 528]$$

$$= 16405.4 \text{ KPa.}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(19339, 16405)$$

$$= 16405.428 \text{ KPa.}$$

Average Primary Membrane Stress [σAvg]:

$$= (f_N + f_S + f_Y) / A_T$$

$$= (65190 + 217271 + 155612) / 240$$

$$= 179.359 \text{ N./mm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 16000 \cdot 528 / 53.5$$

$$= 157.9 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$



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Nozzle Calcs.: N3- Shell inlet Nozl: 11 9:40am Aug 12,2026

$$= \max(2 \cdot 179 - 158, 158)$$

$$= 200.9 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	242.78	N./mm ²
Local Primary Membrane Stress	PL	200.86	N./mm ²
Maximum Allowable Working Pressure	Pmax	16405.43	KPa.

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (181 + 88.5) / 181$$

$$= 1.490 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [L_T]:

$$= \pi/2 \cdot (R_n + t_n)$$

$$= \pi/2 \cdot (181 + 88.5)$$

$$= 422.858 \text{ mm}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$t_c, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 88.500, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$$

$$= \min(155612 \cdot 1.49, 1.5 \cdot 154(159+0), \pi/4 \cdot 16000 \cdot 181^2 \cdot 1.49^2)$$

$$= 92865.953 \text{ Kg}$$

Weld Stress Value [T]:

$$= \text{fwelds} / (L_T(0.49 \cdot L41T + 0.6 \cdot t_{w1} + 0.49 \cdot L43T))$$

$$= 92866 / (423(0.49 \cdot 7.07 + 0.6 \cdot 53.5 + 0.49 \cdot 0))$$

$$= 60.558 < \text{or } = \text{to } 161.855 \text{ Weld Stress Passed}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	6 °C
User entered Minimum Design Temperature	:	-33 °C
The Curve for this Material is	:	B
From Figure 3.8M- with PWHT, the MDMT is	:	11 °C
Temperature reduction per Graph 3.13	:	5 °C
The computed stress ratio is	:	0.898

The computed thickness ratio [sr]:

$$= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$$

$$= (48 \cdot 1) / (55 - 1.5 - 0)$$

$$= 0.898$$

Warning: Impact Testing Is Mandatory!

Gov. MDMT of the nozzle to shell joint welded assembly : 6 °C

The Drop for this Nozzle is : 78.6553 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 383.6553 mm

Input Echo, WRC297 Item 1, Description: N3- Shell inle:

Diameter Basis for Cylindrical Shell	ID
Shell Internal Corrosion Allowance	1.5000 mm
Shell External Corrosion Allowance	0.0000 mm



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Nozzle Calcs.: N3- Shell inlet Nozl: 11 9:40am Aug 12,2026

Shell Diameter		1000.000	mm
Shell Thickness		55.0000	mm
Shell Stress Concentration Factor		1.000	
Vessel Material		SA-516 70	
Vessel Cold S.I. Allowable	Smc	175.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	161.86	N./mm ²
Diameter Basis for Nozzle		ID	
Nozzle Corrosion Allowance		1.5000	mm
Nozzle Diameter		358.400	mm
Nozzle Thickness		90.0000	mm
Nozzle Stress Concentration Factor		1.000	
Nozzle Material		SA-350 LF2	
Nozzle Cold S.I. Allowable	SNmc	165.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	153.86	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	16000.00	KPa.
Radial Load	P	-5098.92	Kgf
Circumferential Shear	Vc	4079.14	Kgf
Longitudinal Shear	Vl	4079.14	Kgf
Circumferential Moment	Mc	4893.08	Kg-m.
Longitudinal Moment	Ml	4893.08	Kg-m.
Torsional Moment	Mt	5708.59	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tn (5.982) is < 10
check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 538.4 + 2 \cdot 1.65 \cdot \sqrt{528.25(55.0 - 1.5)}$$

$$= 1093.167 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1056.500	mm
Nozzle Outside Diameter	(d) =	538.400	mm
Vessel Thickness used	(T) =	53.500	mm
Nozzle Thickness used	(t) =	88.500	mm
T / t	=	0.605	
d / t	=	10.000	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.265$$

Nr/P	=	0.070
Mr/P	=	0.127
M0/P	=	0.039
N0/P	=	0.096
MrD/Mc	=	0.271
NrDL/Mc	=	0.110
M0d/Mc	=	0.086
N0DL/Mc	=	0.071
MrD/Ml	=	0.108



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Nozzle Calcs.: N3- Shell inlet

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NrDL/M1 = 0.062

MØD/M1 = 0.034

NØDL/M1 = 0.118

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-3	-3	3	3
Outplane Bending (ML)	-6	6	6	-6
Normal Pressure Stress	142	158	142	158
Outplane Stress Summary	138	158	158	153

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	13	-13	13	-13
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-20	20	20	-20
Inplane Pressure Stress	71	71	71	71
Inplane Stress Summary	63	77	107	40

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	3	3	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	138	158	158	153

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	-2	-2	2	2
Outplane Bending (Mc)	-16	16	16	-16
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	71	71	71	71



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Nozzle Calcs.: N3- Shell inlet Nozl: 11 9:40am Aug 12,2026

Outplane Stress Summary	58	82	95	54
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	13	-13	13	-13
Inplane Membrane (Mc)	-3	-3	3	3
Inplane Bending (Mc)	-50	50	50	-50
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	142	158	142	158

Inplane Stress Summary	102	193	210	99
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	2	2	2	2

Shear Stress Summary	1	1	3	3
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	102	193	211	99
------------------------	-----	-----	-----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-3	-3	3	3
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	31	50	31	50

Hoop Stress Summary	29	48	36	56
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	3	-3	3	-3
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-3	-3	3	3
Axial Bending (ML)	-5	5	5	-5
Axial Pressure Stress	15	15	15	15

Axial Stress Summary	10	13	28	11
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Nozzle Calcs.: N3- Shell inlet

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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1
Shear Stress Summary	1	1	0	0

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	29	48	36	56

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-2	-2	2	2
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	31	50	31	50
Hoop Stress Summary	30	50	35	54

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	3	-3	3	-3
Axial Membrane (Mc)	-3	-3	3	3
Axial Bending (Mc)	-14	14	14	-14
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	15	15	15	15
Axial Stress Summary	0	23	38	1

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1
Shear Stress Summary	0	0	1	1

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	30	50	39	54

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):



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Nozzle Calcs.: N3- Shell inlet

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Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	142	158	142	158	142	158	142	158
Circ. Pl (SUS)	-2	-2	4	4	-2	-2	4	4
Circ. Q (SUS)	-2	2	10	-10	-37	37	63	-63
Long. Pm (SUS)	71	71	71	71	71	71	71	71
Long. Pl (SUS)	0	0	2	2	-1	-1	3	3
Long. Q (SUS)	-7	7	33	-33	-12	12	20	-20
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	142.0	158.0	142.0	158.0	142.0	158.0	142.0	158.0
Pm+Pl (SUS)	140.0	156.0	146.0	162.0	140.0	156.0	146.0	162.0
Pm+Pl+Q (Total)	138.1	158.0	156.1	152.0	103.1	193.0	209.0	99.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	158.00	161.86	Passed
Pm+Pl (SUS)	162.00	242.78	Passed
Pm+Pl+Q (TOTAL)	209.03	485.57	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	31	50	31	50	31	50	31	50
Circ. Pl (SUS)	-2	-2	4	4	-1	-1	3	3
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	15	15	15	15	15	15	15	15
Long. Pl (SUS)	-3	-3	3	3	-3	-3	3	3
Long. Q (SUS)	-2	2	8	-8	-11	11	17	-17
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	31.0	50.0	31.0	50.0	31.0	50.0	31.0	50.0
Pm+Pl (SUS)	29.0	48.0	35.0	54.0	30.0	49.0	34.0	53.0
Pm+Pl+Q (Total)	29.1	48.0	35.1	54.0	30.0	49.0	35.6	53.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
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Nozzle Calcs.: N3- Shell inlet

Nozl: 11 9:40am Aug 12,2026

Pm (SUS)	50.00	153.86	Passed
Pm+Pl (SUS)	54.00	230.78	Passed
Pm+Pl+Q (TOTAL)	54.02	461.57	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Nozzle Calcs.: N2- Channel out Nozl: 12 9:40am Aug 12,2026

Input, Nozzle Desc: N2- Channel outlet

From: 50

Pressure for Reinforcement Calculations	P	16005.881	KPa.
Temperature for Internal Pressure	Temp	100	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	159.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	175.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1000.00	mm
Shell Finished (Minimum) Thickness	t	56.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		8385.00	mm
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	151.00	N./mm ²
Allowable Stress At Ambient	Sna	165.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		15.9528	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	26.0000	mm
Hub Height of Integral Nozzle	h	190.0000	mm
Height of Beveled Transition	L'	74.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		100.0000	mm
Corrosion Allowance	can	1.5000	mm
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	30.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		30.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N2- Channel outlet

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	15.953	in.
Actual Thickness Used in Calculation	1.024	in.



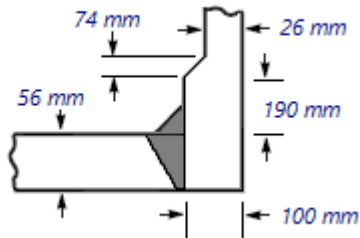
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Nozzle Calcs.: N2- Channel out

Nozl: 12 9:40am Aug 12,2026



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 408 \cdot (\exp(16006/(151 \cdot 1)) - 1) + 1.5 + 0$$

$$= 22.8 + 1.5 + 0 = 24.3 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 151 \cdot 1 \cdot \ln((2 \cdot 24.5 + 408) / 408) - 0$$

$$= 17117 - 0 = 17117 \text{ KPa.}$$

Nozzle/Tubesheet Proximity Warning:

This nozzle opening seems to be too close to the tubesheet (see 4.18.3(h) and Figure 4.18.19).

Nozzle Dimension d, 397.400 mm, is < 1.8 * sqrt(D * t): 420.844 mm.

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm, while the distance from the center of the nozzle to the edge of the diameter limit is: 467.92 mm

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 9.840 mm

Min. required shell thickness, internal p + c : 54.663 mm

Nozzle neck minimum required thickness : 9.840 mm

Minimum nozzle neck thickness 26.010 mm, is greater than required thickness 9.840 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 73.6,	Allowable	: 151.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 321.4 N./mm ²	Passed
Occasional	: 62.9,	Allowable	: 200.8 N./mm ²	Passed
Shear	: 8.3,	Allowable	: 105.7 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[151/159, 1]$$

$$= 0.950$$



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Nozzle Calcs.: N2- Channel out Nozl: 12 9:40am Aug 12, 2026

Thickness of Nozzle at Shell [tn]:

= hub thickness - corrosion allowance
= 100 - 1.5
= 98.500 mm

Thickness of Nozzle at Top [tn2]:

= thickness - corrosion allowance
= 26 - 1.5
= 24.500 mm

Shell Diameter to Thickness ratio [D/t]:

= Di/t
= 1003/54.5
= 18.404 must be less than 400.

Effective Pressure Radius [Reff]:

= Di/2 + corrosion allowance
= 1000/2 + 1.5
= 501.500 mm

Effective Length of Vessel Wall [LR]:

= $\min(\sqrt{Reff \cdot t}, 2 \cdot Rn)$
= $\min(\sqrt{502 \cdot 54.5}, 2 \cdot 204)$
= 165.323 mm

Thickness Limit Candidate [LH1]:

= $\min[1.5t, te] + \sqrt{Rn \cdot tn}$
= $\min[1.5 \cdot 54.5, 0] + \sqrt{204 \cdot 98.5}$
= 141.788 mm

Thickness Limit Candidate [LH2]:

= Lpr1
= 250
= 250.000 mm

Thickness Limit Candidate [LH3]:

= $8(t + te)$
= $8(54.5 + 0)$
= 436.000 mm

Effective Nozzle Wall Length Outside the Vessel [LH]:

= $\min[LH1, LH2, LH3] + t$ for set-in nozzles
= $\min[142, 250, 436] + 54.5$
= 196.288 mm

Effective Vessel Thickness [teff]:

= $t + ((A5 \cdot frp) / LR)$
= $54.5 + ((0 \cdot 1) / 165)$
= 54.500 mm

Parameter [λ]:

= $\min(12, (2 \cdot Rn + tn) / (\sqrt{(Di + teff) \cdot teff}))$
= $\min(12, (2 \cdot 204 + 98.5) / (\sqrt{(1003 + 54.5) \cdot 54.5}))$
= 2.111

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

= $t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$
= $54.5 \cdot 165 \cdot \max((2.11/5)^{0.85}, 1)$
= 90.101 cm²

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

= $tn \cdot LH$
= $98.5 \cdot 196$



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Nozzle Calcs.: N2- Channel out Nozl: 12 9:40am Aug 12,2026

$$= 193.344 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$

$$= 0.5 \cdot 30^2$$

$$= 4.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$

$$= 90.1 + 0.95(193 + 0) + 4.5 + 0 + 0$$

$$= 278.217 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 98.5 / \ln[1 + 98.5/204]$$

$$= 250.126 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 54.5 / \ln[1 + 54.5/502]$$

$$= 528.282 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 159 \cdot 1$$

$$= 238.5 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 16006 \cdot 250 \cdot 196$$

$$= 80136.2 \text{ Kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (L_R + t_n)$$

$$= 16006 \cdot 528 (165 + 98.5)$$

$$= 227486.0 \text{ Kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 16006 \cdot 528 \cdot 204$$

$$= 175988.7 \text{ Kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (227486 + 175989 + 80136) / 16006$$

$$= 2962.9 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / AT - R_{xs} / t_{eff})$$

$$= 238 / (2 \cdot 2963 / 278 - 528 / 54.5)$$

$$= 20548.1 \text{ KPa.}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S / t / R_{xs}$$

$$= 159 [54.5 / 528]$$

$$= 16402.1 \text{ KPa.}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(20548, 16402)$$

$$= 16402.068 \text{ KPa.}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / AT$$

$$= (80136 + 227486 + 175989) / 278$$



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 Nozzle Calcs.: N2- Channel out Nozl: 12 9:40am Aug 12,2026

$$= 170.467 \text{ N./mm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot Rxs / t_{eff}$$

$$= 16006 \cdot 528 / 54.5$$

$$= 155.2 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 170 - 155, 155)$$

$$= 185.8 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	238.50	N./mm ²
Local Primary Membrane Stress	PL	185.78	N./mm ²
Maximum Allowable Working Pressure	Pmax	16402.07	KPa.

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (204 + 98.5) / 204$$

$$= 1.483 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lr]:

$$= \pi / 2 \cdot (R_n + t_n)$$

$$= \pi / 2 \cdot (204 + 98.5)$$

$$= 475.323 \text{ mm}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 21.213, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$t_c, L41T = 21.213, \text{ must be } \geq \min(0.7 \cdot 98.500, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(f_y \cdot ky, 1.5 \cdot S_n(A_2 + A_3), \pi / 4 \cdot P \cdot R_n^2 \cdot ky^2)$$

$$= \min(175989 \cdot 1.48, 1.5 \cdot 151(193+0), \pi / 4 \cdot 16006 \cdot 204^2 \cdot 1.48^2)$$

$$= 117382.625 \text{ Kgf}$$

Weld Stress Value [T]:

$$= fwelds / (L_r(0.49 \cdot L41T + 0.6 \cdot t_{w1} + 0.49 \cdot L43T))$$

$$= 117383 / (475(0.49 \cdot 21.2 + 0.6 \cdot 30 + 0.49 \cdot 0))$$

$$= 85.293 < \text{ or } = \text{ to } 158.998 \text{ Weld Stress Passed}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	5 °C
User entered Minimum Design Temperature	:	-33 °C
The Curve for this Material is	:	B
From Figure 3.8M- with PWHT, the MDMT is	:	11 °C
Temperature reduction per Graph 3.13	:	6 °C
The computed stress ratio is	:	0.881

The computed thickness ratio [sr]:

$$= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$$

$$= (48 \cdot 1) / (56 - 1.5 - 0)$$

$$= 0.881$$

Warning: Impact Testing Is Mandatory!

Gov. MDMT of the nozzle to shell joint welded assembly : 5 °C



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Nozzle Calcs.: N2- Channel out Nozl: 12 9:40am Aug 12,2026

The Drop for this Nozzle is : 101.9632 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 407.9633 mm

Input Echo, WRC297 Item 1, Description: N2- Channel ou:

Diameter Basis for Cylindrical Shell	ID		
Shell Internal Corrosion Allowance	1.5000	mm	
Shell External Corrosion Allowance	0.0000	mm	
Shell Diameter	1000.000	mm	
Shell Thickness	56.0000	mm	
Shell Stress Concentration Factor	1.000		
Vessel Material	SA-516 70		
Vessel Cold S.I. Allowable	Smc	175.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	159.00	N./mm ²
Diameter Basis for Nozzle	ID		
Nozzle Corrosion Allowance	1.5000	mm	
Nozzle Diameter	405.200	mm	
Nozzle Thickness	100.0000	mm	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-350 LF2		
Nozzle Cold S.I. Allowable	SNmc	165.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	151.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	16005.88	KPa.
Radial Load	P	-3441.77	Kgf
Circumferential Shear	Vc	2294.51	Kgf
Longitudinal Shear	Vl	2294.51	Kgf
Circumferential Moment	Mc	2889.98	Kg-m.
Longitudinal Moment	Ml	2889.98	Kg-m.
Torsional Moment	Mt	4128.54	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, D_v/T_v (19.857) is < 20
check the Limitations of WRC bulletin 297.

Warning - The ratio, D_n/D_v (.544) is > 0.5
check the Limitations of WRC bulletin 297.

Warning - The ratio, D_n/T_n (6.052) is < 10
check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 605.2 + 2 \cdot 1.65 \cdot \sqrt{528.75(56.0 - 1.5)}$$

$$= 1165.393 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1057.500 mm
Nozzle Outside Diameter	(d) =	605.200 mm
Vessel Thickness used	(T) =	54.500 mm
Nozzle Thickness used	(t) =	98.500 mm
T / t	=	0.553



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Nozzle Calcs.: N2- Channel out Nozl: 12 9:40am Aug 12,2026

$$d / t = 10.000$$

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.521$$

Nr/P	=	0.066
Mr/P	=	0.119
M0/P	=	0.037
N0/P	=	0.091
MrD/Mc	=	0.270
NrDL/Mc	=	0.106
M0d/Mc	=	0.084
N0DL/Mc	=	0.077
MrD/M1	=	0.103
NrDL/M1	=	0.056
M0D/M1	=	0.031
N0DL/M1	=	0.110

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	2	-2	2	-2
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-1	-1	1	1
Outplane Bending (ML)	-2	2	2	-2
Normal Pressure Stress	139	155	139	155
Outplane Stress Summary	138	155	147	153

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	8	-8	8	-8
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	-9	9	9	-9
Inplane Pressure Stress	69	69	69	69
Inplane Stress Summary	68	71	89	53

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	1	1	0	0

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	138	155	147	153



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Nozzle Calcs.: N2- Channel out

Nozl: 12 9:40am Aug 12,2026

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	2	-2	2	-2
Outplane Membrane (Mc)	-1	-1	1	1
Outplane Bending (Mc)	-7	7	7	-7
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	69	69	69	69
Outplane Stress Summary	64	75	82	61

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	8	-8	8	-8
Inplane Membrane (Mc)	-1	-1	1	1
Inplane Bending (Mc)	-25	25	25	-25
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	139	155	139	155
Inplane Stress Summary	121	172	175	124

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1
Shear Stress Summary	0	0	1	1

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	121	172	175	124

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-1	-1	1	1
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	32	51	32	51
Hoop Stress Summary	31	50	34	54

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
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Nozzle Calcs.: N2- Channel out Nozl: 12 9:40am Aug 12,2026

(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	1	-1	1	-1
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-1	-1	1	1
Axial Bending (ML)	-2	2	2	-2
Axial Pressure Stress	16	16	16	16
Axial Stress Summary	14	14	21	13

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	31	50	34	54

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-1	-1	1	1
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	32	51	32	51
Hoop Stress Summary	31	51	34	53

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	1	-1	1	-1
Axial Membrane (Mc)	-1	-1	1	1
Axial Bending (Mc)	-6	6	6	-6
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	16	16	16	16
Axial Stress Summary	10	19	25	9

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0



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Nozzle Calcs.: N2- Channel out Nozl: 12 9:40am Aug 12,2026

Shear due to Torsion	0	0	0	0
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Shear Stress Summary	0	0	0	0
----------------------	---	---	---	---

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	31	51	34	53

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	139	155	139	155	139	155	139	155
Circ. Pl (SUS)	0	0	2	2	-1	-1	1	1
Circ. Q (SUS)	0	0	4	-4	-17	17	33	-33
Long. Pm (SUS)	69	69	69	69	69	69	69	69
Long. Pl (SUS)	0	0	0	0	0	0	2	2
Long. Q (SUS)	-1	1	17	-17	-5	5	9	-9
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	139.0	155.0	139.0	155.0	139.0	155.0	139.0	155.0
Pm+Pl (SUS)	139.0	155.0	141.0	157.0	138.0	154.0	140.0	156.0
Pm+Pl+Q (Total)	139.0	155.0	145.0	153.0	121.0	171.0	173.0	123.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	155.00	159.00	Passed
Pm+Pl (SUS)	157.00	238.50	Passed
Pm+Pl+Q (TOTAL)	173.01	477.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	32	51	32	51	32	51	32	51
Circ. Pl (SUS)	0	0	2	2	0	0	2	2
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	16	16	16	16	16	16	16	16
Long. Pl (SUS)	-1	-1	1	1	-1	-1	1	1
Long. Q (SUS)	-1	1	3	-3	-5	5	7	-7
Shear Pm (SUS)	0	0	0	0	0	0	0	0



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Nozzle Calcs.: N2- Channel out Nozl: 12 9:40am Aug 12,2026

Shear P1 (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	32.0	51.0	32.0	51.0	32.0	51.0	32.0	51.0
Pm+P1 (SUS)	32.0	51.0	34.0	53.0	32.0	51.0	34.0	53.0
Pm+P1+Q (Total)	32.0	51.0	34.0	53.0	32.0	51.0	34.0	53.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	51.00	151.00	Passed
Pm+P1 (SUS)	53.00	226.50	Passed
Pm+P1+Q (TOTAL)	53.00	453.00	Passed

*Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.*

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ASME TS Calc: DESIGN CASE

Case: 1 9:40a Aug 12, 2026

Input Echo, Tubesheet Number 1, Description: Fixed tubesheet

Shell Data:

Main Shell Description: Shell

Shell Maximum Design Pressure	Psd,max	16000.00 KPa.
Shell Maximum Operating Pressure	Psox,max	16000.00 KPa.
Shell Minimum Operating Pressure	Psox,min	0.00 KPa.
Shell Thickness	ts	55.0000 mm
Shell Internal Corrosion Allowance	cas	1.5000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	1000.000 mm
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	80.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	161.86 N./mm ²
Shell Allowable Stress at Ambient		175.00 N./mm ²

Channel Data:

Channel Description: Chnl. rear

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	16000.00 KPa.
Channel Maximum Operating Pressure	Ptox,max	16000.00 KPa.
Channel Minimum Operating Pressure	Ptox,min	0.00 KPa.
Channel Thickness	tc	56.0000 mm
Channel Internal Corrosion Allowance	cac	1.5000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	1000.000 mm
Channel Design Temperature	TEMPC	100.00 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	159.00 N./mm ²
Channel Allowable Stress at Ambient		175.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	1015
Tube Wall Thickness	et	2.5000 mm
Tube Outside Diameter	D	19.0000 mm
Total Straight Tube Length	Lt	6000.00 mm
Straight Tube Length (bet. inner tubsht faces) L		5610.00 mm
Design Temperature of the Tubes		100.00 °C
Tube Material		SA-334 1
Tube Material UNS Number		K03008
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		126.00 N./mm ²
Tube Allowable Stress At Ambient		138.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	189.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	25.0000 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.0000 mm
Groove Weld Leg	ag	2.0000 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		4.21.2
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	454.384 mm



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ASME TS Calc: DESIGN CASE

Case: 1 9:40a Aug 12, 2026

Largest Center-to-Center Tube Distance	U1	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	5190.7 cm ²
Perimeter of Tube Layout	Cp	2786.6 mm

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf A

Tubesheet Design Metal Temperature	T	100.00 °C
Tubesheet Material		SA-350 LF2

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Tubesheet Material UNS Number		K03011
Tubesheet Allowable Stress at Temperature	S	151.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	165.00 N./mm ²
Thickness of Tubesheet	h	195.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	1.5000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	1.5000 mm
Tubesheet Outside Diameter	A	1110.000 mm

Area of the Untubed Lanes	AL	0.0 cm ²
---------------------------	----	---------------------

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	0.000 mm
Tube End condition corresponding to Span (l)	k	0.00
Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)		Yes
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Mean Shell Metal Temp. along Shell len.	Tsm	0.00 °C
Mean Tube Metal Temp. along Tube length	Ttm	0.00 °C
Junction Stress Reduction option	Perform Plastic Calculation	
Tubesheet Integral with		Both
Tubesheet Extended as Flange		No
Is Exchanger in Creep range (skip EP, Use 3S for Sps)		No



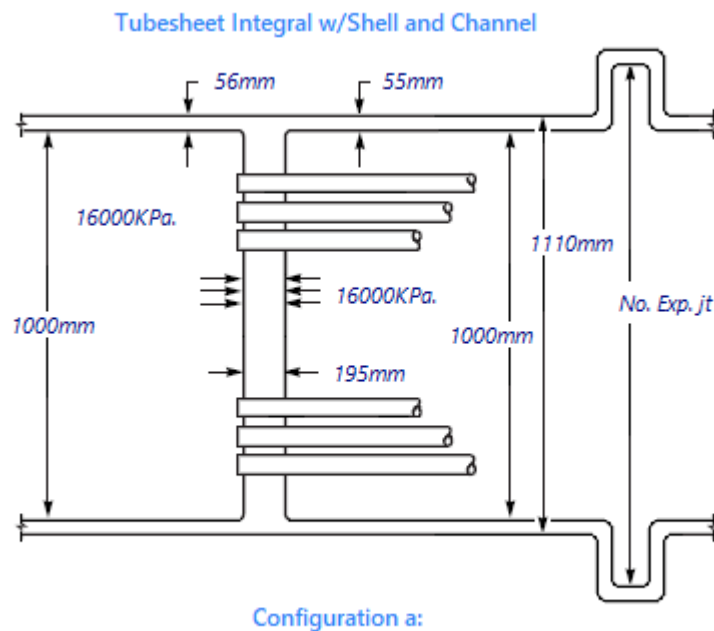
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ASME TS Calc: DESIGN CASE

Case: 1 9:40a Aug 12, 2026



Messages regarding the Tubesheet & Exchanger Input Data:

Warning - Tube end condition k should be bet 0 - 1 (.00).

ASME Division 2 Tubesheet Results per Part 4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000112153 /°C
Elastic Mod. at Design Temperature	80.0 °C	0.19933E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.20409E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Design Temperature	100.0 °C	0.19815E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
Tubes - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000112153 /°C
Elastic Mod. at Design Temperature	100.0 °C	0.19815E+09 KPa.
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.20409E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	100.0 °C	0.19815E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
TubeSheet - TM-1 Carbon Steels with C<= 0.3%



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ASME TS Calc: DESIGN CASE

Case: 1 9:40a Aug 12, 2026

Elastic Mod. at Design Temperature	100.0 °C	0.19815E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.
Please insure these properties are consistent with the
type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c$ Equation 4.3.1
 $= 0.5 \cdot 14 \cdot (\exp(16000/(126 \cdot 1)) - 1) + 0$
 $= 0.95 + 0$
 $= 0.95 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

Determine Parameter [Mx]:

$= L / \sqrt{R_o \cdot t}$
 $= 5610 / \sqrt{9.5 \cdot 2.5}$
 $= 1151.148$

Compute the Factor [Ch]:

Since $Mx > 2 \cdot (Do/t) \cdot 0.94$, $Ch = 0.55 / (Do/t)$
 $= 0.55 \cdot (19/2.5)$
 $= 0.0724$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$= 1.6 \cdot Ch \cdot E_y (t / Do)$
 $= 1.6 \cdot 0.072 \cdot 198151376 (2.5/19)$
 $= 3019.102 \text{ N./mm}^2$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$= Fhe / F_y$
 $= 3019.1/189.0$
 $= 15.974$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$= \sigma_{ys} / \sigma_{uts}$
 $= 189/379$
 $= 0.498$

Ratio [AE]:

$= Fhe / E$
 $= 3019/198151376$
 $= 0.015235$

Factor [m1]:

$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$
 $= (\ln(0.5) + (0.20000E-04 - 0.002)) / \ln(\ln(1 + 0.20000E-04) / \ln(1 + 0.002))$
 $= 0.152$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$
 $= 189 (1 + 0.0020) / \ln(1 + 0.0020)^{0.15168217}$
 $= 486.157 \text{ N./mm}^2$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$= m2f / (m2f2 - R)$
 $= 0.6 (1 - 0.5)$
 $= 0.301$



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ASME TS Calc: DESIGN CASE

Case: 1 9:40a Aug 12, 2026

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m_2)) / (m_2 m_2)$$

$$= (379 \cdot \exp(0.3)) / (0.3 \cdot 0.3)$$

$$= 735.456 \text{ N./mm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.51.5 - 0.5 \cdot 0.52.5 - 0.53.5$$

$$= 0.353$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	189	-1.982	0.350E-07	0.315E-07	0.970E-07	-0.813E-07	0.115E+08	0.131E-02
2	94	-4.808	0.726E-09	0.726E-09	0.194E-07	-0.194E-07	0.154E+09	0.146E-01
3	142	-3.395	0.701E-08	0.697E-08	0.497E-07	-0.493E-07	0.515E+08	0.125E-01
4	165	-2.688	0.166E-07	0.162E-07	0.711E-07	-0.685E-07	0.247E+08	0.852E-02
5	177	-2.335	0.244E-07	0.232E-07	0.835E-07	-0.770E-07	0.169E+08	0.472E-02
6	183	-2.158	0.293E-07	0.273E-07	0.901E-07	-0.800E-07	0.139E+08	0.206E-02
7	186	-2.070	0.321E-07	0.294E-07	0.935E-07	-0.809E-07	0.126E+08	0.472E-03
8	188	-2.026	0.335E-07	0.305E-07	0.952E-07	-0.812E-07	0.120E+08	0.394E-03
9	187	-2.048	0.328E-07	0.299E-07	0.944E-07	-0.810E-07	0.123E+08	0.451E-04
10	187	-2.037	0.332E-07	0.302E-07	0.948E-07	-0.811E-07	0.122E+08	0.173E-03
11	187	-2.042	0.330E-07	0.301E-07	0.946E-07	-0.811E-07	0.123E+08	0.635E-04
12	187	-2.045	0.329E-07	0.300E-07	0.945E-07	-0.811E-07	0.123E+08	0.913E-05
13	187	-2.047	0.328E-07	0.300E-07	0.944E-07	-0.811E-07	0.123E+08	0.180E-04
14	187	-2.046	0.329E-07	0.300E-07	0.944E-07	-0.811E-07	0.123E+08	0.444E-05
15	187	-2.046	0.329E-07	0.300E-07	0.944E-07	-0.811E-07	0.123E+08	0.234E-05
16	187	-2.046	0.329E-07	0.300E-07	0.944E-07	-0.811E-07	0.123E+08	0.105E-05

Factor of Safety [FS]:
 Since $Fic > 0.55 F_y$ and $Fic < F_y$

$$= 2.407 - 0.741 \cdot Fic / F_y$$

$$= 2.407 - 0.741 \cdot 187/189$$

$$= 1.672$$

Allowable Circumferential Compressive Stress [Fha]:

$$= Fic/FS$$

$$= 187/1.67$$

$$= 112.120 \text{ N./mm}^2$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot Fha(t / Do)$$

$$= 2 \cdot 112114 (2.5/19)$$

$$= 29503.564 \text{ KPa.}$$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	1.6244 mm
Allowable Internal Pressure at Corroded thickness	38475.49 KPa.
Required Internal Design Pressure	16000.00 KPa.
Allowable External Pressure at Corroded thickness	29503.56 KPa.
Required External Design Pressure	16000.00 KPa.
Required Thickness due to Shell Side pressure	1.6244 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME Fixed Tubesheet Calculations for Configuration a:

Results for Tubesheet Calculations Original Thickness:



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ASME TS Calc: DESIGN CASE

Case: 1 9:40a Aug 12, 2026

ASME Section VIII-2 Part 4.18.8.4 Step 1:

Tube expansion depth ratio $[\rho]$:

$$= l_{tx} / h \text{ (modified for corrosion if present)}$$

$$= 0/195$$

$$= 0.0000 \text{ (must be } 0 \leq \rho \leq 1 \text{)}$$

Effective tube hole diameter $[d^*]$:

$$= \max(dt - 2 \cdot tt(EtT/E) \cdot (StT/S) \cdot (\rho); dt - 2 \cdot tt)$$

$$= \max(19 - 2 \cdot 2.5 (198151376/198151376) \cdot$$

$$(126/151) \cdot (0); 19 - 2 \cdot 2.5)$$

$$= 19.0000 \text{ mm}$$

Equivalent outer tube limit circle diameter $[Do]$:

$$= 2 \cdot r_o + dt$$

$$= 2 \cdot 454 + 19$$

$$= 927.768 \text{ mm}$$

Determine the basic ligament efficiency for shear $[\mu]$:

$$= (p - dt) / p$$

$$= (25 - 19) / 25$$

$$= 0.240$$

Equivalent outer tube limit radius $[ao]$:

$$= Do / 2$$

$$= 928/2$$

$$= 463.8840 \text{ mm}$$

Effective tube pitch $[p^*]$:

$$= p / \sqrt{ 1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot Do \cdot p) / (\pi \cdot Do^2) }$$

$$= 25 / \sqrt{ 1 - 4 \cdot \min(0 \cdot 100 ; 4 \cdot 928 \cdot 25) / (\pi \cdot 928^2) }$$

$$= 25.0000 \text{ mm}$$

Effective ligament efficiency for bending $[\mu^*]$:

$$= (p^* - d^*) / p^*$$

$$= (25 - 19) / 25$$

$$= 0.2400$$

Ratio $[ps]$:

$$= as / ao$$

$$= 500/464$$

$$= 1.0779$$

Ratio $[pc]$:

$$= ac / ao$$

$$= 500/464$$

$$= 1.0779$$

Compute parameter $[xt]$:

$$= 1 - Nt((dt - 2 \cdot tt) / (2 \cdot ao))^2$$

$$= 1 - 1015 ((19 - 2 \cdot 2.5) / (2 \cdot 464))^2$$

$$= 0.7689$$

Determine parameter $[xs]$:

$$= 1 - Nt(dt / (2 \cdot ao))^2$$

$$= 1 - 1015 (19 / (2 \cdot 464))^2$$

$$= 0.5743$$

Determine the value $[h'g]$:

$$= \max((hg - CATC); 0) \text{ (For pressure only cases)}$$

$$= \max((0 - 0); 0)$$

$$= 0.000 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 2:



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ASME TS Calc: DESIGN CASE

Case: 1 9:40a Aug 12, 2026

Axial Shell Stiffness [Ks]:

$$= \pi \cdot ts (Ds + ts) Es / L$$

$$= \pi \cdot 55 (1000 + 55) 199328640 / 5610$$

$$= 660493.3125 \text{ Kg/mm}$$

Axial Tube Stiffness [Kt]:

$$= \pi \cdot tt (Dt - tt) Et / L$$

$$= \pi \cdot 2.5 (19 - 2.5) 198151376 / 5610$$

$$= 466.7717 \text{ Kg/mm}$$

Stiffness Factor [Ks,t]:

$$= Ks / (Nt \cdot Kt)$$

$$= 660493 / (1015 \cdot 467)$$

$$= 1.39411$$

Rigidity Ratio [J]:

$$= 1 / (1 + Ks/Kt)$$

$$= 1 / (1 + 660493/0)$$

$$= 1.0000 (= 1 \text{ if No Exp. Jt.})$$

Shell coefficient [β_s]:

$$= ((12(1 - \nu_s^2))^{1/4}) / ((Ds + ts)ts)$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((1000 + 55)55)$$

$$= 0.0075 \text{ 1/mm}$$

Shell coefficient [ks]:

$$= \beta_s \cdot Es \cdot ts^3 / (6(1 - \nu_s^2))$$

$$= 0.0075 \cdot 199328640 \cdot 55^3 / (6(1 - 0.3^2))$$

$$= 45836677120.0000 \text{ KPa} \cdot \text{mm}^2$$

Shell coefficient [λ_s]:

$$= 6(Ds + ts) \cdot ks / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(1000 + 55) \cdot 45.84\text{E}+09 / 195^3 \cdot (1 + 195 \cdot 0.0075 + (195 \cdot 0.0075)^2 / 2)$$

$$= 139082480.0000 \text{ KPa}$$

Shell coefficient [δ_s]:

$$= Ds^2 / (4 \cdot Es \cdot ts) \cdot (1 - Ds \cdot \nu_s / (2(Ds + ts)))$$

$$= 1000^2 / (4 \cdot 199328640 \cdot 55) \cdot (1 - 1000 \cdot 0.3 / (2(1000 + 55)))$$

$$= 0.19560\text{E}-01 \text{ mm/N./mm}^2$$

Channel coefficient [β_c]:

$$= ((12(1 - \nu_c^2))^{1/4}) / ((Dc + tc)tc)$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((1000 + 56)56)$$

$$= 0.0075 \text{ 1/mm}$$

Channel coefficient [kc]:

$$= \beta_c \cdot Ec \cdot tc^3 / (6(1 - \nu_c^2))$$

$$= 0.0075 \cdot 198151376 \cdot 56^3 / (6(1 - 0.3^2))$$

$$= 47642906624.0000 \text{ KPa} \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(Dc + tc / \cos(\theta)) \cdot kc / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (Cm\Delta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot Cq\Delta)$$

$$= 6(1000 + 56 / \cos(0)) \cdot 47.64\text{E}+09 / 195^3 \cdot (1 + 195 \cdot 0.0075 \cdot (1 + 1) / 2 + (195 \cdot 0.0075)^2 / 2 \cdot 1)$$

$$= 143306576.0 \text{ KPa}$$

Channel coefficient [δ_c]:

$$= Dc^2 / (4 \cdot Ec \cdot Tc) \cdot (1 - Dc \cdot \nu_c / (2(Dc + tc)))$$

$$= 1000^2 / (4 \cdot 198151376 \cdot 56) \cdot (1 - 1000 \cdot 0.3 / (2(1000 + 56)))$$

$$= 0.19328\text{E}-01 \text{ mm/N./mm}^2$$



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ASME TS Calc: DESIGN CASE

Case: 1 9:40a Aug 12, 2026

Channel coefficient [zetac]:

$$\text{zetac} = 0$$

ASME Section VIII-2 Part 4.18.8.4 Step 3:

$h/p = 7.800000$; $\mu^* = 0.240000$; $E^* = 40428684$ KPa.

$E^*/E = 0.204029$; $\nu^* = 0.407183$

Note: As h/p (7.800) is > 2 , data values for $h/p = 2$ were used.

Tube Bundle Stiffness Factor [Xa]:

$$\begin{aligned} &= ((24(1 - \nu^{*2})Nt \cdot Et \cdot tt(dt - tt)ao^2) / (E^* \cdot L \cdot H^3))^{1/4} \\ &= ((24(1 - 0.41^2)1015 \cdot 198151376 \cdot 2.5 \\ &\quad (19 - 2.5)^{464^2} / (40428684 \cdot 5610 \cdot 195^3))^{1/4} \\ &= 2.1471 \end{aligned}$$

Values from Table 4.18.3

$$\begin{aligned} Z_d &= 0.138660 ; Z_v = 0.147370 ; Z_m = 0.620842 \\ Z_a &= 0.961373E+00 ; Z_w = 0.147370 \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 4:

Diameter ratio [K]:

$$\begin{aligned} &= A / Do \\ &= 1110/928 \\ &= 1.1964 \end{aligned}$$

Compute coefficient [F]:

$$\begin{aligned} &= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K)) \\ &= (1 - 0.41) / 40428684 \cdot (139082480 + 143306576 + \\ &\quad 198151376 \cdot \ln(1.2)) \\ &= 4.6618 \end{aligned}$$

Compute parameter [ϕ]:

$$\begin{aligned} &= (1 + \nu^*) F \\ &= (1 + 0.41) 4.66 \\ &= 6.5600 \end{aligned}$$

Compute parameter [Q1]:

$$\begin{aligned} &= (\rho_s - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m) \\ &= (1.08 - 1 - 6.56 \cdot 0.15) / (1 + 6.56 \cdot 0.62) \\ &= -0.175230 \end{aligned}$$

Compute Parameter [Qz1]:

$$\begin{aligned} &= (Z_d + Q1 \cdot Z_w) / 2 \cdot Xa^4 \\ &= (0.14 + -0.18 \cdot 0.15) / 2 \cdot 2.15^4 \\ &= 1.1991 \end{aligned}$$

Compute parameter [Qz2]:

$$\begin{aligned} &= (Z_v + Q1 \cdot Z_m) / 2 \cdot Xa^4 \\ &= (0.15 + -0.18 \cdot 0.62) / 2 \cdot 2.15^4 \\ &= 0.4100 \end{aligned}$$

Compute parameter [U]:

$$\begin{aligned} &= (Z_w + (\rho_s - 1) Z_m) Xa^4 / (1 + \phi \cdot Z_m) \\ &= (0.15 + (1.08 - 1) 0.62) 2.15^4 / (1 + 6.56 \cdot 0.62) \\ &= 0.8200 \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 5:

Determine factor [γ_b]:

$$= 0 \text{ (config a)}$$

Compute Parameter [γ]:

$$= 0.000 \text{ mm (For Pressure only cases)}$$



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Calculate parameter [ws]:

$$= \rho_s \cdot k_s \cdot \beta_s \cdot \delta S (1 + h \cdot \beta_s) \\ = 1.08 \cdot 45.84E+09 \cdot 0.0075 \cdot 0.019560438 (1 + 195 \cdot 0.0075) \\ = 18026.0078 \text{ mm}^2$$

Calculate parameter [ω*s]:

$$= a_o^2 (\rho_s^2 - 1) \cdot (\rho_s - 1) / 4 - \omega_s \\ = 464^2 (1.08^2 - 1) \cdot (1.08 - 1) / 4 - 18026 \\ = -17348.4355 \text{ mm}^2$$

Parameter [ωc]:

$$= \rho_c \cdot k_c [\delta C (C_m \theta + \beta C + C_q \Delta \cdot h \cdot \beta C^2) - \\ \zeta c (C_m \theta + h \cdot C_q \theta + h \cdot \beta C / 2)] \\ = 1.08 \cdot 47.64E+09 [0.019328313 (1 + 0.0075 \cdot 1 \cdot 195) - \\ 0.000 (1/195 + 1 \cdot 0.0075/2)] \\ = 18236.1973 \text{ mm}^2$$

Parameter [ω*c]:

$$= a_o^2 [(\rho_c^2 + 1) \cdot (\rho_c - 1) / 4 - (\rho_s - 1) / 2] - \omega_c \\ = 464^2 [(1.08^2 + 1) \cdot (1.08 - 1) / 4 - (1.08 - 1) / 2] - 18236 \\ = -17558.6250 \text{ mm}^2$$

Pressure [P*S]:

$$= 0 \text{ For Pressure only cases or Configurations d,e,f,A,B,C,D}$$

Pressure [P*C]:

$$= 0 \text{ For Pressure only cases or Configurations b,c,d,B,C,D}$$

ASME Section VIII-2 Part 4.18.8.4 Step 6:

Pressure [P's]:

$$= P_s \{ x_s + 2(1 - x_s) v_t + [2/K_s t \cdot (D_s/D_o)^2] v_s - \\ [(\rho_s^2 - 1)/(J \cdot K_s t)] - [(1 - J)/(2J \cdot K_s t)] [(D_j^2 - (D_s)^2)/D_o^2] \} \\ = 16000 \{ 0.57 + 2(1 - 0.57) 0.3 + [2/1.39 \cdot \\ (1000/928)^2] 0.3 - [(1.08^2 - 1)/(1 \cdot 1.39)] - \\ [(1 - 1)/(2 \cdot 1 \cdot 1.39)] [(0^2 - (1000)^2)/928^2] \} \\ = 19419.0234 \text{ KPa.}$$

Pressure [P't]:

$$= [x_t + 2(1 - x_t) v_t + 1/(J \cdot K_s t)] P_t \\ = [0.77 + 2(1 - 0.77) 0.3 + 1/(1 \cdot 1.39)] 16000 \\ = 25997.6484 \text{ KPa.}$$

Pressure [P_y]:

$$= N_t \cdot K_t \cdot \gamma / (\pi \cdot a_o^2) \\ = 1015 \cdot 467 \cdot 0 / (3.14 \cdot 464^2) \\ = 0.0000 \text{ KPa.}$$

Pressure [P_w]:

$$= -\gamma_b \cdot U \cdot W^* / (2 \cdot \pi \cdot a_o^2) \\ = -0 \cdot 0.82 \cdot 0 / (2 \cdot \pi \cdot 464^2) \\ = -0.0000 \text{ KPa.}$$

Calculate the pressure [Prim]:

$$= -(U/a_o^2) \cdot (\omega^* s \cdot P_s - \omega^* c \cdot P_t) \\ = -(0.82/464^2) \cdot (-17348 \cdot 16000 - -17559 \cdot 16000) \\ = -12.8146 \text{ KPa.}$$

Calculate the pressure [P_w]:

$$= (U/a_o^2) \cdot (\omega_s \cdot P^* s - \omega_c \cdot P^* c) \\ = (0.82/464^2) \cdot (18026 \cdot 0 - 18236 \cdot 0) \\ = 0.0000 \text{ KPa.}$$

Determine the effective pressure [P_e]:



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$$\begin{aligned}
 &= J \cdot Kst / ((1 + J \cdot Kst(Qz1 + (ps - 1)Qz2)) \cdot \\
 &\quad (P's - P't + P\gamma + P\omega + Pw + Prim)) \\
 &= 0.1000E+01 \cdot 1.39 / (1 + 1 \cdot 1.39 (1.2 + (1.08 - \\
 &\quad 1) 0.41)) \cdot (19419 - 25998 + 0 + 0 + -0 + -12.8) \\
 &= -3383.1643 \text{ KPa.}
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 7:

Determine factor [Q2]:

$$\begin{aligned}
 &= [((\omega * s \cdot P_s - \omega * c \cdot P_t) - (\omega s \cdot P * s - \omega c \cdot P * c)) + \\
 &\quad W * \gamma_b / (2 \cdot \pi)] / (1 + \phi \cdot Z_m) \\
 &= [((-17348 \cdot 16000 - -17559 \cdot 16000) - \\
 &\quad (18026 \cdot 0 - 18236 \cdot 0)) \cdot 0.0001 + \\
 &\quad 0 \cdot 0 / (2 \cdot \pi)] / (1 + 6.56 \cdot 0.62) \\
 &= 67.6057 \text{ Kgf}
 \end{aligned}$$

Calculate factor [Q3]:

$$\begin{aligned}
 &= Q1 + 2 \cdot Q2 / (Pe \cdot a o^2) \\
 &= -0.18 + 2 \cdot 67.6 / (-3383 \cdot 464^2) \\
 &= -0.17705
 \end{aligned}$$

Fm Value from Table 4.18.3 = 0.098725

Tubesheet Bending Stress - original thickness [σ]:

$$\begin{aligned}
 &= (1.5 \cdot F_m / \mu *) \cdot (2 \cdot a o / (H - h'g))^2 \cdot Pe \\
 &= (1.5 \cdot 0.099 / 0.24) \cdot (2 \cdot 464 / (195 - 0))^2 \cdot -3383 \\
 &= -47.2568 \text{ N./mm}^2
 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ allowed]:

$$= 1.5 \cdot S = 1.5 \cdot 151 = 226 \text{ N./mm}^2$$

Tubesheet Bending Stress - final thickness [σ_f]:

$$\begin{aligned}
 &= (1.5 \cdot F_m / \mu *) \cdot (2 \cdot a o / (h - h'g))^2 \cdot Pe \\
 &= (1.5 \cdot 0.061 / 0.24) \cdot (2 \cdot 464 / (55.5 - 0))^2 \cdot -2141 \\
 &= -226.4956 \text{ N./mm}^2
 \end{aligned}$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$\begin{aligned}
 &= h + Cats + Catc \\
 &= 55.5 + 0 + 0 \\
 &= 55.5370 \text{ mm}
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 8:

Shear Stress check [Tlimit]:

$$\begin{aligned}
 &= (2 \cdot \mu \cdot h / a o) \cdot \min(0.8S; 0.533Sy) \\
 &= (2 \cdot 0.24 \cdot 195 / 464) \cdot 121 \\
 &= 24.3741 \text{ N./mm}^2
 \end{aligned}$$

Shear Stress is not required to be computed; [Pe] <= Taulimit

Tubesheet Shear Stress is probably low, use the following req. thk:

$$\begin{aligned}
 \text{Tubesheet thickness (Incl. Corr.)} &= 3.8100 \text{ mm} \\
 \text{Tubesheet Shear Stress} &= 93.5833 \text{ N./mm}^2
 \end{aligned}$$

Reqd Tubesheet Thickness for Given Loadings (including CA) [Hreqd] :

$$\begin{aligned}
 &= \max(HreqB; HreqS) \\
 &= \max(55.5 ; 3.81) \\
 &= 55.5370 \text{ mm}
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 9:

The F_{tmin} and F_{tmax} Coefficients from Table 4.18.4:

$$F_{tmin} = 0.71, F_{tmax} = 1.2$$



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First Extreme Tube Axial Stress from among all the tubes [σ_{t1}]:

$$= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmin}) / (X_t - X_s)$$

$$= ((16000 \cdot 0.57 - 16000 \cdot 0.77) - (-3383) \cdot 0.71) /$$

$$(0.77 - 0.57)$$

$$= -3.6108 \text{ N./mm}^2$$

Second Extreme value of Tube Axial Stress from among all the tubes [σ_{t2}]:

$$= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmax}) / (X_t - X_s)$$

$$= ((16000 \cdot 0.57 - 16000 \cdot 0.77) - (-3383) \cdot 1.2) /$$

$$(0.77 - 0.57)$$

$$= 4.8005 \text{ N./mm}^2$$

Maximum Tube Axial Stress [$\sigma_{t,max}$]:

$$= \max(| \sigma_{t,1} | ; | \sigma_{t,2} |) = 4.8 \text{ N./mm}^2$$

Allowable Tube Stress [$\sigma_{t,A}$]:

$$= S_{ot} = 125.9988 \text{ N./mm}^2$$

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [Fs]:

$$= \max((3.25 - 0.25(Z_d + Q_3 \cdot Z_w) X_a^4); 1.25)$$

$$= \max((3.25 - 0.25(0.14 + -0.18 \cdot 0.15) 2.15^4); 1.25)$$

$$= 2.0000 \text{ (Should be } \leq 2 \text{)}$$

Determine the factor [rt]:

$$= (\sqrt{ (dt^2 + (dt - 2 \cdot tt)^2) }) / 4$$

$$= (\sqrt{ (19^2 + (19 - 2 \cdot 2.5)^2) }) / 4$$

$$= 5.9002 \text{ mm}$$

Determine the factor [Ct]:

$$= \sqrt{ (2 \cdot \pi^2 \cdot E_t / S_{yt})}$$

$$= \sqrt{ (2 \cdot \pi^2 \cdot 198151376 / 189)}$$

$$= 143.8624$$

Determine the factor [Ft]:

$$= k \cdot L / r$$

$$= 0 \cdot 0 / 5.9$$

$$= 0.0000$$

Buckling Allowable Stress [Stb]:

$$= (S_{y,t} / F_s) \cdot (1 - F_t / (2 \cdot C_t))$$

$$= (189 / 2) \cdot (1 - 0 / (2 \cdot 144))$$

$$= 94.499 \text{ N./mm}^2 \text{ (Never greater than } S_{ot} \text{)}$$

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load [Wt]:

$$= \sigma_{t,max} \cdot \text{Tube Area}$$

$$= 4.8 \cdot 1.3$$

$$= 63.43 \text{ Kg}$$

Tube Metal Area [At]:

$$= \pi (d_o - t) t$$

$$= \pi (19 - 2.5) 2.5$$

$$= 1.296 \text{ cm}^2$$

Tube Weld Size Results per VIII-2 4.21.2:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 2.5 (19 - 2.5) 126$$

$$= 1664.990 \text{ Kg}$$

Groove Weld Strength [Fg]:



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$$\begin{aligned}
 &= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t) \\
 &= 0.85 \cdot \pi \cdot 2 (19 + 0.67 \cdot 2) 126 \\
 &= 1395.6853 \text{ Kgf}
 \end{aligned}$$

Weld Strength Factor [fw]:

$$\begin{aligned}
 &= S / S_w \text{ where } S_w = \min(S \text{ and } S_t) \\
 &= 126/126 \\
 &= 1.000
 \end{aligned}$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$\begin{aligned}
 &= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t) \\
 &= 0.55 \cdot \pi \cdot 2 (19 + 0.67 \cdot 2) 126 \\
 &= 903.0905 \text{ Kgf}
 \end{aligned}$$

Minimum Length of Weld [ar]:

$$\begin{aligned}
 &= 2 \left(\sqrt{((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d)} - 0.75 \cdot d_o \right) \\
 &= 2 \left(\sqrt{((0.75 \cdot 19)^2 + 1.07 \cdot 2.5 \cdot (19 - 2.5) 1 \cdot 1)} - 0.75 \cdot 19 \right) \\
 &= 2.945 \text{ mm}
 \end{aligned}$$

Minimum Required Fillet Weld Leg [afr]: 1.5000 mm

Minimum Required Groove Weld Leg [agr]: 1.5000 mm

Tube-Tubesheet Jt allowable, 1665 is $\geq$ tube strength 1665 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

ASME Section VIII-2 Part 4.18.8.4 Step 10:

Shell Axial Membrane Allowable Stress:

$$= S_s \cdot E_{sw} = 162 \cdot 1 = 162 \text{ N./mm}^2$$

Axial Membrane Stress in Shell [σ_s, m]:

$$\begin{aligned}
 &= a_o^2 / ((D_s + t_s) t_s) \cdot [P_e + (p_s^2 - 1)(P_s - P_t)] + a_s^2 \cdot P_t / ((D_s + t_s) t_s) \\
 &= 464^2 / ((1000 + 55) 55) \cdot [-3383 + (1.08^2 - 1) \\
 &\quad (16000 - 16000)] + 500^2 \cdot 16000 / ((1000 + 55) 55) \\
 &= 56.3925 \text{ N./mm}^2
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 11:

Note:

For a given Shell thickness of 55 mm, the minimum Shell length adjacent to the tubesheet should be 422.137 mm

Shell Membrane Stress due to Joint Interaction [σ_s, m]:

$$\begin{aligned}
 &= a_o^2 / ((D_s + t_s) t_s) [P_e + (p_s^2 - 1)(P_s - P_t)] + a_s^2 \cdot P_t / ((D_s + t_s) t_s) \\
 &= 464^2 / ((1000 + 55) 55) [-3383 + (1.08^2 - 1) \\
 &\quad (16000 - 16000)] + 500^2 \cdot 16000 / ((1000 + 55) 55) \\
 &= 56.3925 \text{ N./mm}^2
 \end{aligned}$$

Shell Bending Stress due to Joint Interaction [σ_s, b]:

$$\begin{aligned}
 &= 6 \cdot k_s / t_s^2 \{ \beta_s [\delta_s \cdot P_s + a_s^2 \cdot P \cdot s / (E \cdot t_s)] + \\
 &\quad 6(1 - \nu^2) / (E \cdot h) (a_o / h)^3 (1 + h \cdot \beta_s / 2) [P_e (Z_v + Z_m \cdot Q_1) + 2/a_o^2 \cdot Z_m \cdot Q_2] \} \\
 &= 6 \cdot 45.84 \text{E}+09 / 55^2 \{ 0.0075 [0.019560438 \cdot 16000 + 500^2 \\
 &\quad \cdot 0 / (10.96 \text{E}+09)] + \\
 &\quad 6(1 - 0.41^2) / (40428684) (464/195)^3 (1 + 195 \cdot \\
 &\quad 0.0075/2) [-3383 (0.15 + 0.62 \cdot -0.18) + \\
 &\quad 2/464^2 \cdot 0.62 \cdot 67.6] \} \\
 &= 181.4253 \text{ N./mm}^2
 \end{aligned}$$

Shell Stress Summation vs. Allowable



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$||\sigma_{s,m}| + |\sigma_{s,b}| \leq 1.5 \cdot S_s$
 $||56.4| + |181| \leq 243 \text{ N./mm}^2$
 $238, \text{ must be } \leq 243 \text{ N./mm}^2$

Note:

For a given Channel thickness of 56 mm, the minimum Channel length adjacent to the tubesheet should be 425.958 mm

Channel Membrane Stress due to Joint Interaction $[\sigma_{c,m}]$:

$= a c^2 \cdot P_t / ((D_c + t_c) t_c \cdot \cos(\theta))$
 $= 500^2 \cdot 16000 / ((1000 + 56) 56 \cdot \cos(0))$
 $= 67.645 \text{ N./mm}^2$

Channel Bending Stress due to Joint Interaction $[\sigma_{c,b}]$:

$= 6 \cdot k c / t c^2 \cdot \{ (\delta_c \cdot C_m \Delta \cdot \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t$
 $- 6 ((1 - \nu^*) / E^*) \cdot (a o / h)^3 \cdot (C_m \theta +$
 $h \cdot \beta_c / 2 \cdot C_m \Delta) \cdot [P_e (Z_v + Z_m \cdot Q_1) + 2 / a o^2 \cdot Z_m \cdot Q_2] \}$
 $= 6 \cdot 47.64 E + 09 / 56^2 \cdot \{ (0.019328313 \cdot 1 \cdot 0.0075 - 0.000 \cdot 1) \cdot 16000$
 $- 6 ((1 - 0.41^2) / 40428684) \cdot$
 $(464 / 195)^3 \cdot (1 + 195 \cdot 0.0075 / 2 \cdot 1) \cdot [-3383 (0.15 + 0.62$
 $\cdot -0.18) + 2 / 464^2 \cdot 0.62 \cdot 67.6] \}$
 $= 244.0296 \text{ N./mm}^2$

Channel Stress Summation vs. Allowable

$||\sigma_{c,m}| + |\sigma_{c,b}| \leq 1.5 \cdot S_c$
 $||67.6| + |244| \leq 238 \text{ N./mm}^2$
 $312 \text{ must be } \leq 238 \text{ N./mm}^2 \text{ (See Elast-Plast iteration below)}$

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	47.3 <=	226.5 N./mm ²	Pass
Tubesheet Shear Stress	13.5 <=	120.8 N./mm ²	Pass
Maximum Tube Stress	4.8 <=	126.0 N./mm ²	Pass
Minimum Tube Stress (Buckling)	-3.6 <=	-94.5 N./mm ²	Pass
Maximum Force on any one Tube	63.4 <=	1665.0 Kg _f	Pass
Axial Membrane Stress in Shell	56.4 <=	161.9 N./mm ²	Pass
Shell Stress (jt. inter.)	237.8 <=	242.8 N./mm ²	Pass
Channel Stress (jt. inter.)	311.7 <=	238.5 N./mm ²	See below

Additional Results for the 2nd Elastic-Plastic Iteration.



These calculations are not intended to be used in the creep range. Check Part 4.18 in Section VIII Division 2.

Compute Allowable $[S_{stars}]$:

$= \min(S_{ys}, S_{PS}, s/2)$
 $= \min(243, 486/2) = 243 \text{ N./mm}^2$

Compute Allowable $[S^*c]$:

$= \min(S_{yc}, S_{PS}, c/2)$
 $= \min(239, 478/2) = 239 \text{ N./mm}^2$

Compute Factor $[facts]$:

$= \min([1.4 - 0.4 \cdot \text{abs}(\sigma_{s,b}) / S_{stars}], 1)$
 $= \min([1.4 - 0.4 \cdot \text{abs}(181) / 243], 1) = 1$

Compute Factor $[factc]$:

$= \min([1.4 - 0.4 \cdot \text{abs}(\sigma_{c,b}) / S_{starc}], 1)$
 $= \min([1.4 - 0.4 \cdot \text{abs}(244) / 239], 1) = 0.99$



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Compute Modified Modulus [Ec]:

= Ec • factc

= 198151376 • 0.99 = 196486384 KPa.

Tubesheet Bending Stress - final thickness [σf]:

= (1.5 • Fm / μ*) • (2 • ao / (h - h'g))² • Pe

= (1.5 • 0.061 / 0.24) • (2 • 464 / (55.5 - 0))² • -2140

= -226.4955 N./mm²

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

= h + Cats + Catc

= 55.5 + 0 + 0

= 55.5106 mm

For the Elastic-Plastic iteration, Shell Allowable = SPS,s

ShellAllow = 486.0172 N./mm²

ASME Section VIII-2 Part 4.18.8.4 Step 12:

For the Elastic-Platic iteration, the channel allowable stress = SPS,c

Challow = 478.0176 N./mm²

Computations completed for ASME tubesheet configuration: a

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	47.3 <=	226.5 N./mm ²	Pass
Tubesheet Shear Stress	13.5 <=	120.8 N./mm ²	Pass
Maximum Tube Stress	4.8 <=	126.0 N./mm ²	Pass
Minimum Tube Stress (Buckling)	-3.6 <=	-94.5 N./mm ²	Pass
Maximum Force on any one Tube	63.4 <=	1665.0 Kgf	Pass
Axial Membrane Stress in Shell	56.4 <=	161.9 N./mm ²	Pass
Shell Stress (jt. inter.)	237.8 <=	486.0 N./mm ²	Pass
Channel Stress (jt. inter.)	311.7 <=	478.0 N./mm ²	Pass

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 14 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend	Stresses Allwd	Case Type	Pass/ Fail
D1uc	185.943	...	208	226	55 121 Ps+Pt-Th	D1 Ok
D2uc	161.136	...	166	226	42 121 Ps+Pt-Th	D2 Ok
D3uc	55.537	...	47	226	13 121 Ps+Pt-Th	D3 Ok
O1uc	117.801	...	209	454	55 121 Ps+Pt+Th	O1 Ok
O2uc	95.092	...	166	454	42 121 Ps+Pt+Th	O2 Ok
O3uc	25.153	...	47	454	13 121 Ps+Pt+Th	O3 Ok
O4uc	3.810	...	...	454	... 121 Ps+Pt+Th	O4 Ok
D1c	187.687	...	211	226	56 121 Ps+Pt-Th-c	D1 Ok
D2c	162.790	...	168	226	42 121 Ps+Pt-Th-c	D2 Ok
D3c	59.153	...	49	226	14 121 Ps+Pt-Th-c	D3 Ok
O1c	120.016	...	212	454	56 121 Ps+Pt+Th-c	O1 Ok
O2c	97.242	...	168	454	42 121 Ps+Pt+Th-c	O2 Ok
O3c	28.714	...	49	454	14 121 Ps+Pt+Th-c	O3 Ok
O4c	6.810	...	...	454	... 121 Ps+Pt+Th-c	O4 Ok
Max:	187.6868	... mm	0.932	0.463	(Str. Ratio)	

Load Case Definitions:



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[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures
derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
Load Case Tables

[(+)-Th]:

With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:

Operating Load Cases using the Maximum and Minimum Operating Pressures and
Operating Temperatures

Shell Axial Membrane Stress Summary:

---- Shell Stresses ----					:	---- Shell Band Stress ----					:	Pass
Case#	Ten	Allwd	Cmp	Allwd	:	Ten	Allwd	Cmp	Allwd	:	Fail	
D1uc	8	161	...	...	:	...	...	...	...	:		Ok
D2uc	49	161	...	...	:	...	...	...	...	:		Ok
D3uc	56	161	...	...	:	...	...	...	...	:		Ok
O1uc	8	486	...	...	:	...	...	...	...	:		Ok
O2uc	49	486	...	...	:	...	...	...	...	:		Ok
O3uc	56	486	...	...	:	...	...	...	...	:		Ok
O4uc	...	486	...	...	:	...	...	...	...	:		Ok
D1c	8	161	...	...	:	...	...	...	...	:		Ok
D2c	50	161	...	...	:	...	...	...	...	:		Ok
D3c	58	161	...	...	:	...	...	...	...	:		Ok
O1c	8	486	...	...	:	...	...	...	...	:		Ok
O2c	50	486	...	...	:	...	...	...	...	:		Ok
O3c	58	486	...	...	:	...	...	...	...	:		Ok
O4c	...	486	...	...	:	...	...	...	...	:		Ok
Max RATIO	0.360		...		:	...		...		:		

Tube, Shell and Channel Stress Summary:

---- Tube Stresses ----					Tube Loads		Shell Stress		Channel Stress		Pass
Case#	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	Fail
D1uc	18	125	-9	-94	241	1664	65	486	335	478	Ok
D2uc	13	125	-13	-94	178	1664	287	486	24	238	Ok
D3uc	5	125	-4	-94	63	1664	238	486	312	478	Ok
O1uc	18	251	-9	-94	241	3329	65	486	335	478	Ok
O2uc	13	251	-13	-94	178	3329	287	486	24	478	Ok
O3uc	5	251	-4	-94	63	3329	238	486	312	478	Ok
O4uc	...	251	...	...	...	3329	...	486	...	478	Ok
D1c	19	125	-10	-94	245	1664	71	486	350	478	Ok
D2c	13	125	-13	-94	176	1664	299	486	27	478	Ok
D3c	5	125	-4	-94	68	1664	244	486	323	478	Ok
O1c	19	251	-10	-94	245	3329	71	486	350	478	Ok
O2c	13	251	-13	-94	176	3329	299	486	27	478	Ok
O3c	5	251	-4	-94	68	3329	244	486	323	478	Ok
O4c	...	251	...	...	...	3329	...	486	...	478	Ok
Max RATIO	0.147			0.142		0.147		0.614		0.732	



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ASME TS Calc: DESIGN CASE

Case: 1 9:40a Aug 12,2026

Summary of Thickness Comparisons for 14 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	187.687	195.000	Ok
Tube thickness :	1.624	2.500	Ok
Tube-tubesheet fillet weld leg :	1.500	2.000	Ok
Tube-tubesheet groove weld leg :	1.500	2.000	Ok

Min Shell length of thk, (55.000) adj. to tubesheet: 422.137 mm

Min Channel length of thk, (56.000) adj. to tubesheet: 425.958 mm

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Extends by : +.000 mm
 Due to Pressure Shell Compresses by : -0.475 mm
 Due to Pressure + Thermal Shell Compresses by : -0.475 mm

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	0 shellside Stress Rat.	Shellside MAWP	0 tubeside Stress Rat.
Tubesheet Bending Stress	17210.96	1.000	21631.48	1.000
Tubesheet Shear Stress	34561.90	1.000	45885.37	1.000
Tube Tensile Stress	108753.22	1.000	149875.66	1.000
Tube Compressive Stress	34290.61	0.223	112406.73	1.000
Tube-Tubesheet Joint load	108751.69	1.000	149875.64	1.000
Shell Stress (Axial, Junction)	17210.96	1.000	21631.48	1.000
Tubesheet-Channel Junction	17210.96	1.000	21631.48	1.000
Tube Pressure Stress	38475.48	1.000	29503.55	1.000
Minimum MAWP	17210.96		21631.48	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	0 shellside Stress Rat.	Shellside MAPnc	0 tubeside Stress Rat.
Tubesheet Bending Stress	18991.84	0.996	23828.33	0.996
Tubesheet Shear Stress	38179.57	1.000	50505.40	1.000
Tube Tensile Stress	120779.52	1.000	164462.58	1.000
Tube Compressive Stress	188961.52	1.000	122155.16	1.000
Tube-Tubesheet Joint load	120779.55	1.000	164462.58	1.000
Shell Stress (Axial, Junction)	18991.84	0.996	23828.33	0.996
Tubesheet-Channel Junction	18991.84	0.996	23828.33	0.996
Tube Pressure Stress	42139.45	1.000	31403.66	1.000
Minimum MAPnc	18991.84		23828.33	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.0896	0.0897	...	...	...	...
2	-0.4747	-0.4749	...	...	...	...
3	-0.3850	-0.3852	...	...	...	...
4	...	...	...	...	...	...



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ASME TS Calc: DESIGN CASE

Case: 1 9:40a Aug 12, 2026

5	0.0896	0.0897	...	...	...	...
6	-0.4747	-0.4749	...	...	...	...
7	-0.3850	-0.3852	...	...	...	...
8	...	...	...	...	...	...

Tubesheet MDMT Calculations:

Shell Side MDMT calculation:

Governing thickness on the shell side per figure 3.10:

$$= \max(\text{tubesheet thk}/4, \min(\text{tubesheet thk}, \text{shell thickness}))$$

$$= \max(195/4, \min(195, 55))$$

$$= 55.000 \text{ mm}$$

MDMT reduction ratio:

$$= \min(\text{ps}/\text{Shellside MAPnc}, 1)$$

$$= \min(16000/23828, 1)$$

$$= 0.671$$

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature : -8 °C
 User entered Minimum Design Temperature : -33 °C
 The Curve for this Material is : B
 From Figure 3.8M- with PWHT, the MDMT is : 11 °C
 Temperature reduction per Graph 3.13 : 19 °C
 The computed stress ratio is : 0.671

Warning: Impact Testing Is Mandatory!

Channel Side MDMT calculation:

Governing thickness on the shell side per figure 3.10:

$$= \max(\text{tubesheet thk}/4, \min(\text{tubesheet thk}, \text{channel thickness}))$$

$$= \max(195/4, \min(195, 56))$$

$$= 56.000 \text{ mm}$$

MDMT reduction ratio:

$$= \min(\text{pt}/\text{Tubeside MAPnc}, 1)$$

$$= \min(16000/18992, 1)$$

$$= 0.842$$

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature : 3 °C
 User entered Minimum Design Temperature : -33 °C
 The Curve for this Material is : B
 From Figure 3.8M- with PWHT, the MDMT is : 11 °C
 Temperature reduction per Graph 3.13 : 8 °C
 The computed stress ratio is : 0.842

Warning: Impact Testing Is Mandatory!

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ASME TS Calc: 100% SOR

Case: 2 9:40a Aug 12, 2026

Input Echo, Tubesheet Number 2, Description: Fixed tubesheet

Shell Data:

Main Shell Description: Shell

Shell Maximum Design Pressure	Psd,max	13020.00 KPa.
Shell Maximum Operating Pressure	Psox,max	13020.00 KPa.
Shell Minimum Operating Pressure	Psox,min	0.00 KPa.
Shell Thickness	ts	55.0000 mm
Shell Internal Corrosion Allowance	cas	1.5000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	1000.000 mm
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	80.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	161.86 N./mm ²
Shell Allowable Stress at Ambient		175.00 N./mm ²

Channel Data:

Channel Description: Chnl. rear

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	13200.00 KPa.
Channel Maximum Operating Pressure	Ptox,max	13200.00 KPa.
Channel Minimum Operating Pressure	Ptox,min	0.00 KPa.
Channel Thickness	tc	56.0000 mm
Channel Internal Corrosion Allowance	cac	1.5000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	1000.000 mm
Channel Design Temperature	TEMPC	100.00 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	159.00 N./mm ²
Channel Allowable Stress at Ambient		175.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	1015
Tube Wall Thickness	et	2.5000 mm
Tube Outside Diameter	D	19.0000 mm
Total Straight Tube Length	Lt	6000.00 mm
Straight Tube Length (bet. inner tubsht faces) L		5610.00 mm
Design Temperature of the Tubes		100.00 °C
Tube Material		SA-334 1
Tube Material UNS Number		K03008
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		126.00 N./mm ²
Tube Allowable Stress At Ambient		138.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	189.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	25.0000 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.0000 mm
Groove Weld Leg	ag	2.0000 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		4.21.2
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	454.384 mm



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ASME TS Calc: 100% SOR

Case: 2 9:40a Aug 12, 2026

Largest Center-to-Center Tube Distance	U1	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	5190.7 cm ²
Perimeter of Tube Layout	Cp	2786.6 mm

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf A

Tubesheet Design Metal Temperature	T	100.00 °C
Tubesheet Material		SA-350 LF2

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Tubesheet Material UNS Number		K03011
Tubesheet Allowable Stress at Temperature	S	151.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	165.00 N./mm ²
Thickness of Tubesheet	h	195.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	1.5000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	1.5000 mm
Tubesheet Outside Diameter	A	1110.000 mm

Area of the Untubed Lanes	AL	0.0 cm ²
---------------------------	----	---------------------

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	0.000 mm
Tube End condition corresponding to Span (l)	k	0.00
Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)		Yes
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Mean Shell Metal Temp. along Shell len.	Tsm	20.00 °C
Mean Tube Metal Temp. along Tube length	Ttm	39.00 °C
Junction Stress Reduction option		Perform Plastic Calculation
Tubesheet Integral with		Both
Tubesheet Extended as Flange		No
Is Exchanger in Creep range (skip EP, Use 3S for Sps)		No



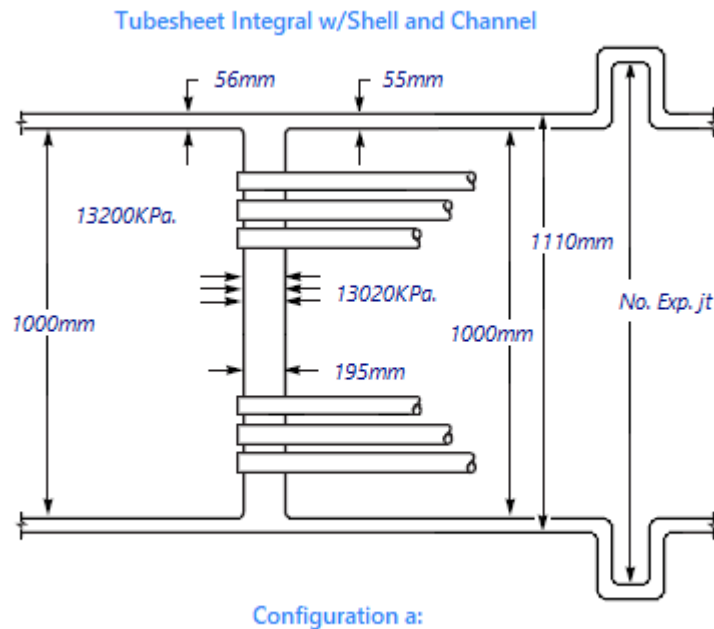
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ASME TS Calc: 100% SOR

Case: 2 9:40a Aug 12, 2026



Messages regarding the Tubesheet & Exchanger Input Data:

Warning - Tube end condition k should be bet 0 - 1 (.00).

ASME Division 2 Tubesheet Results per Part 4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	20.0 °C	0.0000115030 /°C
Elastic Mod. at Design Temperature	80.0 °C	0.19933E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Metal Temp. along Len	20.0 °C	0.20277E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Design Temperature	100.0 °C	0.19815E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
Tubes - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	39.0 °C	0.0000117070 /°C
Elastic Mod. at Design Temperature	100.0 °C	0.19815E+09 KPa.
Elastic Mod. at Metal Temp. along Len	39.0 °C	0.20168E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	100.0 °C	0.19815E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
TubeSheet - TM-1 Carbon Steels with C> 0.3%



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ASME TS Calc: 100% SOR

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Elastic Mod. at Design Temperature	100.0 °C	0.19677E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20132E+09 KPa.

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.
Please insure these properties are consistent with the
type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i$ Equation 4.3.1
 $= 0.5 \cdot 14 \cdot (\exp(13200/(126 \cdot 1)) - 1) + 0$
 $= 0.77 + 0$
 $= 0.77 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

Determine Parameter [Mx]:

$= L / \sqrt{(R_o \cdot t)}$
 $= 5610 / \sqrt{(9.5 \cdot 2.5)}$
 $= 1151.148$

Compute the Factor [Ch]:

Since $Mx > 2 \cdot (Do/t)^{0.94}$, $Ch = 0.55 / (Do/t)$
 $= 0.55 \cdot (19/2.5)$
 $= 0.0724$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$= 1.6 \cdot Ch \cdot E_y (t / Do)$
 $= 1.6 \cdot 0.072 \cdot 198151376 (2.5/19)$
 $= 3019.102 \text{ N./mm}^2$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$= Fhe / F_y$
 $= 3019.1/189.0$
 $= 15.974$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$= \sigma_{ys} / \sigma_{uts}$
 $= 189/379$
 $= 0.498$

Ratio [AE]:

$= Fhe / E$
 $= 3019/198151376$
 $= 0.015235$

Factor [m1]:

$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$
 $= (\ln(0.5) + (0.20000E-04 - 0.002)) / \ln(\ln(1 + 0.20000E-04) / \ln(1 + 0.002))$
 $= 0.152$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$
 $= 189 (1 + 0.0020) / \ln(1 + 0.0020)^{0.15168217}$
 $= 486.157 \text{ N./mm}^2$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$= m2f / (m2f2 - R)$
 $= 0.6 (1 - 0.5)$
 $= 0.301$



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ASME TS Calc: 100% SOR

Case: 2 9:40a Aug 12, 2026

Curve Fitting Constant [A2]:
 $= (\sigma_{uts} \cdot \exp(m_2)) / (m_2 m_2)$
 $= (379 \cdot \exp(0.3)) / (0.3 \cdot 0.3)$
 $= 735.456 \text{ N./mm}^2$

Material Parameter for Stress-Strain Curve Model [K]:
 $= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$
 $= 1.5 \cdot 0.51.5 - 0.5 \cdot 0.52.5 - 0.53.5$
 $= 0.353$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	189	-1.982	0.350E-07	0.315E-07	0.970E-07	-0.813E-07	0.115E+08	0.131E-02
2	94	-4.808	0.726E-09	0.726E-09	0.194E-07	-0.194E-07	0.154E+09	0.146E-01
3	142	-3.395	0.701E-08	0.697E-08	0.497E-07	-0.493E-07	0.515E+08	0.125E-01
4	165	-2.688	0.166E-07	0.162E-07	0.711E-07	-0.685E-07	0.247E+08	0.852E-02
5	177	-2.335	0.244E-07	0.232E-07	0.835E-07	-0.770E-07	0.169E+08	0.472E-02
6	183	-2.158	0.293E-07	0.273E-07	0.901E-07	-0.800E-07	0.139E+08	0.206E-02
7	186	-2.070	0.321E-07	0.294E-07	0.935E-07	-0.809E-07	0.126E+08	0.472E-03
8	188	-2.026	0.335E-07	0.305E-07	0.952E-07	-0.812E-07	0.120E+08	0.394E-03
9	187	-2.048	0.328E-07	0.299E-07	0.944E-07	-0.810E-07	0.123E+08	0.451E-04
10	187	-2.037	0.332E-07	0.302E-07	0.948E-07	-0.811E-07	0.122E+08	0.173E-03
11	187	-2.042	0.330E-07	0.301E-07	0.946E-07	-0.811E-07	0.123E+08	0.635E-04
12	187	-2.045	0.329E-07	0.300E-07	0.945E-07	-0.811E-07	0.123E+08	0.913E-05
13	187	-2.047	0.328E-07	0.300E-07	0.944E-07	-0.811E-07	0.123E+08	0.180E-04
14	187	-2.046	0.329E-07	0.300E-07	0.944E-07	-0.811E-07	0.123E+08	0.444E-05
15	187	-2.046	0.329E-07	0.300E-07	0.944E-07	-0.811E-07	0.123E+08	0.234E-05
16	187	-2.046	0.329E-07	0.300E-07	0.944E-07	-0.811E-07	0.123E+08	0.105E-05

Factor of Safety [FS]:
 Since $Fic > 0.55 F_y$ and $Fic < F_y$
 $= 2.407 - 0.741 \cdot Fic / F_y$
 $= 2.407 - 0.741 \cdot 187/189$
 $= 1.672$

Allowable Circumferential Compressive Stress [Fha]:
 $= Fic/FS$
 $= 187/1.67$
 $= 112.120 \text{ N./mm}^2$

Allowable External Pressure at the given Thickness [Pa]:
 $= 2 \cdot Fha(t / Do)$
 $= 2 \cdot 112114 (2.5/19)$
 $= 29503.564 \text{ KPa.}$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	1.4087 mm
Allowable Internal Pressure at Corroded thickness	38475.49 KPa.
Required Internal Design Pressure	13200.00 KPa.
Allowable External Pressure at Corroded thickness	29503.56 KPa.
Required External Design Pressure	13020.00 KPa.
Required Thickness due to Shell Side pressure	1.4087 mm

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 14 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend Allwd	Stresses Shear Allwd	Case Type	Pass/ Fail
D1uc 165.714	...	172	226	46	121	Ps+Pt-Th D1 Ok
D2uc 140.420	...	135	226	34	121	Ps+Pt-Th D2 Ok



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ASME TS Calc: 100% SOR

Case: 2 9:40a Aug 12, 2026

D3uc	47.185	...	40	226	12	121	Ps+Pt-Th	D3	Ok
O1uc	63.575	...	118	454	28	121	Ps+Pt+Th	O1	Ok
O2uc	110.136	...	190	454	52	121	Ps+Pt+Th	O2	Ok
O3uc	7.628	...	21	454	6	121	Ps+Pt+Th	O3	Ok
O4uc	34.425	...	62	454	17	121	Ps+Pt+Th	O4	Ok
D1c	167.601	...	175	226	46	121	Ps+Pt-Th-c	D1	Ok
D2c	142.213	...	137	226	34	121	Ps+Pt-Th-c	D2	Ok
D3c	50.746	...	42	226	12	121	Ps+Pt-Th-c	D3	Ok
O1c	67.048	...	120	454	29	121	Ps+Pt+Th-c	O1	Ok
O2c	111.786	...	191	454	52	121	Ps+Pt+Th-c	O2	Ok
O3c	10.003	...	21	454	6	121	Ps+Pt+Th-c	O3	Ok
O4c	36.973	...	62	454	17	121	Ps+Pt+Th-c	O4	Ok

Max: 167.6008 ... mm 0.772 0.428 (Str. Ratio)

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[(+)-Th]:

With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:

Operating Load Cases using the Maximum and Minimum Operating Pressures and Operating Temperatures

Shell Axial Membrane Stress Summary:

Case#	---- Shell Stresses ----				:	---- Shell Band Stress ----				: Pass
	Ten	Allwd	Cmp	Allwd		Ten	Allwd	Cmp	Allwd	
D1uc	6	161	...	...		...	...	...	...	Ok
D2uc	40	161	...	...		...	...	...	...	Ok
D3uc	46	161	...	...		...	...	...	...	Ok
O1uc	23	486	...	...		...	...	...	...	Ok
O2uc	56	486	...	...		...	...	...	...	Ok
O3uc	62	486	...	...		...	...	...	...	Ok
O4uc	16	486	...	...		...	...	...	...	Ok
D1c	7	161	...	...		...	...	...	...	Ok
D2c	41	161	...	...		...	...	...	...	Ok
D3c	47	161	...	...		...	...	...	...	Ok
O1c	23	486	...	...		...	...	...	...	Ok
O2c	57	486	...	...		...	...	...	...	Ok
O3c	64	486	...	...		...	...	...	...	Ok
O4c	16	486	...	...		...	...	...	...	Ok
Max RATIO	0.293		...			...		...		

Tube, Shell and Channel Stress Summary:

Case#	---- Tube Stresses ----				Tube Loads		Shell Stress		Channel Stress		Pass
	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	



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ASME TS Calc: 100% SOR

Case: 2 9:40a Aug 12, 2026

D1uc	15	125	-7	-94	199	1664	53	242	277	478	Ok
D2uc	11	125	-11	-94	145	1664	233	242	19	238	Ok
D3uc	4	125	-3	-94	54	1664	193	242	257	478	Ok
O1uc	23	251	-23	-94	307	3329	25	486	232	478	Ok
O2uc	38	251	-38	-94	503	3329	294	486	64	478	Ok
O3uc	23	251	-23	-94	304	3329	254	486	213	478	Ok
O4uc	27	251	-27	-94	358	3329	61	486	45	478	Ok
D1c	15	125	-8	-94	202	1664	58	242	289	478	Ok
D2c	11	125	-11	-94	144	1664	243	486	22	238	Ok
D3c	4	125	-3	-94	59	1664	198	242	267	478	Ok
O1c	24	251	-24	-94	312	3329	28	486	242	478	Ok
O2c	38	251	-38	-94	499	3329	306	486	68	478	Ok
O3c	22	251	-22	-94	297	3329	261	486	220	478	Ok
O4c	27	251	-27	-94	355	3329	63	486	46	478	Ok

Max RATIO 0.151 0.403 0.151 0.962 0.604

Summary of Thickness Comparisons for 14 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	167.601	195.000	Ok
Tube thickness :	1.409	2.500	Ok
Tube-tubesheet fillet weld leg :	1.500	2.000	Ok
Tube-tubesheet groove weld leg :	1.500	2.000	Ok

Min Shell length of thk, (55.000) adj. to tubesheet: 422.137 mm

Min Channel length of thk, (56.000) adj. to tubesheet: 425.958 mm

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Extends by : +1.247 mm

Due to Pressure Shell Compresses by : -0.386 mm

Due to Pressure + Thermal Shell Extends by : +1.321 mm

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	0 shellside Stress Rat.	Shellside MAWP	0 tubeside Stress Rat.
Tubesheet Bending Stress	17212.10	1.000	21637.27	1.000
Tubesheet Shear Stress	34574.79	1.000	39265.95	1.000
Tube Tensile Stress	108639.05	1.000	149723.94	1.000
Tube Compressive Stress	38412.84	0.415	80083.12	1.000
Tube-Tubesheet Joint load	108639.70	1.000	149724.14	1.000
Shell Stress (Axial, Junction)	17212.10	1.000	21637.27	1.000
Tubesheet-Channel Junction	17212.10	1.000	21637.27	1.000
Tube Pressure Stress	38475.48	1.000	29503.55	1.000
Minimum MAWP	17212.10		21637.27	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	0 shellside Stress Rat.	Shellside MAPnc	0 tubeside Stress Rat.
Tubesheet Bending Stress	18981.93	0.995	23835.15	0.996
Tubesheet Shear Stress	38193.05	1.000	50522.26	1.000
Tube Tensile Stress	120661.51	1.000	164299.58	1.000
Tube Compressive Stress	186665.52	1.000	122034.02	1.000
Tube-Tubesheet Joint load	120661.41	1.000	164299.36	1.000
Shell Stress (Axial, Junction)	18981.93	0.995	23835.15	0.996



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ASME TS Calc: 100% SOR

Case: 2 9:40a Aug 12,2026

Tubesheet-Channel Junction	18981.93	0.995	23835.15	0.996
Tube Pressure Stress	42139.45	1.000	31403.66	1.000
Minimum MAPnc	18981.93		23835.15	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.0740	0.0740	...	...	...	...
2	-0.3863	-0.3865	...	...	...	...
3	-0.3123	-0.3125	...	...	...	...
4	...	...	...	...	...	...
5	0.0740	0.0740	1.2466	1.2472	...	...
6	-0.3863	-0.3865	1.2466	1.2472	...	...
7	-0.3123	-0.3125	1.2466	1.2472	...	...
8	...	...	1.2466	1.2472	...	...

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ASME TS Calc: 110% EOR

Case: 3 9:40a Aug 12, 2026

Input Echo, Tubesheet Number 3, Description: Fixed tubesheet

Shell Data:

Main Shell Description: Shell

Shell Maximum Design Pressure	Psd,max	13090.00 KPa.
Shell Maximum Operating Pressure	Psox,max	13090.00 KPa.
Shell Minimum Operating Pressure	Psox,min	0.00 KPa.
Shell Thickness	ts	55.0000 mm
Shell Internal Corrosion Allowance	cas	1.5000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	1000.000 mm
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	80.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	161.86 N./mm ²
Shell Allowable Stress at Ambient		175.00 N./mm ²

Channel Data:

Channel Description: Chnl. rear

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	13370.00 KPa.
Channel Maximum Operating Pressure	Ptox,max	13370.00 KPa.
Channel Minimum Operating Pressure	Ptox,min	0.00 KPa.
Channel Thickness	tc	56.0000 mm
Channel Internal Corrosion Allowance	cac	1.5000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	1000.000 mm
Channel Design Temperature	TEMPC	100.00 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	159.00 N./mm ²
Channel Allowable Stress at Ambient		175.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	1015
Tube Wall Thickness	et	2.5000 mm
Tube Outside Diameter	D	19.0000 mm
Total Straight Tube Length	Lt	6000.00 mm
Straight Tube Length (bet. inner tubsht faces) L		5610.00 mm
Design Temperature of the Tubes		100.00 °C
Tube Material		SA-334 1
Tube Material UNS Number		K03008
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		126.00 N./mm ²
Tube Allowable Stress At Ambient		138.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	189.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	25.0000 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.0000 mm
Groove Weld Leg	ag	2.0000 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		4.21.2
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	454.384 mm



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ASME TS Calc: 110% EOR

Case: 3 9:40a Aug 12,2026

Largest Center-to-Center Tube Distance	U1	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	5190.7 cm ²
Perimeter of Tube Layout	Cp	2786.6 mm

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf A

Tubesheet Design Metal Temperature	T	100.00 °C
Tubesheet Material		SA-350 LF2

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Tubesheet Material UNS Number		K03011
Tubesheet Allowable Stress at Temperature	S	151.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	165.00 N./mm ²
Thickness of Tubesheet	h	195.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	1.5000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	1.5000 mm
Tubesheet Outside Diameter	A	1110.000 mm

Area of the Untubed Lanes	AL	0.0 cm ²
---------------------------	----	---------------------

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	0.000 mm
Tube End condition corresponding to Span (l)	k	0.00
Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)		Yes
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Mean Shell Metal Temp. along Shell len.	Tsm	18.00 °C
Mean Tube Metal Temp. along Tube length	Ttm	37.00 °C
Junction Stress Reduction option		Perform Plastic Calculation
Tubesheet Integral with		Both
Tubesheet Extended as Flange		No
Is Exchanger in Creep range (skip EP, Use 3S for Sps)		No



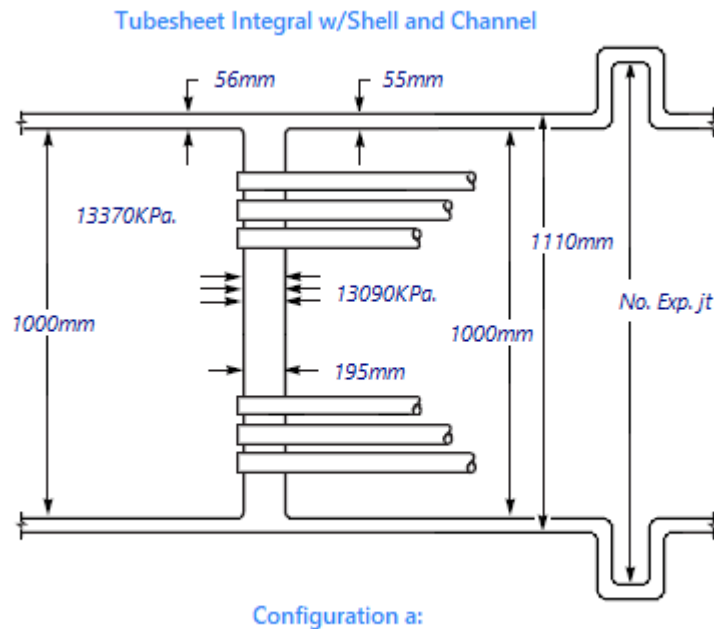
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Messages regarding the Tubesheet & Exchanger Input Data:

Warning - Tube end condition k should be bet 0 - 1 (.00).

ASME Division 2 Tubesheet Results per Part 4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	18.0 °C	0.0000114743 /°C
Elastic Mod. at Design Temperature	80.0 °C	0.19933E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Metal Temp. along Len	18.0 °C	0.20291E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Design Temperature	100.0 °C	0.19815E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
Tubes - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	37.0 °C	0.0000116906 /°C
Elastic Mod. at Design Temperature	100.0 °C	0.19815E+09 KPa.
Elastic Mod. at Metal Temp. along Len	37.0 °C	0.20179E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	100.0 °C	0.19815E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
TubeSheet - TM-1 Carbon Steels with C> 0.3%



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ASME TS Calc: 110% EOR

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Elastic Mod. at Design Temperature	100.0 °C	0.19677E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20132E+09 KPa.

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.
Please insure these properties are consistent with the
type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c$ Equation 4.3.1
 $= 0.5 \cdot 14 \cdot (\exp(13370/(126 \cdot 1)) - 1) + 0$
 $= 0.78 + 0$
 $= 0.78 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

Determine Parameter [Mx]:

$= L / \sqrt{R_o \cdot t}$
 $= 5610 / \sqrt{9.5 \cdot 2.5}$
 $= 1151.148$

Compute the Factor [Ch]:

Since $Mx > 2 \cdot (Do/t)^{0.94}$, $Ch = 0.55 / (Do/t)$
 $= 0.55 \cdot (19/2.5)$
 $= 0.0724$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$= 1.6 \cdot Ch \cdot E_y (t / Do)$
 $= 1.6 \cdot 0.072 \cdot 198151376 (2.5/19)$
 $= 3019.102 \text{ N./mm}^2$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$= Fhe / F_y$
 $= 3019.1/189.0$
 $= 15.974$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$= \sigma_{ys} / \sigma_{uts}$
 $= 189/379$
 $= 0.498$

Ratio [AE]:

$= Fhe / E$
 $= 3019/198151376$
 $= 0.015235$

Factor [m1]:

$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$
 $= (\ln(0.5) + (0.20000E-04 - 0.002)) / \ln(\ln(1 + 0.20000E-04) / \ln(1 + 0.002))$
 $= 0.152$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$
 $= 189 (1 + 0.0020) / \ln(1 + 0.0020)^{0.15168217}$
 $= 486.157 \text{ N./mm}^2$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$= m2f / (m2f2 - R)$
 $= 0.6 (1 - 0.5)$
 $= 0.301$



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Curve Fitting Constant [A2]:
 $= (\sigma_{uts} \cdot \exp(m_2)) / (m_2 m_2)$
 $= (379 \cdot \exp(0.3)) / (0.3 \cdot 0.3)$
 $= 735.456 \text{ N./mm}^2$

Material Parameter for Stress-Strain Curve Model [K]:
 $= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$
 $= 1.5 \cdot 0.51.5 - 0.5 \cdot 0.52.5 - 0.53.5$
 $= 0.353$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	189	-1.982	0.350E-07	0.315E-07	0.970E-07	-0.813E-07	0.115E+08	0.131E-02
2	94	-4.808	0.726E-09	0.726E-09	0.194E-07	-0.194E-07	0.154E+09	0.146E-01
3	142	-3.395	0.701E-08	0.697E-08	0.497E-07	-0.493E-07	0.515E+08	0.125E-01
4	165	-2.688	0.166E-07	0.162E-07	0.711E-07	-0.685E-07	0.247E+08	0.852E-02
5	177	-2.335	0.244E-07	0.232E-07	0.835E-07	-0.770E-07	0.169E+08	0.472E-02
6	183	-2.158	0.293E-07	0.273E-07	0.901E-07	-0.800E-07	0.139E+08	0.206E-02
7	186	-2.070	0.321E-07	0.294E-07	0.935E-07	-0.809E-07	0.126E+08	0.472E-03
8	188	-2.026	0.335E-07	0.305E-07	0.952E-07	-0.812E-07	0.120E+08	0.394E-03
9	187	-2.048	0.328E-07	0.299E-07	0.944E-07	-0.810E-07	0.123E+08	0.451E-04
10	187	-2.037	0.332E-07	0.302E-07	0.948E-07	-0.811E-07	0.122E+08	0.173E-03
11	187	-2.042	0.330E-07	0.301E-07	0.946E-07	-0.811E-07	0.123E+08	0.635E-04
12	187	-2.045	0.329E-07	0.300E-07	0.945E-07	-0.811E-07	0.123E+08	0.913E-05
13	187	-2.047	0.328E-07	0.300E-07	0.944E-07	-0.811E-07	0.123E+08	0.180E-04
14	187	-2.046	0.329E-07	0.300E-07	0.944E-07	-0.811E-07	0.123E+08	0.444E-05
15	187	-2.046	0.329E-07	0.300E-07	0.944E-07	-0.811E-07	0.123E+08	0.234E-05
16	187	-2.046	0.329E-07	0.300E-07	0.944E-07	-0.811E-07	0.123E+08	0.105E-05

Factor of Safety [FS]:
 Since $Fic > 0.55 F_y$ and $Fic < F_y$
 $= 2.407 - 0.741 \cdot Fic / F_y$
 $= 2.407 - 0.741 \cdot 187/189$
 $= 1.672$

Allowable Circumferential Compressive Stress [Fha]:
 $= Fic/FS$
 $= 187/1.67$
 $= 112.120 \text{ N./mm}^2$

Allowable External Pressure at the given Thickness [Pa]:
 $= 2 \cdot Fha(t / Do)$
 $= 2 \cdot 112114 (2.5/19)$
 $= 29503.564 \text{ KPa.}$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	1.4139 mm
Allowable Internal Pressure at Corroded thickness	38475.49 KPa.
Required Internal Design Pressure	13370.00 KPa.
Allowable External Pressure at Corroded thickness	29503.56 KPa.
Required External Design Pressure	13090.00 KPa.
Required Thickness due to Shell Side pressure	1.4139 mm

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 14 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend Allwd	Stresses Shear Allwd	Case Type	Pass/ Fail
D1uc 167.017	...	174	226	46	121	Ps+Pt-Th D1 Ok
D2uc 140.943	...	136	226	34	121	Ps+Pt-Th D2 Ok



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Case: 3 9:40a Aug 12, 2026

D3uc	48.950	...	42	226	12	121	Ps+Pt-Th	D3	Ok
O1uc	65.218	...	120	454	29	121	Ps+Pt+Th	O1	Ok
O2uc	110.371	...	190	454	52	121	Ps+Pt+Th	O2	Ok
O3uc	6.734	...	20	454	5	121	Ps+Pt+Th	O3	Ok
O4uc	34.289	...	61	454	17	121	Ps+Pt+Th	O4	Ok
D1c	168.896	...	177	226	47	121	Ps+Pt-Th-c	D1	Ok
D2c	142.732	...	138	226	34	121	Ps+Pt-Th-c	D2	Ok
D3c	52.492	...	43	226	12	121	Ps+Pt-Th-c	D3	Ok
O1c	68.664	...	123	454	29	121	Ps+Pt+Th-c	O1	Ok
O2c	112.021	...	192	454	52	121	Ps+Pt+Th-c	O2	Ok
O3c	9.117	...	19	454	5	121	Ps+Pt+Th-c	O3	Ok
O4c	36.840	...	62	454	17	121	Ps+Pt+Th-c	O4	Ok

Max: 168.8959 ... mm 0.781 0.429 (Str. Ratio)

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[(+)-Th]:

With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:

Operating Load Cases using the Maximum and Minimum Operating Pressures and Operating Temperatures

Shell Axial Membrane Stress Summary:

Case#	---- Shell Stresses ----				:	---- Shell Band Stress ----				: Pass
	Ten	Allwd	Cmp	Allwd		Ten	Allwd	Cmp	Allwd	
D1uc	7	161	...	...		...	...	...	...	Ok
D2uc	40	161	...	...		...	...	...	...	Ok
D3uc	46	161	...	...		...	...	...	...	Ok
O1uc	23	486	...	...		...	...	...	...	Ok
O2uc	56	486	...	...		...	...	...	...	Ok
O3uc	63	486	...	...		...	...	...	...	Ok
O4uc	16	486	...	...		...	...	...	...	Ok
D1c	7	161	...	...		...	...	...	...	Ok
D2c	41	161	...	...		...	...	...	...	Ok
D3c	48	161	...	...		...	...	...	...	Ok
O1c	23	486	...	...		...	...	...	...	Ok
O2c	57	486	...	...		...	...	...	...	Ok
O3c	64	486	...	...		...	...	...	...	Ok
O4c	16	486	...	...		...	...	...	...	Ok
Max RATIO	0.295		...			...		...		

Tube, Shell and Channel Stress Summary:

Case#	---- Tube Stresses ----				Tube Loads		Shell Stress		Channel Stress		Pass
	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	



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Case: 3 9:40a Aug 12, 2026

D1uc	15	125	-7	-94	202	1664	54	242	280	478	Ok
D2uc	11	125	-11	-94	146	1664	235	242	19	238	Ok
D3uc	4	125	-3	-94	56	1664	194	242	261	478	Ok
O1uc	23	251	-23	-94	308	3329	26	486	236	478	Ok
O2uc	38	251	-38	-94	503	3329	295	486	64	478	Ok
O3uc	23	251	-23	-94	301	3329	254	486	216	478	Ok
O4uc	27	251	-27	-94	357	3329	61	486	44	478	Ok
D1c	16	125	-8	-94	205	1664	59	242	292	478	Ok
D2c	11	125	-11	-94	145	1664	244	486	22	238	Ok
D3c	5	125	-3	-94	60	1664	199	242	270	478	Ok
O1c	24	251	-24	-94	313	3329	29	486	246	478	Ok
O2c	38	251	-38	-94	499	3329	307	486	68	478	Ok
O3c	22	251	-22	-94	294	3329	262	486	224	478	Ok
O4c	27	251	-27	-94	354	3329	63	486	46	478	Ok

Max RATIO 0.151 0.402 0.151 0.967 0.611

Summary of Thickness Comparisons for 14 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	168.896	195.000	Ok
Tube thickness :	1.414	2.500	Ok
Tube-tubesheet fillet weld leg :	1.500	2.000	Ok
Tube-tubesheet groove weld leg :	1.500	2.000	Ok

Min Shell length of thk, (55.000) adj. to tubesheet: 422.137 mm

Min Channel length of thk, (56.000) adj. to tubesheet: 425.958 mm

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Extends by : +1.243 mm

Due to Pressure Shell Compresses by : -0.389 mm

Due to Pressure + Thermal Shell Extends by : +1.318 mm

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	17212.10	1.000	21637.27	1.000
Tubesheet Shear Stress	34574.79	1.000	39288.54	1.000
Tube Tensile Stress	108639.05	1.000	149723.94	1.000
Tube Compressive Stress	38398.81	0.415	80192.84	1.000
Tube-Tubesheet Joint load	108639.70	1.000	149724.14	1.000
Shell Stress (Axial, Junction)	17212.10	1.000	21637.27	1.000
Tubesheet-Channel Junction	17212.10	1.000	21637.27	1.000
Tube Pressure Stress	38475.48	1.000	29503.55	1.000
Minimum MAWP	17212.10		21637.27	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	18981.94	0.995	23835.15	0.996
Tubesheet Shear Stress	38193.05	1.000	50522.26	1.000
Tube Tensile Stress	120661.51	1.000	164299.58	1.000
Tube Compressive Stress	186665.52	1.000	122034.02	1.000
Tube-Tubesheet Joint load	120661.41	1.000	164299.36	1.000
Shell Stress (Axial, Junction)	18981.94	0.995	23835.15	0.996



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ASME TS Calc: 110% EOR

Case: 3 9:40a Aug 12,2026

Tubesheet-Channel Junction	18981.94	0.995	23835.15	0.996
Tube Pressure Stress	42139.45	1.000	31403.66	1.000
Minimum MAPnc	18981.94		23835.15	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.0749	0.0749	...	...	...	...
2	-0.3883	-0.3886	...	...	...	...
3	-0.3134	-0.3136	...	...	...	...
4	...	...	...	...	...	...
5	0.0749	0.0749	1.2423	1.2430	...	...
6	-0.3883	-0.3886	1.2423	1.2430	...	...
7	-0.3134	-0.3136	1.2423	1.2430	...	...
8	...	...	1.2423	1.2430	...	...

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ASME TS Calc: 10% SOR

Case: 4 9:41a Aug 12, 2026

Input Echo, Tubesheet Number 4, Description: Fixed tubesheet

Shell Data:

Main Shell Description: Shell

Shell Maximum Design Pressure	Psd,max	13830.00 KPa.
Shell Maximum Operating Pressure	Psox,max	13830.00 KPa.
Shell Minimum Operating Pressure	Psox,min	0.00 KPa.
Shell Thickness	ts	55.0000 mm
Shell Internal Corrosion Allowance	cas	1.5000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	1000.000 mm
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	80.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	161.86 N./mm ²
Shell Allowable Stress at Ambient		175.00 N./mm ²

Channel Data:

Channel Description: Chnl. rear

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	13830.00 KPa.
Channel Maximum Operating Pressure	Ptox,max	13830.00 KPa.
Channel Minimum Operating Pressure	Ptox,min	0.00 KPa.
Channel Thickness	tc	56.0000 mm
Channel Internal Corrosion Allowance	cac	1.5000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	1000.000 mm
Channel Design Temperature	TEMPC	100.00 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	159.00 N./mm ²
Channel Allowable Stress at Ambient		175.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	1015
Tube Wall Thickness	et	2.5000 mm
Tube Outside Diameter	D	19.0000 mm
Total Straight Tube Length	Lt	6000.00 mm
Straight Tube Length (bet. inner tubsht faces) L		5610.00 mm
Design Temperature of the Tubes		100.00 °C
Tube Material		SA-334 1
Tube Material UNS Number		K03008
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		126.00 N./mm ²
Tube Allowable Stress At Ambient		138.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	189.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	25.0000 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.0000 mm
Groove Weld Leg	ag	2.0000 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		4.21.2
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	454.384 mm



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ASME TS Calc: 10% SOR

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Largest Center-to-Center Tube Distance	U1	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	5190.7 cm ²
Perimeter of Tube Layout	Cp	2786.6 mm

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf A

Tubesheet Design Metal Temperature	T	100.00 °C
Tubesheet Material		SA-350 LF2

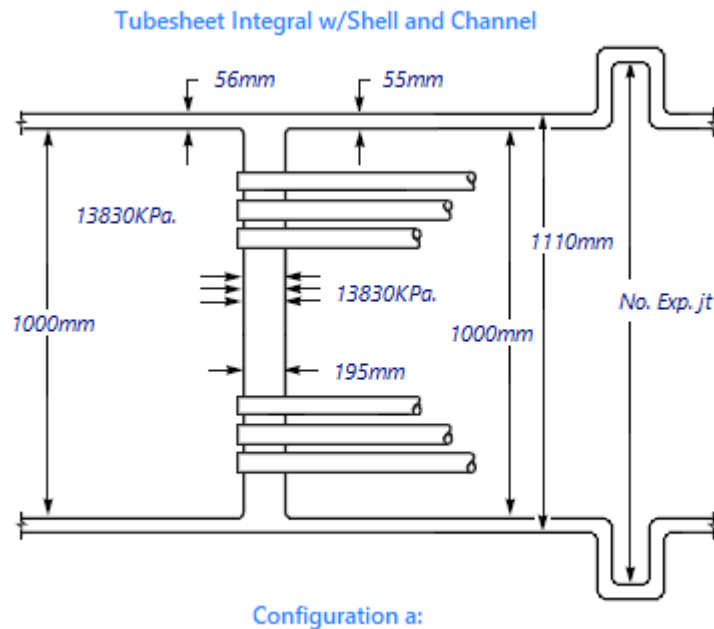
See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Tubesheet Material UNS Number		K03011
Tubesheet Allowable Stress at Temperature	S	151.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	165.00 N./mm ²
Thickness of Tubesheet	h	195.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	1.5000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	1.5000 mm
Tubesheet Outside Diameter	A	1110.000 mm

Area of the Untubed Lanes	AL	0.0 cm ²
---------------------------	----	---------------------

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	0.000 mm
Tube End condition corresponding to Span (l)	k	0.00
Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)		Yes
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Mean Shell Metal Temp. along Shell len.	Tsm	21.00 °C
Mean Tube Metal Temp. along Tube length	Ttm	37.00 °C
Junction Stress Reduction option		Perform Plastic Calculation
Tubesheet Integral with		Both
Tubesheet Extended as Flange		No
Is Exchanger in Creep range (skip EP, Use 3S for Sps)		No



Messages regarding the Tubesheet & Exchanger Input Data:

Warning - Tube end condition k should be bet 0 - 1 (.00).

ASME Division 2 Tubesheet Results per Part 4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	21.0 °C	0.0000115174 /°C
Elastic Mod. at Design Temperature	80.0 °C	0.19933E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Metal Temp. along Len	21.0 °C	0.20271E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Design Temperature	100.0 °C	0.19815E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
Tubes - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	37.0 °C	0.0000116906 /°C
Elastic Mod. at Design Temperature	100.0 °C	0.19815E+09 KPa.
Elastic Mod. at Metal Temp. along Len	37.0 °C	0.20179E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	100.0 °C	0.19815E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
TubeSheet - TM-1 Carbon Steels with C> 0.3%



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ASME TS Calc: 10% SOR

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Elastic Mod. at Design Temperature	100.0 °C	0.19677E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20132E+09 KPa.

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.
Please insure these properties are consistent with the
type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i$ Equation 4.3.1
 $= 0.5 \cdot 14 \cdot (\exp(13830/(126 \cdot 1)) - 1) + 0$
 $= 0.81 + 0$
 $= 0.81 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

Determine Parameter [Mx]:

$= L / \sqrt{(R_o \cdot t)}$
 $= 5610 / \sqrt{(9.5 \cdot 2.5)}$
 $= 1151.148$

Compute the Factor [Ch]:

Since $M_x > 2 \cdot (D_o/t) \cdot 0.94$, $Ch = 0.55 / (D_o/t)$
 $= 0.55 \cdot (19/2.5)$
 $= 0.0724$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$= 1.6 \cdot Ch \cdot E_y (t / D_o)$
 $= 1.6 \cdot 0.072 \cdot 198151376 (2.5/19)$
 $= 3019.102 \text{ N./mm}^2$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$= Fhe / F_y$
 $= 3019.1/189.0$
 $= 15.974$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$= \sigma_{ys} / \sigma_{uts}$
 $= 189/379$
 $= 0.498$

Ratio [AE]:

$= Fhe / E$
 $= 3019/198151376$
 $= 0.015235$

Factor [m1]:

$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$
 $= (\ln(0.5) + (0.20000E-04 - 0.002)) / \ln(\ln(1 + 0.20000E-04) / \ln(1 + 0.002))$
 $= 0.152$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$
 $= 189 (1 + 0.0020) / \ln(1 + 0.0020)^{0.15168217}$
 $= 486.157 \text{ N./mm}^2$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$= m2f / (m2f2 - R)$
 $= 0.6 (1 - 0.5)$
 $= 0.301$



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Curve Fitting Constant [A2]:
 $= (\sigma_{uts} \cdot \exp(m_2)) / (m_2 m_2)$
 $= (379 \cdot \exp(0.3)) / (0.3 \cdot 0.3)$
 $= 735.456 \text{ N./mm}^2$

Material Parameter for Stress-Strain Curve Model [K]:
 $= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$
 $= 1.5 \cdot 0.51.5 - 0.5 \cdot 0.52.5 - 0.53.5$
 $= 0.353$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	189	-1.982	0.350E-07	0.315E-07	0.970E-07	-0.813E-07	0.115E+08	0.131E-02
2	94	-4.808	0.726E-09	0.726E-09	0.194E-07	-0.194E-07	0.154E+09	0.146E-01
3	142	-3.395	0.701E-08	0.697E-08	0.497E-07	-0.493E-07	0.515E+08	0.125E-01
4	165	-2.688	0.166E-07	0.162E-07	0.711E-07	-0.685E-07	0.247E+08	0.852E-02
5	177	-2.335	0.244E-07	0.232E-07	0.835E-07	-0.770E-07	0.169E+08	0.472E-02
6	183	-2.158	0.293E-07	0.273E-07	0.901E-07	-0.800E-07	0.139E+08	0.206E-02
7	186	-2.070	0.321E-07	0.294E-07	0.935E-07	-0.809E-07	0.126E+08	0.472E-03
8	188	-2.026	0.335E-07	0.305E-07	0.952E-07	-0.812E-07	0.120E+08	0.394E-03
9	187	-2.048	0.328E-07	0.299E-07	0.944E-07	-0.810E-07	0.123E+08	0.451E-04
10	187	-2.037	0.332E-07	0.302E-07	0.948E-07	-0.811E-07	0.122E+08	0.173E-03
11	187	-2.042	0.330E-07	0.301E-07	0.946E-07	-0.811E-07	0.123E+08	0.635E-04
12	187	-2.045	0.329E-07	0.300E-07	0.945E-07	-0.811E-07	0.123E+08	0.913E-05
13	187	-2.047	0.328E-07	0.300E-07	0.944E-07	-0.811E-07	0.123E+08	0.180E-04
14	187	-2.046	0.329E-07	0.300E-07	0.944E-07	-0.811E-07	0.123E+08	0.444E-05
15	187	-2.046	0.329E-07	0.300E-07	0.944E-07	-0.811E-07	0.123E+08	0.234E-05
16	187	-2.046	0.329E-07	0.300E-07	0.944E-07	-0.811E-07	0.123E+08	0.105E-05

Factor of Safety [FS]:
 Since $Fic > 0.55 F_y$ and $Fic < F_y$
 $= 2.407 - 0.741 \cdot Fic / F_y$
 $= 2.407 - 0.741 \cdot 187/189$
 $= 1.672$

Allowable Circumferential Compressive Stress [Fha]:
 $= Fic/FS$
 $= 187/1.67$
 $= 112.120 \text{ N./mm}^2$

Allowable External Pressure at the given Thickness [Pa]:
 $= 2 \cdot Fha(t / Do)$
 $= 2 \cdot 112114 (2.5/19)$
 $= 29503.564 \text{ KPa.}$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	1.4685 mm
Allowable Internal Pressure at Corroded thickness	38475.49 KPa.
Required Internal Design Pressure	13830.00 KPa.
Allowable External Pressure at Corroded thickness	29503.56 KPa.
Required External Design Pressure	13830.00 KPa.
Required Thickness due to Shell Side pressure	1.4685 mm

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 14 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend Allwd	Stresses Shear Allwd	Case Type	Pass/ Fail
D1uc	170.490 ...	180	226	48 121	Ps+Pt-Th	D1 Ok
D2uc	146.343 ...	144	226	36 121	Ps+Pt-Th	D2 Ok



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Case: 4 9:41a Aug 12, 2026

D3uc	47.337	...	41	226	12	121	Ps+Pt-Th	D3	Ok
O1uc	75.500	...	134	454	33	121	Ps+Pt+Th	O1	Ok
O2uc	109.542	...	190	454	51	121	Ps+Pt+Th	O2	Ok
O3uc	3.810	...	11	454	3	121	Ps+Pt+Th	O3	Ok
O4uc	27.998	...	52	454	15	121	Ps+Pt+Th	O4	Ok
D1c	172.345	...	183	226	48	121	Ps+Pt-Th-c	D1	Ok
D2c	148.096	...	145	226	36	121	Ps+Pt-Th-c	D2	Ok
D3c	50.927	...	42	226	12	121	Ps+Pt-Th-c	D3	Ok
O1c	78.679	...	137	454	34	121	Ps+Pt+Th-c	O1	Ok
O2c	111.250	...	191	454	51	121	Ps+Pt+Th-c	O2	Ok
O3c	6.810	...	10	454	3	121	Ps+Pt+Th-c	O3	Ok
O4c	30.661	...	53	454	15	121	Ps+Pt+Th-c	O4	Ok

Max: 172.3447 ... mm 0.808 0.423 (Str. Ratio)

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[(+)-Th]:

With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:

Operating Load Cases using the Maximum and Minimum Operating Pressures and Operating Temperatures

Shell Axial Membrane Stress Summary:

Case#	---- Shell Stresses ----				:	---- Shell Band Stress ----				: Pass
	Ten	Allwd	Cmp	Allwd		Ten	Allwd	Cmp	Allwd	
D1uc	7	161	...	...		...	...	...	...	Ok
D2uc	42	161	...	...		...	...	...	...	Ok
D3uc	49	161	...	...		...	...	...	...	Ok
O1uc	20	486	...	...		...	...	...	...	Ok
O2uc	56	486	...	...		...	...	...	...	Ok
O3uc	62	486	...	...		...	...	...	...	Ok
O4uc	14	486	...	...		...	...	...	...	Ok
D1c	7	161	...	...		...	...	...	...	Ok
D2c	43	161	...	...		...	...	...	...	Ok
D3c	50	161	...	...		...	...	...	...	Ok
O1c	21	486	...	...		...	...	...	...	Ok
O2c	57	486	...	...		...	...	...	...	Ok
O3c	64	486	...	...		...	...	...	...	Ok
O4c	14	486	...	...		...	...	...	...	Ok
Max RATIO	0.311		...			...		...		

Tube, Shell and Channel Stress Summary:

Case#	---- Tube Stresses ----				Tube Loads		Shell Stress		Channel Stress		Pass
	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	



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ASME TS Calc: 10% SOR

Case: 4 9:41a Aug 12, 2026

D1uc	16	125	-8	-94	209	1664	56	242	290	478	Ok
D2uc	12	125	-12	-94	154	1664	248	486	20	238	Ok
D3uc	4	125	-3	-94	55	1664	206	242	269	478	Ok
O1uc	21	251	-21	-94	279	3329	32	486	252	478	Ok
O2uc	34	251	-34	-94	455	3329	299	486	58	478	Ok
O3uc	19	251	-19	-94	247	3329	257	486	232	478	Ok
O4uc	23	251	-23	-94	301	3329	51	486	38	478	Ok
D1c	16	125	-9	-94	212	1664	61	486	302	478	Ok
D2c	12	125	-12	-94	153	1664	258	486	23	238	Ok
D3c	4	125	-3	-94	59	1664	211	242	279	478	Ok
O1c	22	251	-22	-94	285	3329	36	486	263	478	Ok
O2c	34	251	-34	-94	452	3329	311	486	62	478	Ok
O3c	18	251	-18	-94	240	3329	264	486	240	478	Ok
O4c	23	251	-23	-94	299	3329	53	486	39	478	Ok

Max RATIO 0.137 0.365 0.137 0.870 0.632

Summary of Thickness Comparisons for 14 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	172.345	195.000	Ok
Tube thickness :	1.468	2.500	Ok
Tube-tubesheet fillet weld leg :	1.500	2.000	Ok
Tube-tubesheet groove weld leg :	1.500	2.000	Ok

Min Shell length of thk, (55.000) adj. to tubesheet: 422.137 mm

Min Channel length of thk, (56.000) adj. to tubesheet: 425.958 mm

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Extends by : +1.050 mm

Due to Pressure Shell Compresses by : -0.411 mm

Due to Pressure + Thermal Shell Extends by : +1.127 mm

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	0 shellside Stress Rat.	Shellside MAWP	0 tubeside Stress Rat.
Tubesheet Bending Stress	17212.10	1.000	21637.27	1.000
Tubesheet Shear Stress	34574.79	1.000	40316.35	1.000
Tube Tensile Stress	108639.05	1.000	149723.94	1.000
Tube Compressive Stress	37760.72	0.385	85181.98	1.000
Tube-Tubesheet Joint load	108639.70	1.000	149724.14	1.000
Shell Stress (Axial, Junction)	17212.10	1.000	21637.27	1.000
Tubesheet-Channel Junction	17212.10	1.000	21637.27	1.000
Tube Pressure Stress	38475.48	1.000	29503.55	1.000
Minimum MAWP	17212.10		21637.27	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	0 shellside Stress Rat.	Shellside MAPnc	0 tubeside Stress Rat.
Tubesheet Bending Stress	18981.93	0.995	23835.15	0.996
Tubesheet Shear Stress	38193.05	1.000	50522.26	1.000
Tube Tensile Stress	120661.51	1.000	164299.58	1.000
Tube Compressive Stress	186665.52	1.000	122034.02	1.000
Tube-Tubesheet Joint load	120661.41	1.000	164299.36	1.000
Shell Stress (Axial, Junction)	18981.93	0.995	23835.15	0.996



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ASME TS Calc: 10% SOR

Case: 4 9:41a Aug 12,2026

Tubesheet-Channel Junction	18981.93	0.995	23835.15	0.996
Tube Pressure Stress	42139.45	1.000	31403.66	1.000
Minimum MAPnc	18981.93		23835.15	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.0775	0.0775	...	...	...	...
2	-0.4103	-0.4105	...	...	...	...
3	-0.3328	-0.3330	...	...	...	...
4	...	...	...	...	...	...
5	0.0775	0.0775	1.0492	1.0498	...	...
6	-0.4103	-0.4105	1.0492	1.0498	...	...
7	-0.3328	-0.3330	1.0492	1.0498	...	...
8	...	...	1.0492	1.0498	...	...

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ASME TS Calc: START UP

Case: 5 9:41a Aug 12, 2026

Input Echo, Tubesheet Number 5, Description: Fixed tubesheet

Shell Data:

Main Shell Description: Shell

Shell Maximum Design Pressure	Psd,max	10000.00 KPa.
Shell Maximum Operating Pressure	Psox,max	10000.00 KPa.
Shell Minimum Operating Pressure	Psox,min	0.00 KPa.
Shell Thickness	ts	55.0000 mm
Shell Internal Corrosion Allowance	cas	1.5000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	1000.000 mm
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	80.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	161.86 N./mm ²
Shell Allowable Stress at Ambient		175.00 N./mm ²

Channel Data:

Channel Description: Chnl. rear

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	10000.00 KPa.
Channel Maximum Operating Pressure	Ptox,max	10000.00 KPa.
Channel Minimum Operating Pressure	Ptox,min	0.00 KPa.
Channel Thickness	tc	56.0000 mm
Channel Internal Corrosion Allowance	cac	1.5000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	1000.000 mm
Channel Design Temperature	TEMPC	100.00 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	159.00 N./mm ²
Channel Allowable Stress at Ambient		175.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	1015
Tube Wall Thickness	et	2.5000 mm
Tube Outside Diameter	D	19.0000 mm
Total Straight Tube Length	Lt	6000.00 mm
Straight Tube Length (bet. inner tubsht faces) L		5610.00 mm
Design Temperature of the Tubes		100.00 °C
Tube Material		SA-334 1
Tube Material UNS Number		K03008
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		126.00 N./mm ²
Tube Allowable Stress At Ambient		138.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	189.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	25.0000 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.0000 mm
Groove Weld Leg	ag	2.0000 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		4.21.2
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	454.384 mm



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ASME TS Calc: START UP

Case: 5 9:41a Aug 12,2026

Largest Center-to-Center Tube Distance	U1	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	5190.7 cm ²
Perimeter of Tube Layout	Cp	2786.6 mm

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf A

Tubesheet Design Metal Temperature	T	100.00 °C
Tubesheet Material		SA-350 LF2

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Tubesheet Material UNS Number		K03011
Tubesheet Allowable Stress at Temperature	S	151.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	165.00 N./mm ²
Thickness of Tubesheet	h	195.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	1.5000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	1.5000 mm
Tubesheet Outside Diameter	A	1110.000 mm

Area of the Untubed Lanes	AL	0.0 cm ²
---------------------------	----	---------------------

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	0.000 mm
Tube End condition corresponding to Span (l)	k	0.00
Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)		Yes
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Mean Shell Metal Temp. along Shell len.	Tsm	40.00 °C
Mean Tube Metal Temp. along Tube length	Ttm	40.00 °C
Junction Stress Reduction option		Perform Plastic Calculation
Tubesheet Integral with		Both
Tubesheet Extended as Flange		No
Is Exchanger in Creep range (skip EP, Use 3S for Sps)		No



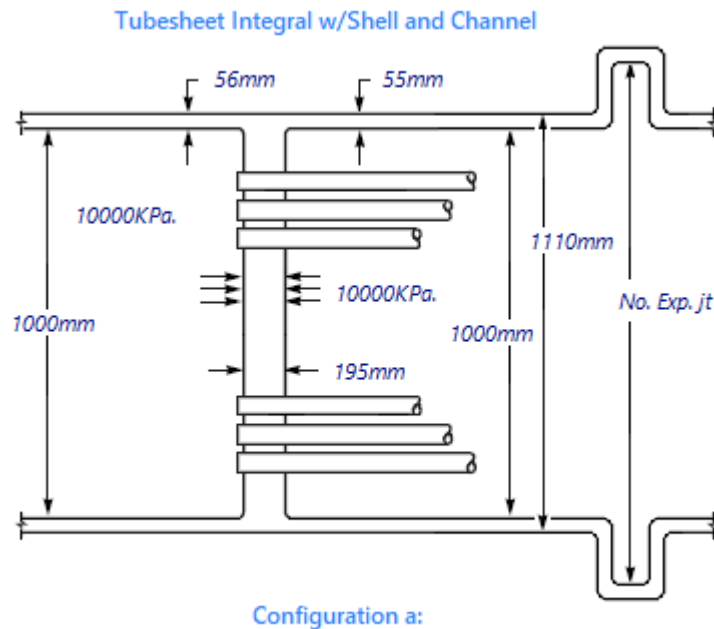
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ASME TS Calc: START UP

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Messages regarding the Tubesheet & Exchanger Input Data:

Warning - Tube end condition k should be bet 0 - 1 (.00).

ASME Division 2 Tubesheet Results per Part 4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	40.0 °C	0.0000117134 /°C
Elastic Mod. at Design Temperature	80.0 °C	0.19933E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Metal Temp. along Len	40.0 °C	0.20162E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Design Temperature	100.0 °C	0.19815E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
Tubes - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	40.0 °C	0.0000117134 /°C
Elastic Mod. at Design Temperature	100.0 °C	0.19815E+09 KPa.
Elastic Mod. at Metal Temp. along Len	40.0 °C	0.20162E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	100.0 °C	0.19815E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
TubeSheet - TM-1 Carbon Steels with C> 0.3%



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ASME TS Calc: START UP

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Elastic Mod. at Design Temperature	100.0 °C	0.19677E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20132E+09 KPa.

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.
Please insure these properties are consistent with the
type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c$ Equation 4.3.1
 $= 0.5 \cdot 14 \cdot (\exp(10000/(126 \cdot 1)) - 1) + 0$
 $= 0.58 + 0$
 $= 0.58 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

Determine Parameter [Mx]:

$= L / \sqrt{R_o \cdot t}$
 $= 5610 / \sqrt{9.5 \cdot 2.5}$
 $= 1151.148$

Compute the Factor [Ch]:

Since $Mx > 2 \cdot (Do/t)^{0.94}$, $Ch = 0.55 / (Do/t)$
 $= 0.55 \cdot (19/2.5)$
 $= 0.0724$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$= 1.6 \cdot Ch \cdot E_y (t / Do)$
 $= 1.6 \cdot 0.072 \cdot 198151376 (2.5/19)$
 $= 3019.102 \text{ N./mm}^2$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$= Fhe / F_y$
 $= 3019.1/189.0$
 $= 15.974$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$= \sigma_{ys} / \sigma_{uts}$
 $= 189/379$
 $= 0.498$

Ratio [AE]:

$= Fhe / E$
 $= 3019/198151376$
 $= 0.015235$

Factor [m1]:

$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$
 $= (\ln(0.5) + (0.20000E-04 - 0.002)) / \ln(\ln(1 + 0.20000E-04) / \ln(1 + 0.002))$
 $= 0.152$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$
 $= 189 (1 + 0.0020) / \ln(1 + 0.0020)^{0.15168217}$
 $= 486.157 \text{ N./mm}^2$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$= m2f / (m2f2 - R)$
 $= 0.6 (1 - 0.5)$
 $= 0.301$



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Curve Fitting Constant [A2]:
 $= (\sigma_{uts} \cdot \exp(m_2)) / (m_2 m_2)$
 $= (379 \cdot \exp(0.3)) / (0.3 \cdot 0.3)$
 $= 735.456 \text{ N./mm}^2$

Material Parameter for Stress-Strain Curve Model [K]:
 $= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$
 $= 1.5 \cdot 0.51.5 - 0.5 \cdot 0.52.5 - 0.53.5$
 $= 0.353$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	189	-1.982	0.350E-07	0.315E-07	0.970E-07	-0.813E-07	0.115E+08	0.131E-02
2	94	-4.808	0.726E-09	0.726E-09	0.194E-07	-0.194E-07	0.154E+09	0.146E-01
3	142	-3.395	0.701E-08	0.697E-08	0.497E-07	-0.493E-07	0.515E+08	0.125E-01
4	165	-2.688	0.166E-07	0.162E-07	0.711E-07	-0.685E-07	0.247E+08	0.852E-02
5	177	-2.335	0.244E-07	0.232E-07	0.835E-07	-0.770E-07	0.169E+08	0.472E-02
6	183	-2.158	0.293E-07	0.273E-07	0.901E-07	-0.800E-07	0.139E+08	0.206E-02
7	186	-2.070	0.321E-07	0.294E-07	0.935E-07	-0.809E-07	0.126E+08	0.472E-03
8	188	-2.026	0.335E-07	0.305E-07	0.952E-07	-0.812E-07	0.120E+08	0.394E-03
9	187	-2.048	0.328E-07	0.299E-07	0.944E-07	-0.810E-07	0.123E+08	0.451E-04
10	187	-2.037	0.332E-07	0.302E-07	0.948E-07	-0.811E-07	0.122E+08	0.173E-03
11	187	-2.042	0.330E-07	0.301E-07	0.946E-07	-0.811E-07	0.123E+08	0.635E-04
12	187	-2.045	0.329E-07	0.300E-07	0.945E-07	-0.811E-07	0.123E+08	0.913E-05
13	187	-2.047	0.328E-07	0.300E-07	0.944E-07	-0.811E-07	0.123E+08	0.180E-04
14	187	-2.046	0.329E-07	0.300E-07	0.944E-07	-0.811E-07	0.123E+08	0.444E-05
15	187	-2.046	0.329E-07	0.300E-07	0.944E-07	-0.811E-07	0.123E+08	0.234E-05
16	187	-2.046	0.329E-07	0.300E-07	0.944E-07	-0.811E-07	0.123E+08	0.105E-05

Factor of Safety [FS]:
 Since $Fic > 0.55 F_y$ and $Fic < F_y$
 $= 2.407 - 0.741 \cdot Fic / F_y$
 $= 2.407 - 0.741 \cdot 187/189$
 $= 1.672$

Allowable Circumferential Compressive Stress [Fha]:
 $= Fic/FS$
 $= 187/1.67$
 $= 112.120 \text{ N./mm}^2$

Allowable External Pressure at the given Thickness [Pa]:
 $= 2 \cdot Fha(t / Do)$
 $= 2 \cdot 112114 (2.5/19)$
 $= 29503.564 \text{ KPa.}$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	1.1767 mm
Allowable Internal Pressure at Corroded thickness	38475.49 KPa.
Required Internal Design Pressure	10000.00 KPa.
Allowable External Pressure at Corroded thickness	29503.56 KPa.
Required External Design Pressure	10000.00 KPa.
Required Thickness due to Shell Side pressure	1.1767 mm

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 14 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend Allwd	Stresses Shear Allwd	Case Type	Pass/ Fail
D1uc	138.549 ...	130	226	35 121	Ps+Pt-Th	D1 Ok
D2uc	115.419 ...	104	226	26 121	Ps+Pt-Th	D2 Ok



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Case: 5 9:41a Aug 12, 2026

D3uc	33.074	...	29	226	8	121	Ps+Pt-Th	D3	Ok
O1uc	76.717	...	130	454	35	121	Ps+Pt+Th	O1	Ok
O2uc	57.240	...	104	454	26	121	Ps+Pt+Th	O2	Ok
O3uc	12.694	...	29	454	8	121	Ps+Pt+Th	O3	Ok
O4uc	3.810	...	...	454	...	121	Ps+Pt+Th	O4	Ok
D1c	140.617	...	132	226	35	121	Ps+Pt-Th-c	D1	Ok
D2c	117.386	...	105	226	26	121	Ps+Pt-Th-c	D2	Ok
D3c	36.669	...	31	226	9	121	Ps+Pt-Th-c	D3	Ok
O1c	79.367	...	132	454	35	121	Ps+Pt+Th-c	O1	Ok
O2c	59.930	...	105	454	26	121	Ps+Pt+Th-c	O2	Ok
O3c	16.109	...	31	454	9	121	Ps+Pt+Th-c	O3	Ok
O4c	6.810	...	...	454	...	121	Ps+Pt+Th-c	O4	Ok

Max: 140.6172 ... mm 0.584 0.289 (Str. Ratio)

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[(+)-Th]:

With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:

Operating Load Cases using the Maximum and Minimum Operating Pressures and Operating Temperatures

Shell Axial Membrane Stress Summary:

Case#	---- Shell Stresses ----				:	---- Shell Band Stress ----				: Pass
	Ten	Allwd	Cmp	Allwd		Ten	Allwd	Cmp	Allwd	
D1uc	5	161	...	...		...	...	...	...	Ok
D2uc	30	161	...	...		...	...	...	...	Ok
D3uc	35	161	...	...		...	...	...	...	Ok
O1uc	5	486	...	...		...	...	...	...	Ok
O2uc	30	486	...	...		...	...	...	...	Ok
O3uc	35	486	...	...		...	...	...	...	Ok
O4uc	...	486	...	...		...	...	...	...	Ok
D1c	5	161	...	...		...	...	...	...	Ok
D2c	31	161	...	...		...	...	...	...	Ok
D3c	36	161	...	...		...	...	...	...	Ok
O1c	5	486	...	...		...	...	...	...	Ok
O2c	31	486	...	...		...	...	...	...	Ok
O3c	36	486	...	...		...	...	...	...	Ok
O4c	...	486	...	...		...	...	...	...	Ok
Max RATIO	0.225		...			...		...		

Tube, Shell and Channel Stress Summary:

Case#	---- Tube Stresses ----				Tube Loads		Shell Stress		Channel Stress		Pass
	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	



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ASME TS Calc: START UP

Case: 5 9:41a Aug 12, 2026

D1uc	11	125	-5	-94	151	1664	40	242	210	238	Ok
D2uc	8	125	-8	-94	111	1664	179	242	15	238	Ok
D3uc	3	125	-2	-94	40	1664	149	242	195	238	Ok
O1uc	11	251	-5	-94	151	3329	40	486	210	478	Ok
O2uc	8	251	-8	-94	111	3329	179	486	15	478	Ok
O3uc	3	251	-2	-94	40	3329	149	486	195	478	Ok
O4uc	...	251	...	...	...	3329	...	486	...	478	Ok
D1c	12	125	-6	-94	153	1664	44	242	219	238	Ok
D2c	8	125	-8	-94	110	1664	187	242	17	238	Ok
D3c	3	125	-2	-94	43	1664	153	242	202	238	Ok
O1c	12	251	-6	-94	153	3329	44	486	219	478	Ok
O2c	8	251	-8	-94	110	3329	187	486	17	478	Ok
O3c	3	251	-2	-94	43	3329	153	486	202	478	Ok
O4c	...	251	...	...	...	3329	...	486	...	478	Ok

Max RATIO 0.092 0.089 0.092 0.769 0.917

Summary of Thickness Comparisons for 14 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	140.617	195.000	Ok
Tube thickness :	1.177	2.500	Ok
Tube-tubesheet fillet weld leg :	1.500	2.000	Ok
Tube-tubesheet groove weld leg :	1.500	2.000	Ok

Min Shell length of thk, (55.000) adj. to tubesheet: 422.137 mm

Min Channel length of thk, (56.000) adj. to tubesheet: 425.958 mm

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Extends by : +.000 mm
 Due to Pressure Shell Compresses by : -0.297 mm
 Due to Pressure + Thermal Shell Compresses by : -0.297 mm

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	0 shellside Stress Rat.	Shellside MAWP	0 tubeside Stress Rat.
Tubesheet Bending Stress	17212.10	1.000	21637.27	1.000
Tubesheet Shear Stress	34574.79	1.000	45901.55	1.000
Tube Tensile Stress	108639.05	1.000	149723.94	1.000
Tube Compressive Stress	34293.31	0.225	112293.09	1.000
Tube-Tubesheet Joint load	108639.70	1.000	149724.14	1.000
Shell Stress (Axial, Junction)	17212.10	1.000	21637.27	1.000
Tubesheet-Channel Junction	17212.10	1.000	21637.27	1.000
Tube Pressure Stress	38475.48	1.000	29503.55	1.000
Minimum MAWP	17212.10		21637.27	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	0 shellside Stress Rat.	Shellside MAPnc	0 tubeside Stress Rat.
Tubesheet Bending Stress	18981.93	0.995	23835.15	0.996
Tubesheet Shear Stress	38193.05	1.000	50522.26	1.000
Tube Tensile Stress	120661.51	1.000	164299.58	1.000
Tube Compressive Stress	186665.52	1.000	122034.02	1.000
Tube-Tubesheet Joint load	120661.41	1.000	164299.36	1.000
Shell Stress (Axial, Junction)	18981.93	0.995	23835.15	0.996



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ASME TS Calc: START UP

Case: 5 9:41a Aug 12,2026

Tubesheet-Channel Junction	18981.93	0.995	23835.15	0.996
Tube Pressure Stress	42139.45	1.000	31403.66	1.000
Minimum MAPnc	18981.93		23835.15	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.0560	0.0561	...	...	...	...
2	-0.2967	-0.2968	...	...	...	...
3	-0.2406	-0.2408	...	...	...	...
4	...	...	...	...	...	...
5	0.0560	0.0561	...	...	...	...
6	-0.2967	-0.2968	...	...	...	...
7	-0.2406	-0.2408	...	...	...	...
8	...	...	...	...	...	...

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ASME TS Calc: SHUT DOWN

Case: 6 9:41a Aug 12, 2026

Input Echo, Tubesheet Number 6, Description: Fixed tubesheet

Shell Data:

Main Shell Description: Shell

Shell Maximum Design Pressure	Psd,max	13020.00 KPa.
Shell Maximum Operating Pressure	Psox,max	13020.00 KPa.
Shell Minimum Operating Pressure	Psox,min	0.00 KPa.
Shell Thickness	ts	55.0000 mm
Shell Internal Corrosion Allowance	cas	1.5000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	1000.000 mm
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	80.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	161.86 N./mm ²
Shell Allowable Stress at Ambient		175.00 N./mm ²

Channel Data:

Channel Description: Chnl. rear

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	13200.00 KPa.
Channel Maximum Operating Pressure	Ptox,max	13200.00 KPa.
Channel Minimum Operating Pressure	Ptox,min	0.00 KPa.
Channel Thickness	tc	56.0000 mm
Channel Internal Corrosion Allowance	cac	1.5000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	1000.000 mm
Channel Design Temperature	TEMPC	100.00 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	159.00 N./mm ²
Channel Allowable Stress at Ambient		175.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	1015
Tube Wall Thickness	et	2.5000 mm
Tube Outside Diameter	D	19.0000 mm
Total Straight Tube Length	Lt	6000.00 mm
Straight Tube Length (bet. inner tubsht faces) L		5610.00 mm
Design Temperature of the Tubes		100.00 °C
Tube Material		SA-334 1
Tube Material UNS Number		K03008
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		126.00 N./mm ²
Tube Allowable Stress At Ambient		138.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	189.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	25.0000 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.0000 mm
Groove Weld Leg	ag	2.0000 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		4.21.2
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	454.384 mm



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Largest Center-to-Center Tube Distance	U1	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	5190.7 cm ²
Perimeter of Tube Layout	Cp	2786.6 mm

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf A

Tubesheet Design Metal Temperature	T	100.00 °C
Tubesheet Material		SA-350 LF2

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Tubesheet Material UNS Number		K03011
Tubesheet Allowable Stress at Temperature	S	151.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	165.00 N./mm ²
Thickness of Tubesheet	h	195.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	1.5000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	1.5000 mm
Tubesheet Outside Diameter	A	1110.000 mm

Area of the Untubed Lanes	AL	0.0 cm ²
---------------------------	----	---------------------

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	0.000 mm
Tube End condition corresponding to Span (l)	k	0.00
Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)		Yes
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Mean Shell Metal Temp. along Shell len.	Tsm	20.00 °C
Mean Tube Metal Temp. along Tube length	Ttm	39.00 °C
Junction Stress Reduction option		Perform Plastic Calculation
Tubesheet Integral with		Both
Tubesheet Extended as Flange		No
Is Exchanger in Creep range (skip EP, Use 3S for Sps)		No



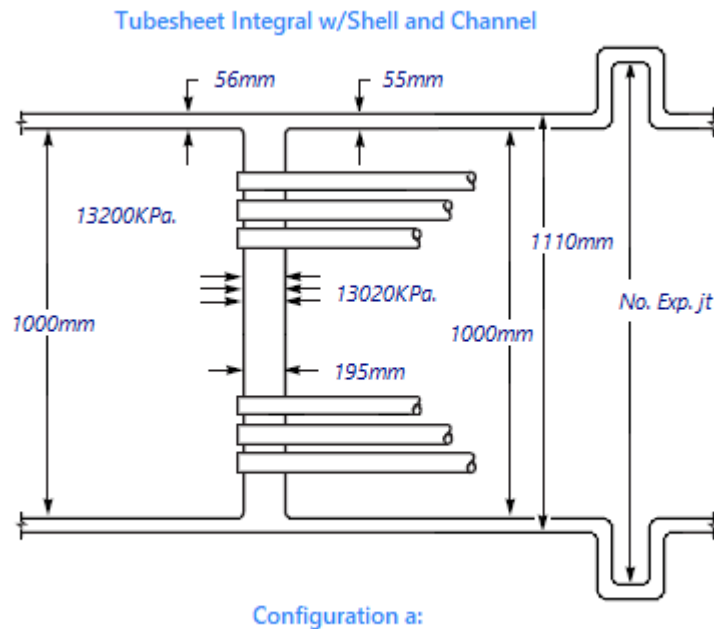
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Messages regarding the Tubesheet & Exchanger Input Data:

Warning - Tube end condition k should be bet 0 - 1 (.00).

ASME Division 2 Tubesheet Results per Part 4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	20.0 °C	0.0000115030 /°C
Elastic Mod. at Design Temperature	80.0 °C	0.19933E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Metal Temp. along Len	20.0 °C	0.20277E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Design Temperature	100.0 °C	0.19815E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
Tubes - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	39.0 °C	0.0000117070 /°C
Elastic Mod. at Design Temperature	100.0 °C	0.19815E+09 KPa.
Elastic Mod. at Metal Temp. along Len	39.0 °C	0.20168E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	100.0 °C	0.19815E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
TubeSheet - TM-1 Carbon Steels with C> 0.3%



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Elastic Mod. at Design Temperature	100.0 °C	0.19677E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20132E+09 KPa.

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.
Please insure these properties are consistent with the
type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c$ Equation 4.3.1
 $= 0.5 \cdot 14 \cdot (\exp(13200/(126 \cdot 1)) - 1) + 0$
 $= 0.77 + 0$
 $= 0.77 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

Determine Parameter [Mx]:

$= L / \sqrt{R_o \cdot t}$
 $= 5610 / \sqrt{9.5 \cdot 2.5}$
 $= 1151.148$

Compute the Factor [Ch]:

Since $Mx > 2 \cdot (Do/t)^{0.94}$, $Ch = 0.55 / (Do/t)$
 $= 0.55 \cdot (19/2.5)$
 $= 0.0724$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$= 1.6 \cdot Ch \cdot E_y (t / Do)$
 $= 1.6 \cdot 0.072 \cdot 198151376 (2.5/19)$
 $= 3019.102 \text{ N./mm}^2$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$= Fhe / F_y$
 $= 3019.1/189.0$
 $= 15.974$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$= \sigma_{ys} / \sigma_{uts}$
 $= 189/379$
 $= 0.498$

Ratio [AE]:

$= Fhe / E$
 $= 3019/198151376$
 $= 0.015235$

Factor [m1]:

$= (\ln(r) + (\epsilon_{ps} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{ps}) / \ln(1 + \epsilon_{sys}))$
 $= (\ln(0.5) + (0.20000E-04 - 0.002)) / \ln(\ln(1 + 0.20000E-04) / \ln(1 + 0.002))$
 $= 0.152$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$
 $= 189 (1 + 0.0020) / \ln(1 + 0.0020)^{0.15168217}$
 $= 486.157 \text{ N./mm}^2$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$= m2f / (m2f2 - R)$
 $= 0.6 (1 - 0.5)$
 $= 0.301$



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Curve Fitting Constant [A2]:
 $= (\sigma_{uts} \cdot \exp(m_2)) / (m_2 m_2)$
 $= (379 \cdot \exp(0.3)) / (0.3 \cdot 0.3)$
 $= 735.456 \text{ N./mm}^2$

Material Parameter for Stress-Strain Curve Model [K]:
 $= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$
 $= 1.5 \cdot 0.51.5 - 0.5 \cdot 0.52.5 - 0.53.5$
 $= 0.353$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	189	-1.982	0.350E-07	0.315E-07	0.970E-07	-0.813E-07	0.115E+08	0.131E-02
2	94	-4.808	0.726E-09	0.726E-09	0.194E-07	-0.194E-07	0.154E+09	0.146E-01
3	142	-3.395	0.701E-08	0.697E-08	0.497E-07	-0.493E-07	0.515E+08	0.125E-01
4	165	-2.688	0.166E-07	0.162E-07	0.711E-07	-0.685E-07	0.247E+08	0.852E-02
5	177	-2.335	0.244E-07	0.232E-07	0.835E-07	-0.770E-07	0.169E+08	0.472E-02
6	183	-2.158	0.293E-07	0.273E-07	0.901E-07	-0.800E-07	0.139E+08	0.206E-02
7	186	-2.070	0.321E-07	0.294E-07	0.935E-07	-0.809E-07	0.126E+08	0.472E-03
8	188	-2.026	0.335E-07	0.305E-07	0.952E-07	-0.812E-07	0.120E+08	0.394E-03
9	187	-2.048	0.328E-07	0.299E-07	0.944E-07	-0.810E-07	0.123E+08	0.451E-04
10	187	-2.037	0.332E-07	0.302E-07	0.948E-07	-0.811E-07	0.122E+08	0.173E-03
11	187	-2.042	0.330E-07	0.301E-07	0.946E-07	-0.811E-07	0.123E+08	0.635E-04
12	187	-2.045	0.329E-07	0.300E-07	0.945E-07	-0.811E-07	0.123E+08	0.913E-05
13	187	-2.047	0.328E-07	0.300E-07	0.944E-07	-0.811E-07	0.123E+08	0.180E-04
14	187	-2.046	0.329E-07	0.300E-07	0.944E-07	-0.811E-07	0.123E+08	0.444E-05
15	187	-2.046	0.329E-07	0.300E-07	0.944E-07	-0.811E-07	0.123E+08	0.234E-05
16	187	-2.046	0.329E-07	0.300E-07	0.944E-07	-0.811E-07	0.123E+08	0.105E-05

Factor of Safety [FS]:
 Since $Fic > 0.55 F_y$ and $Fic < F_y$
 $= 2.407 - 0.741 \cdot Fic / F_y$
 $= 2.407 - 0.741 \cdot 187/189$
 $= 1.672$

Allowable Circumferential Compressive Stress [Fha]:
 $= Fic/FS$
 $= 187/1.67$
 $= 112.120 \text{ N./mm}^2$

Allowable External Pressure at the given Thickness [Pa]:
 $= 2 \cdot Fha(t / Do)$
 $= 2 \cdot 112114 (2.5/19)$
 $= 29503.564 \text{ KPa.}$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	1.4087 mm
Allowable Internal Pressure at Corroded thickness	38475.49 KPa.
Required Internal Design Pressure	13200.00 KPa.
Allowable External Pressure at Corroded thickness	29503.56 KPa.
Required External Design Pressure	13020.00 KPa.
Required Thickness due to Shell Side pressure	1.4087 mm

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 14 Load Cases:

Case#	--Reqd. Thk. + CA		---- Tubesheet Stresses				Case Type	Pass/Fail	
	Tbsht	Extnsn	Bend	Allwd	Shear	Allwd			
D1uc 165.714	...	...	172	226	46	121	Ps+Pt-Th	D1	Ok
D2uc 140.420	...	...	135	226	34	121	Ps+Pt-Th	D2	Ok



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D3uc	47.185	...	40	226	12	121	Ps+Pt-Th	D3	Ok
O1uc	63.575	...	118	454	28	121	Ps+Pt+Th	O1	Ok
O2uc	110.136	...	190	454	52	121	Ps+Pt+Th	O2	Ok
O3uc	7.628	...	21	454	6	121	Ps+Pt+Th	O3	Ok
O4uc	34.425	...	62	454	17	121	Ps+Pt+Th	O4	Ok
D1c	167.601	...	175	226	46	121	Ps+Pt-Th-c	D1	Ok
D2c	142.213	...	137	226	34	121	Ps+Pt-Th-c	D2	Ok
D3c	50.746	...	42	226	12	121	Ps+Pt-Th-c	D3	Ok
O1c	67.048	...	120	454	29	121	Ps+Pt+Th-c	O1	Ok
O2c	111.786	...	191	454	52	121	Ps+Pt+Th-c	O2	Ok
O3c	10.003	...	21	454	6	121	Ps+Pt+Th-c	O3	Ok
O4c	36.973	...	62	454	17	121	Ps+Pt+Th-c	O4	Ok

Max: 167.6008 ... mm 0.772 0.428 (Str. Ratio)

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[(+)-Th]:

With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:

Operating Load Cases using the Maximum and Minimum Operating Pressures and Operating Temperatures

Shell Axial Membrane Stress Summary:

Case#	---- Shell Stresses ----				:	---- Shell Band Stress ----				: Pass
	Ten	Allwd	Cmp	Allwd		Ten	Allwd	Cmp	Allwd	
D1uc	6	161	...	...		...	...	...	...	Ok
D2uc	40	161	...	...		...	...	...	...	Ok
D3uc	46	161	...	...		...	...	...	...	Ok
O1uc	23	486	...	...		...	...	...	...	Ok
O2uc	56	486	...	...		...	...	...	...	Ok
O3uc	62	486	...	...		...	...	...	...	Ok
O4uc	16	486	...	...		...	...	...	...	Ok
D1c	7	161	...	...		...	...	...	...	Ok
D2c	41	161	...	...		...	...	...	...	Ok
D3c	47	161	...	...		...	...	...	...	Ok
O1c	23	486	...	...		...	...	...	...	Ok
O2c	57	486	...	...		...	...	...	...	Ok
O3c	64	486	...	...		...	...	...	...	Ok
O4c	16	486	...	...		...	...	...	...	Ok
Max RATIO	0.293		...			...		...		

Tube, Shell and Channel Stress Summary:

Case#	---- Tube Stresses ----				Tube Loads		Shell Stress		Channel Stress		Pass
	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	



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Case: 6 9:41a Aug 12, 2026

D1uc	15	125	-7	-94	199	1664	53	242	277	478	Ok
D2uc	11	125	-11	-94	145	1664	233	242	19	238	Ok
D3uc	4	125	-3	-94	54	1664	193	242	257	478	Ok
O1uc	23	251	-23	-94	307	3329	25	486	232	478	Ok
O2uc	38	251	-38	-94	503	3329	294	486	64	478	Ok
O3uc	23	251	-23	-94	304	3329	254	486	213	478	Ok
O4uc	27	251	-27	-94	358	3329	61	486	45	478	Ok
D1c	15	125	-8	-94	202	1664	58	242	289	478	Ok
D2c	11	125	-11	-94	144	1664	243	486	22	238	Ok
D3c	4	125	-3	-94	59	1664	198	242	267	478	Ok
O1c	24	251	-24	-94	312	3329	28	486	242	478	Ok
O2c	38	251	-38	-94	499	3329	306	486	68	478	Ok
O3c	22	251	-22	-94	297	3329	261	486	220	478	Ok
O4c	27	251	-27	-94	355	3329	63	486	46	478	Ok

Max RATIO 0.151 0.403 0.151 0.962 0.604

Summary of Thickness Comparisons for 14 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	167.601	195.000	Ok
Tube thickness :	1.409	2.500	Ok
Tube-tubesheet fillet weld leg :	1.500	2.000	Ok
Tube-tubesheet groove weld leg :	1.500	2.000	Ok

Min Shell length of thk, (55.000) adj. to tubesheet: 422.137 mm

Min Channel length of thk, (56.000) adj. to tubesheet: 425.958 mm

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Extends by : +1.247 mm

Due to Pressure Shell Compresses by : -0.386 mm

Due to Pressure + Thermal Shell Extends by : +1.321 mm

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	0 shellside Stress Rat.	Shellside MAWP	0 tubeside Stress Rat.
Tubesheet Bending Stress	17212.10	1.000	21637.27	1.000
Tubesheet Shear Stress	34574.79	1.000	39265.95	1.000
Tube Tensile Stress	108639.05	1.000	149723.94	1.000
Tube Compressive Stress	38412.84	0.415	80083.12	1.000
Tube-Tubesheet Joint load	108639.70	1.000	149724.14	1.000
Shell Stress (Axial, Junction)	17212.10	1.000	21637.27	1.000
Tubesheet-Channel Junction	17212.10	1.000	21637.27	1.000
Tube Pressure Stress	38475.48	1.000	29503.55	1.000
Minimum MAWP	17212.10		21637.27	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	0 shellside Stress Rat.	Shellside MAPnc	0 tubeside Stress Rat.
Tubesheet Bending Stress	18981.93	0.995	23835.15	0.996
Tubesheet Shear Stress	38193.05	1.000	50522.26	1.000
Tube Tensile Stress	120661.51	1.000	164299.58	1.000
Tube Compressive Stress	186665.52	1.000	122034.02	1.000
Tube-Tubesheet Joint load	120661.41	1.000	164299.36	1.000
Shell Stress (Axial, Junction)	18981.93	0.995	23835.15	0.996



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Tubesheet-Channel Junction	18981.93	0.995	23835.15	0.996
Tube Pressure Stress	42139.45	1.000	31403.66	1.000
Minimum MAPnc	18981.93		23835.15	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.0740	0.0740	...	...	...	...
2	-0.3863	-0.3865	...	...	...	...
3	-0.3123	-0.3125	...	...	...	...
4	...	...	...	...	...	...
5	0.0740	0.0740	1.2466	1.2472	...	...
6	-0.3863	-0.3865	1.2466	1.2472	...	...
7	-0.3123	-0.3125	1.2466	1.2472	...	...
8	...	...	1.2466	1.2472	...	...

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PV Elite® Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: